

Enterprise-scale DBMS and SQL basics

Week 8

Dr. Farshid Keivanian

1. Introduction
2. Enterprise-scale DBMS
3. Cloud Models (IaaS, PaaS, SaaS)
4. Provisioning Azure SQL
5. SQL Basics
6. Security in Azure – RBAC (Role-Based Access Control)
7. Week 8 Tutorial Activity (Step-by-Step)
8. Review & Reflection
9. Discussion & Role-Play Questions

- This week, we move from small-scale (like Access) to enterprise-scale databases (like SQL Server & Azure SQL).
- *Analogy: Think of Access like a family car (good for small trips), and Azure SQL like a train network (handles thousands of passengers daily).*
- By the end, you should:
 1. Understand enterprise DBMS basics.
 2. Provision a relational database in Azure.
 3. Write simple SQL queries.

- Enterprise DBMS = handles very large data, many users, high security
- Used in banks, airlines, hospitals, universities
- MS Access → simple, desktop-based
- Azure SQL → cloud-based, scalable

Comparison Table

Feature	MS Access (Small-scale)	Azure SQL (Enterprise)
Data Size	Small (MBs–GBs)	Huge (TBs–PBs)
Users	1–10	1000s
Security	Basic	Advanced (RBAC + encryption)
Cost	One-time	Subscription (monthly)
Example (Aus)	Café loyalty program	Qantas booking system

- **Access DB:** You'll see the file size is small and usually only supports a handful of users.
- **Scaling to Woolworths:** Problems would include slow performance, frequent crashes, limited security, and no ability to handle thousands of users or massive data.
- **Discussion point:** Woolworths would never run on Access because it needs a system that can handle millions of transactions daily, strict security, and high availability — things only enterprise systems like Azure SQL or Oracle can provide.

IaaS, PaaS, SaaS – Simple Explanation

- **IaaS:** You rent a server. You manage everything yourself.
- **PaaS:** You get a managed platform. Example: **Azure SQL** (Microsoft handles backups & security).
- **SaaS:** Software delivered to you, ready to use. Example: **Gmail, Office 365.**

Cloud Models (IaaS, PaaS, SaaS)

- IaaS = Infrastructure as a Service
- PaaS = Platform as a Service
- SaaS = Software as a Service

IaaS (Infrastructure as a Service)

You rent the basic building blocks (servers, storage, network)

You control almost everything

- You manage: OS, apps, data
- Provider manages: hardware

Think: “I get the computer, I do the rest”

PaaS (Platform as a Service)

You get a ready-made environment to build/run apps

- You manage: your app + data
- Provider manages: OS, runtime, updates

Think: “I just build my app”

SaaS (Software as a Service)

You just use the software

- You manage: nothing technical
- Provider manages: everything

Think: “I just use it”

Easy memory trick

- IaaS → Infrastructure → lowest level
- PaaS → Platform → middle
- SaaS → Software → highest (user level)

Cloud Models (IaaS, PaaS, SaaS)

IaaS = control everything

PaaS = build apps only

SaaS = just use the software

Analogy

- **IaaS** = renting a block of land & building your own house.
- **PaaS** = renting an apartment with services included.
- **SaaS** = Airbnb – just move in and use it, no management.

Discussion Question)

Classify these: Office 365, Azure SQL, Google Drive → IaaS, PaaS, SaaS

Model Answer:

- Office 365 → SaaS (ready-to-use software)
- Azure SQL → PaaS (managed database platform)
- Google Drive → SaaS (software/storage service you just use)

Provisioning Azure SQL

Provisioning = setting up a cloud service.

Azure SQL steps (overview):

- Choose subscription (student/edu).
- Pick/create a resource group.
- Database name = StudentIDName.
- Admin login + password.
- Region = *Australia East (Sydney)*.

Example

- DB Name: A00127870_SajanDB
- Resource Group: StudentRG2025

SQL = Structured Query Language

- The common language all databases understand.
- You use it to ask for information.

Example Queries

- `SELECT * FROM Customers;` → shows *all* details of all customers.
- `SELECT FirstName, LastName FROM Employees WHERE Department='IT';` → shows only names of employees in IT.

Analogy

SQL is like ordering food in a restaurant:

- You ask (query) → “I want a pizza.”
- The kitchen (database) prepares and delivers your order.

Security in Azure – RBAC (Role-Based Access Control)

- Security Principal = Who → e.g., user, student, staff
- Role = What they can do → read, write, delete
- Scope = Where → database, table, or resource

Example

A bank teller can *view customer accounts* but cannot *change system passwords*.

Security in Azure – RBAC (Role-Based Access Control)

Hands-on Debate

Question: Should students have full admin rights on Uni DBs?

Answer:

No — giving students full admin rights is risky. They might:

- Accidentally delete or change important data
- Access private information
- Cause security or system issues

Security in Azure – RBAC (Role-Based Access Control)

Better approach:

- Students should only get the role and scope they need (e.g., read-only access for learning).
- Admin rights stay with IT staff to keep the system safe.

Week 8 Tutorial Activity (Step-by-Step)

Week 8+ uses Virtual Lab in browser, no install needed.

- **Topic:** Exploring Azure SQL Database
- **Focus:** How to create (provision), secure, and query a cloud database

- **Why important?**
 - Real-world companies use cloud DBs (banks, airlines, supermarkets)
 - Same SQL language works in *Access and Azure*
 - Learn how to scale from small desktop DBs → enterprise cloud systems

Learning Goals for Week 8

- Understand how to provision an Azure SQL Database
- Practice SQL queries in the cloud
- Compare Access vs Azure results
- Apply security basics (RBAC roles & scope)

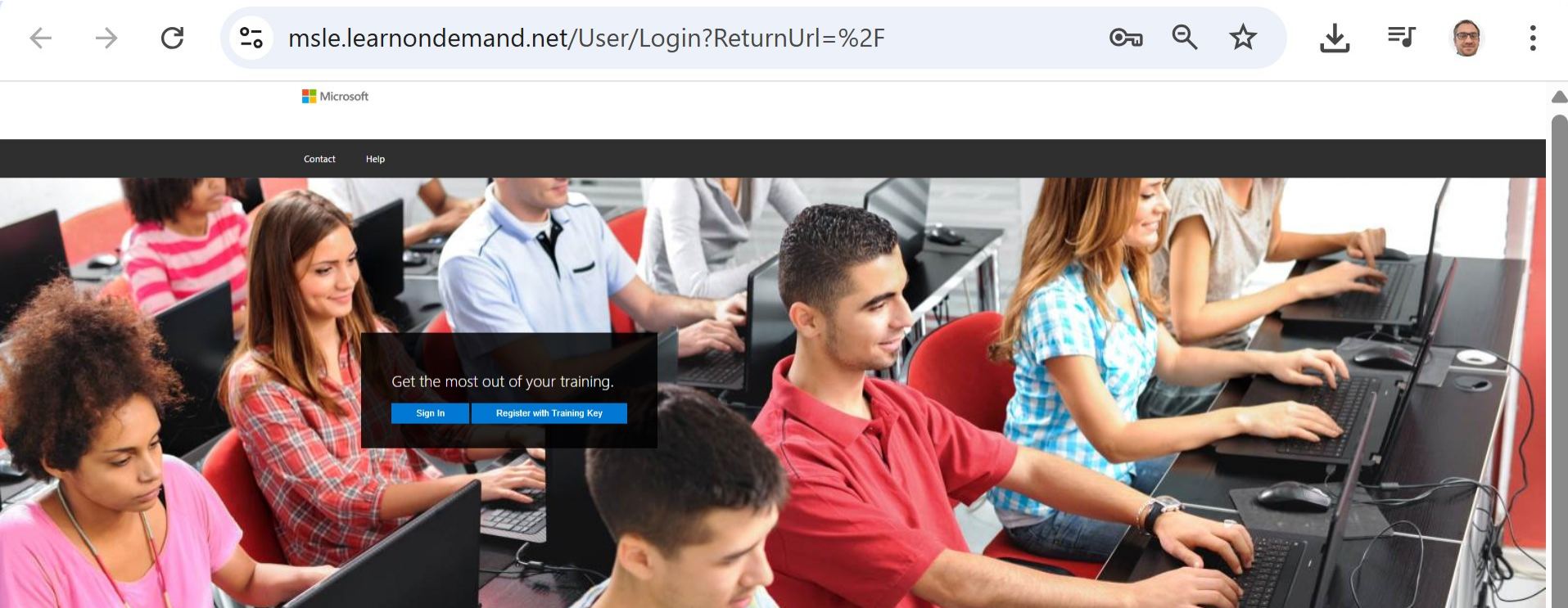
Analogy

Running queries in Access is like using a calculator.

Running queries in Azure is like using a supercomputer that many people can share safely.

Week 8 Tutorial Activity (Step-by-Step)

- Go to <https://msle.learndemand.net/>

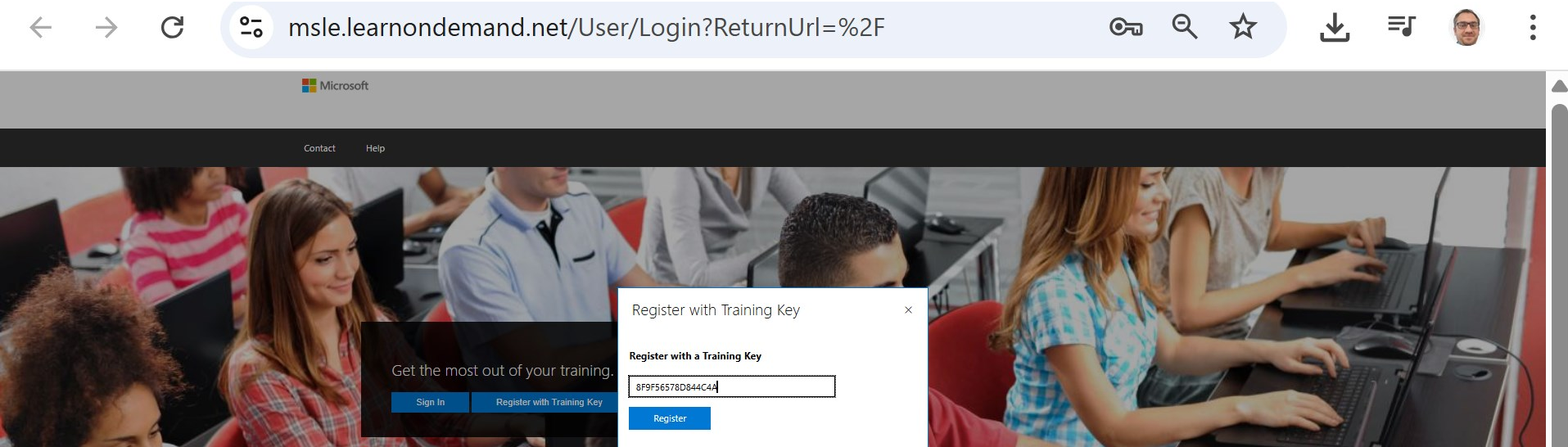


Open this page in your browser

Week 8 Tutorial Activity (Step-by-Step)



- Register with key: **39595D4F29DCCDC7**

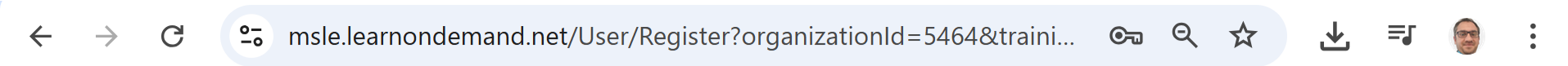


Open this page in your browser

Week 8 Tutorial Activity (Step-by-Step)



- Fill the form



Register

First Name: *

Last Name: *

Primary Email: *

Username:

Password: ☐ Show Characters OK

Confirm Password: [Password Policy](#)

Country: *

Time Zone:

☒ Enable Notifications

Save

Open this page in your browser

Week 8 Tutorial Activity (Step-by-Step)



- If already registered and you want to log in
 1. On the popup (as in your screenshot), choose **Skillable Account**.
 2. Enter the username/email and password you created when you first registered with the key.
 3. Once logged in, you'll see your training dashboard → then launch Explore Azure SQL Database.

How would you like to sign in?

Microsoft Account

Entra ID

Skillable Account

Week 8 Tutorial Activity (Step-by-Step)



- If already registered and you want to log in

Don't pick Microsoft Account or Entra ID unless your lab instructions specifically require it — those are for enterprise logins.

How would you like to sign in?



Microsoft Account

Entra ID

Skillable Account

Week 8 Tutorial Activity (Step-by-Step)



- If already registered and you want to log in

msle.learnondemand.net/User/CurrentTraining

Microsoft Farshid

My Training My Transcript Contact Help

Current Training Farshid Keivanian Details Edit

Transcript Redeem Training Key

All times shown in AUS Eastern Standard Time

Classes (1)

Class	Room	When ↑	Status
Modern Database Design ITEC617 (DP-900)		19 August 2025 18:00 - 31 October 2025 04:00 (AUS Eastern Standard Time)	Enrolled



Redeem Training Key

Training Key:

39595D4F29DCCDC7

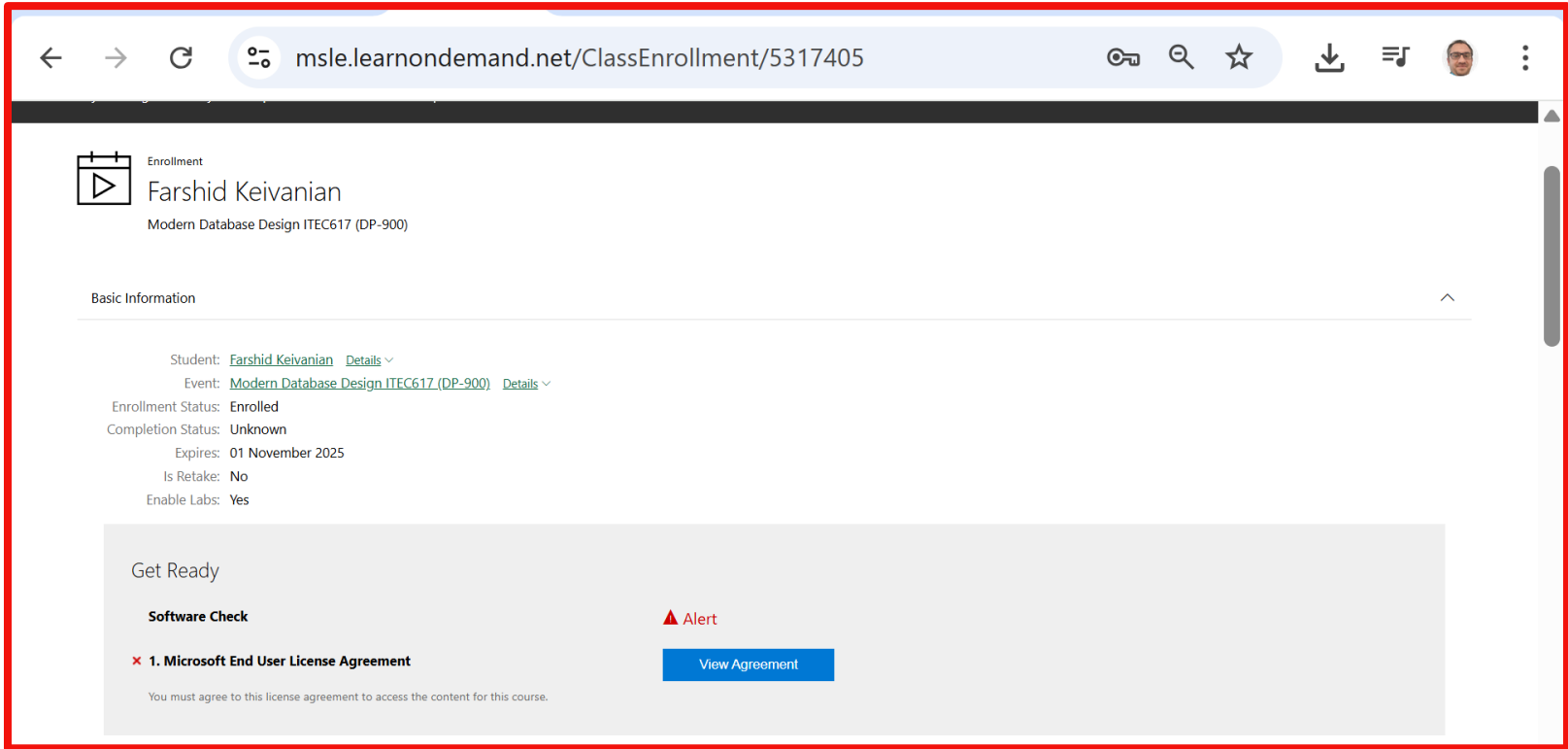
*

Redeem Training Key

Open this page in your browser

Week 8 Tutorial Activity (Step-by-Step)

- View Agreement (either you just registered or logged in, you will see this page)



msle.learnondemand.net/ClassEnrollment/5317405

Enrollment
Farshid Keivanian
Modern Database Design ITEC617 (DP-900)

Basic Information

Student: [Farshid Keivanian](#) Details ▾
Event: [Modern Database Design ITEC617 \(DP-900\)](#) Details ▾
Enrollment Status: **Enrolled**
Completion Status: **Unknown**
Expires: 01 November 2025
Is Retake: **No**
Enable Labs: **Yes**

Get Ready

Software Check

▲ Alert

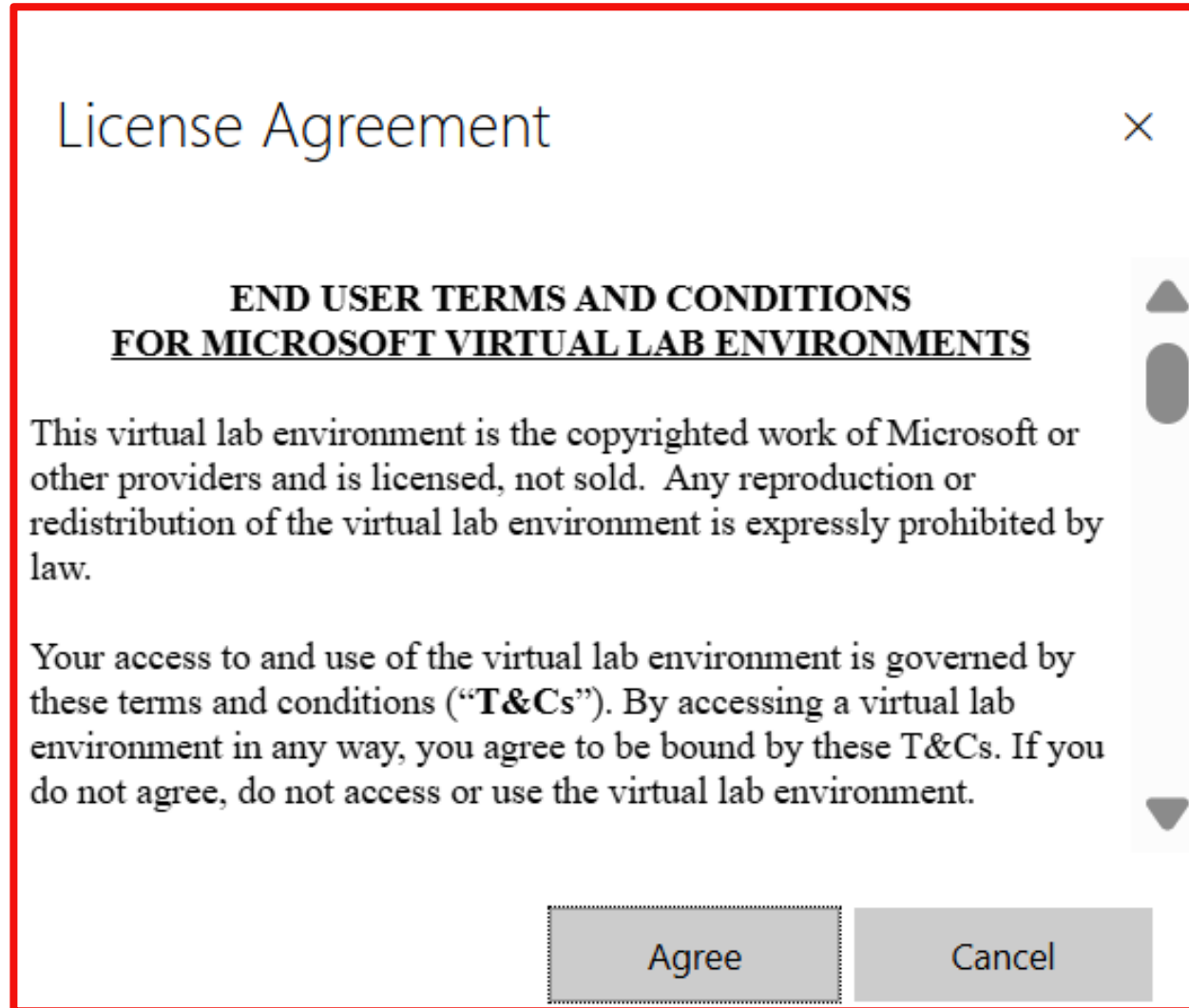
✖ 1. Microsoft End User License Agreement

You must agree to this license agreement to access the content for this course.

View Agreement

Open this page in your
browser

- Accept the license (Agree)



Week 8 Tutorial Activity (Step-by-Step)



- Launch the lab “Explore Azure SQL Database”
- Use the lab’s Azure username/password to sign in to the Azure Portal.
- Click Start → Open Azure Portal and sign in with those credentials.

A screenshot of a web browser displaying the msle.learnondemand.net website. The browser's address bar shows the URL 'msle.learnondemand.net/ClassEnrollment/5317405'. Below the browser, the page has a header with navigation icons. The main content area is titled 'Activities' and features a dark grey banner stating 'Access to your labs will expire on 31 October 2025 17:00 (UTC)'. Below this is a progress bar at 0% completion, with the text '0 of 4 required activities complete'. A list of activities is shown, with the first one selected: 'Explore Azure SQL Database' (Expected Duration 2 hours, 15 minutes). This activity is associated with 'DP-900T00-A Introduction to Microsoft Azure Data [Cloud Slice Provided], Learning Path 02 (CSS)'. It is marked as 'Required: Yes' and 'Status: Not Started'. A blue 'Launch' button is present, and a note at the bottom indicates '10 of 10 launch attempts remaining'.

Week 8 Tutorial Activity (Step-by-Step)

Lab Setup - Google Chrome

labclient.labondemand.com/Setup/509a207a-ac3b-442e-9a78-c31bb7671501



Building



Your lab will be ready in about 1 minute.

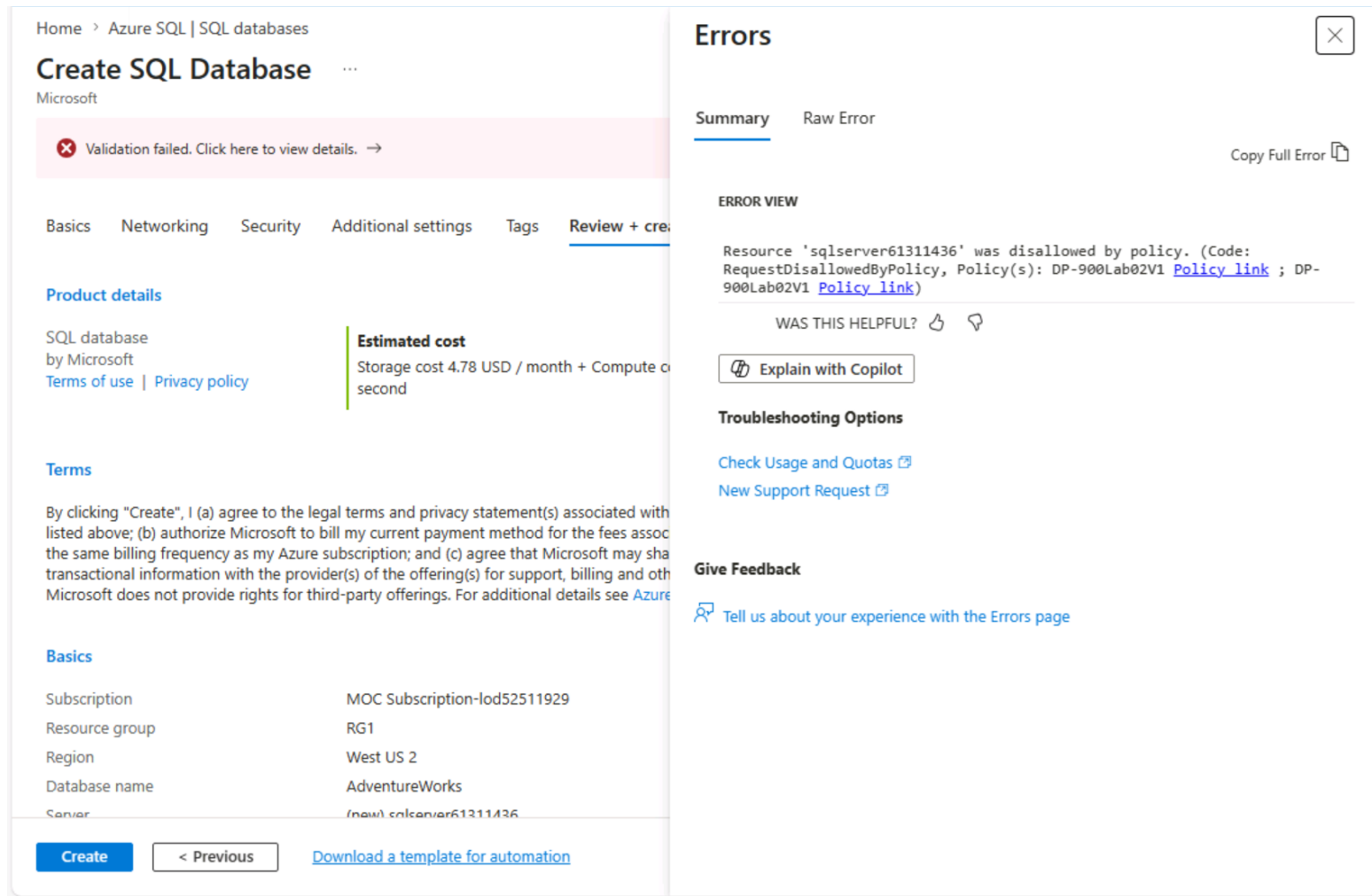
Important:

Follow the on-screen instructions in the right-hand side window of the virtual lab to log in and complete the tasks as instructed.

Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

Common Lab Issue: Azure Policy Validation Error



The screenshot shows the 'Create SQL Database' page in the Azure Portal. The page is divided into two main sections: the left pane for configuration and the right pane for errors and feedback.

Left Pane: Create SQL Database

- Home > Azure SQL | SQL databases
- Create SQL Database
- Microsoft
- Validation failed. Click here to view details. →
- Tabs: Basics, Networking, Security, Additional settings, Tags, Review + create
- Product details: SQL database by Microsoft. Terms of use | Privacy policy
- Estimated cost: Storage cost 4.78 USD / month + Compute cost per second
- Terms: By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share transactional information with the provider(s) of the offering(s) for support, billing and other purposes. Microsoft does not provide rights for third-party offerings. For additional details see [Azure SQL Database Terms of Use](#).
- Basics table:

Subscription	MOC Subscription-lod52511929
Resource group	RG1
Region	West US 2
Database name	AdventureWorks
Server	(new).sqlserver61311436
- Buttons: Create, < Previous, [Download a template for automation](#)

Right Pane: Errors

- Summary (selected), Raw Error
- Copy Full Error
- ERROR VIEW: Resource 'sqlserver61311436' was disallowed by policy. (Code: RequestDisallowedByPolicy, Policy(s): DP-900Lab02V1 [Policy link](#) ; DP-900Lab02V1 [Policy link](#))
- WAS THIS HELPFUL? (thumbs up/down icons)
- Explain with Copilot
- Troubleshooting Options: [Check Usage and Quotas](#), [New Support Request](#)
- Give Feedback: [Tell us about your experience with the Errors page](#)

What does it mean?

- You are working in a controlled lab environment
- Azure applies policies (rules) to restrict configurations
- If settings don't match lab requirements → validation fails

Why this happens

- Incorrect region (e.g., West US 2 instead of West US)
- Wrong or random server name
- Not using the required resource group (RG1)
- Ignoring instructions on the right-hand lab panel

How to fix it

- ✓ Use server name provided on the right side window
- ✓ Avoid double click on server name (only one click)
- ✓ Match region EXACTLY as shown in lab instructions
- ✓ Use RG1 as the resource group
- ✓ Follow lab panel settings (not defaults)
- ✓ If needed → go back and re-enter details or restart lab

Key takeaway

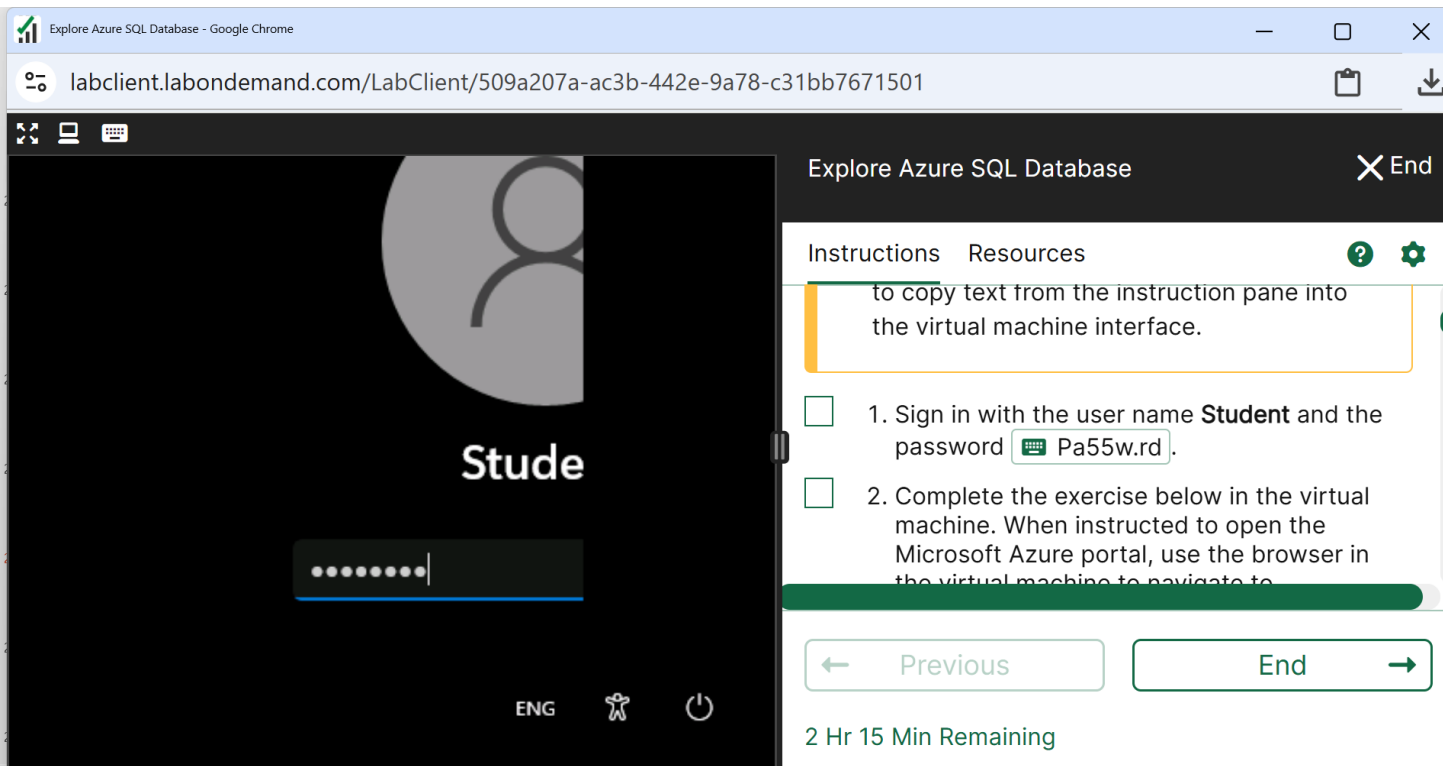
Small configuration differences in cloud environments can cause errors —

Always follow lab instructions exactly.

“This is not a mistake — it’s how real cloud systems enforce security and consistency.”

Week 8 Tutorial Activity (Step-by-Step)

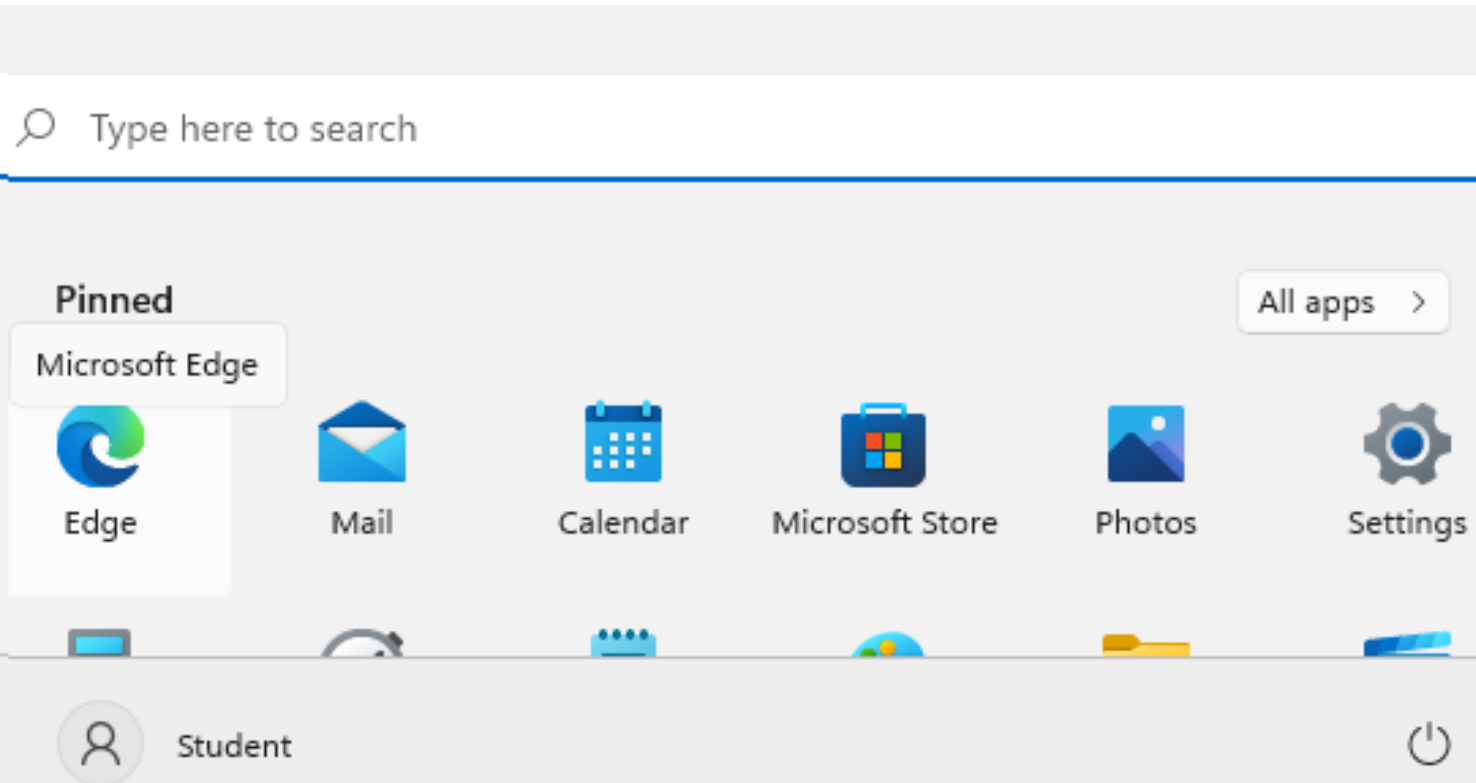
Click on Password and Press Enter



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

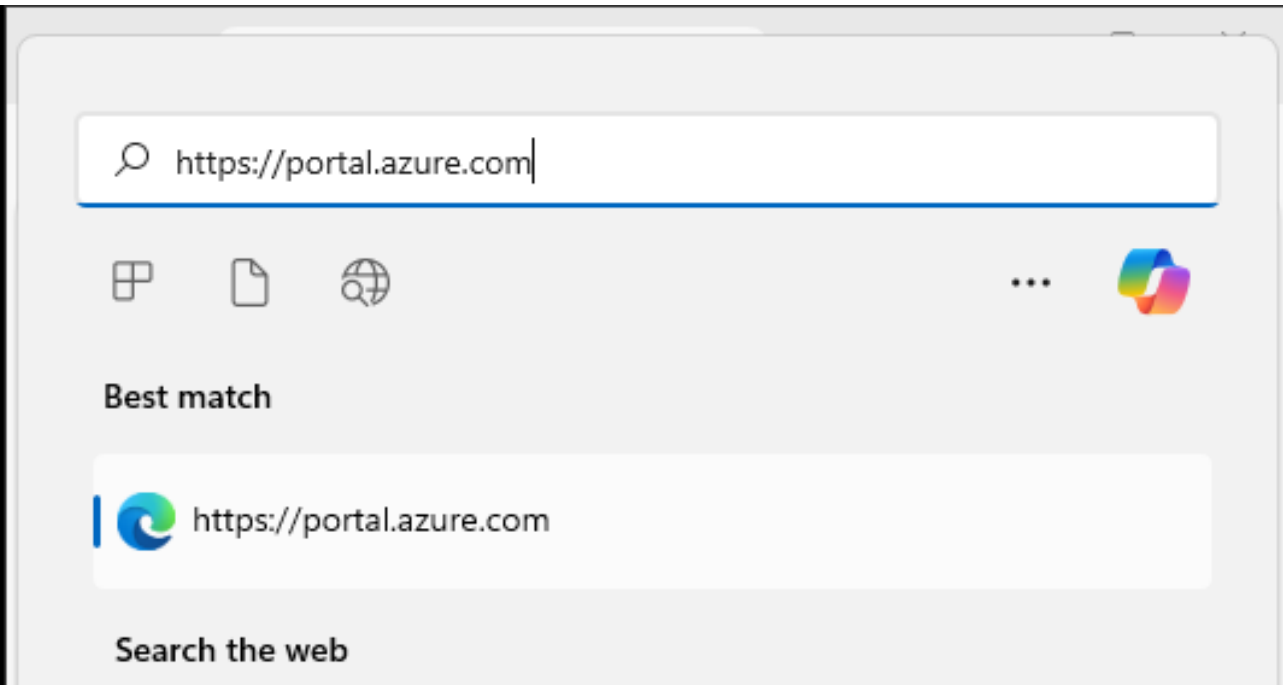
Click Start and Go to Edge Browser



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

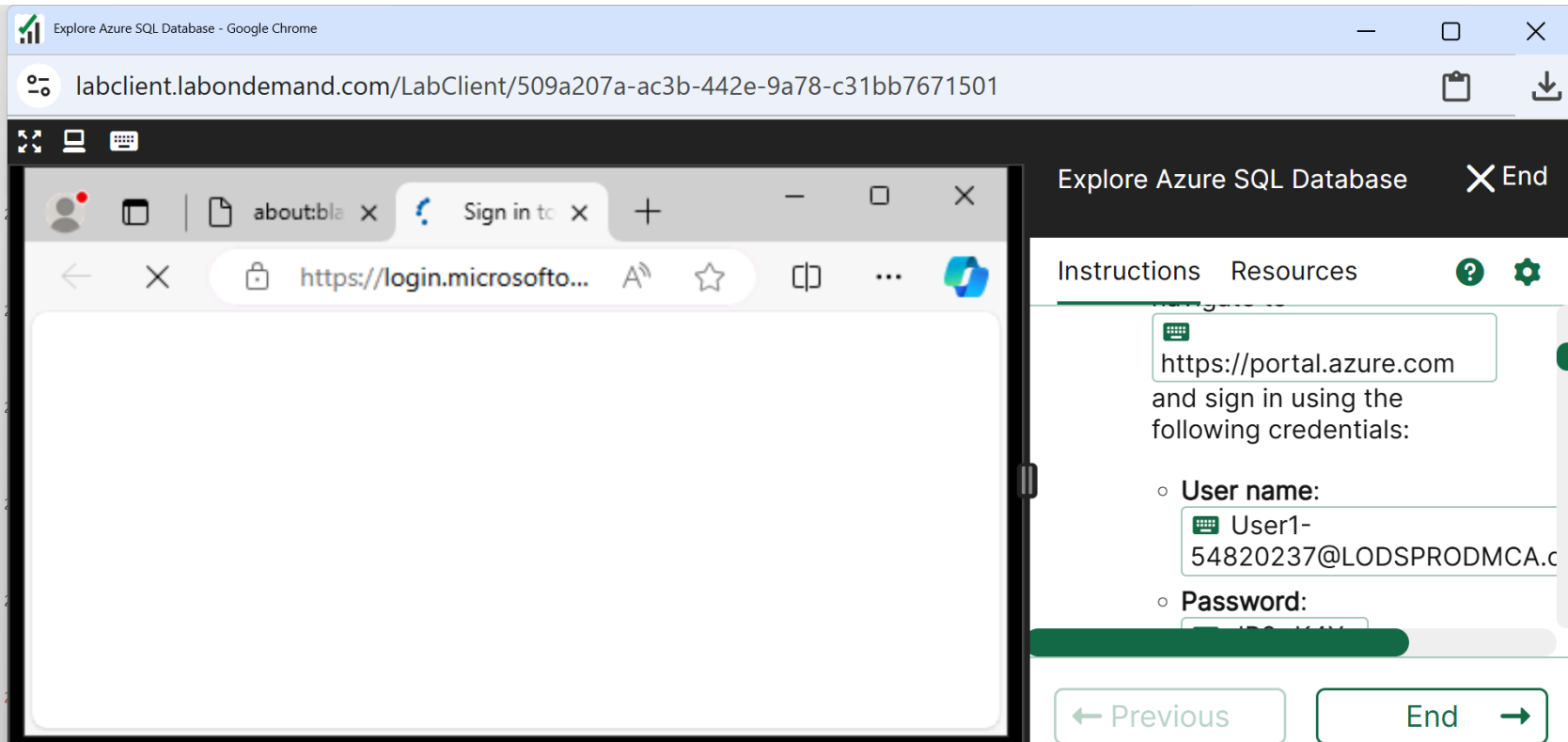
Click on Website (Portal); it is on right sidebar
You will see this



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

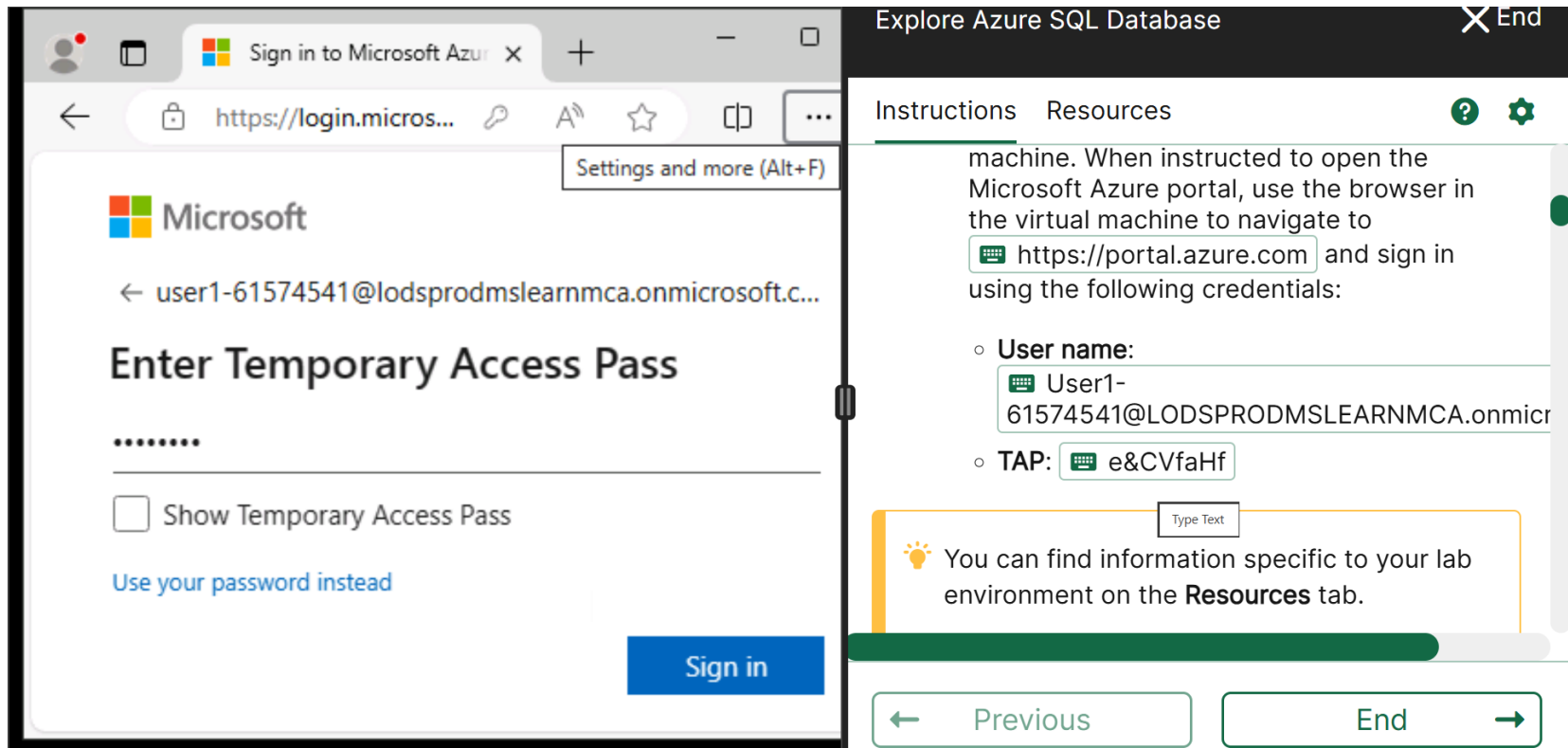
Press Enter; you will see this page



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

Click on User and Pass (on right sidebar) to log in



The screenshot shows a web browser window with the Microsoft Azure login page. The address bar shows `https://login.micros...`. The page title is "Sign in to Microsoft Azure". The main heading is "Enter Temporary Access Pass". Below the heading is a password input field with a masked password ".....". There is a checkbox labeled "Show Temporary Access Pass" and a link "Use your password instead". A blue "Sign in" button is at the bottom right.

On the right, the "Explore Azure SQL Database" sidebar is visible. It has a title bar with "Explore Azure SQL Database" and a close button. Below the title bar are tabs for "Instructions" and "Resources". The "Instructions" tab is active, showing text: "machine. When instructed to open the Microsoft Azure portal, use the browser in the virtual machine to navigate to `https://portal.azure.com` and sign in using the following credentials:". Below this are two items:

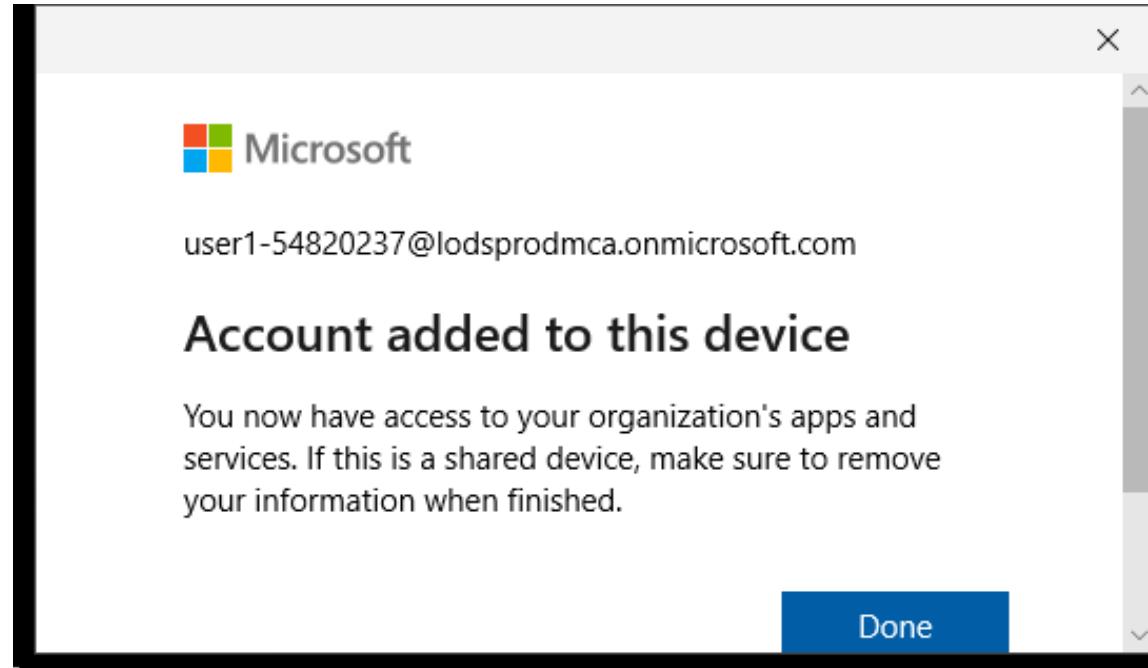
- **User name:** `User1-61574541@LODSPRODMSLEARNMCA.onmicr`
- **TAP:** `e&CVfaHf`

Below the instructions is a "Type Text" input field and a lightbulb icon with the text: "You can find information specific to your lab environment on the **Resources** tab." At the bottom of the sidebar are "Previous" and "End" buttons.

Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

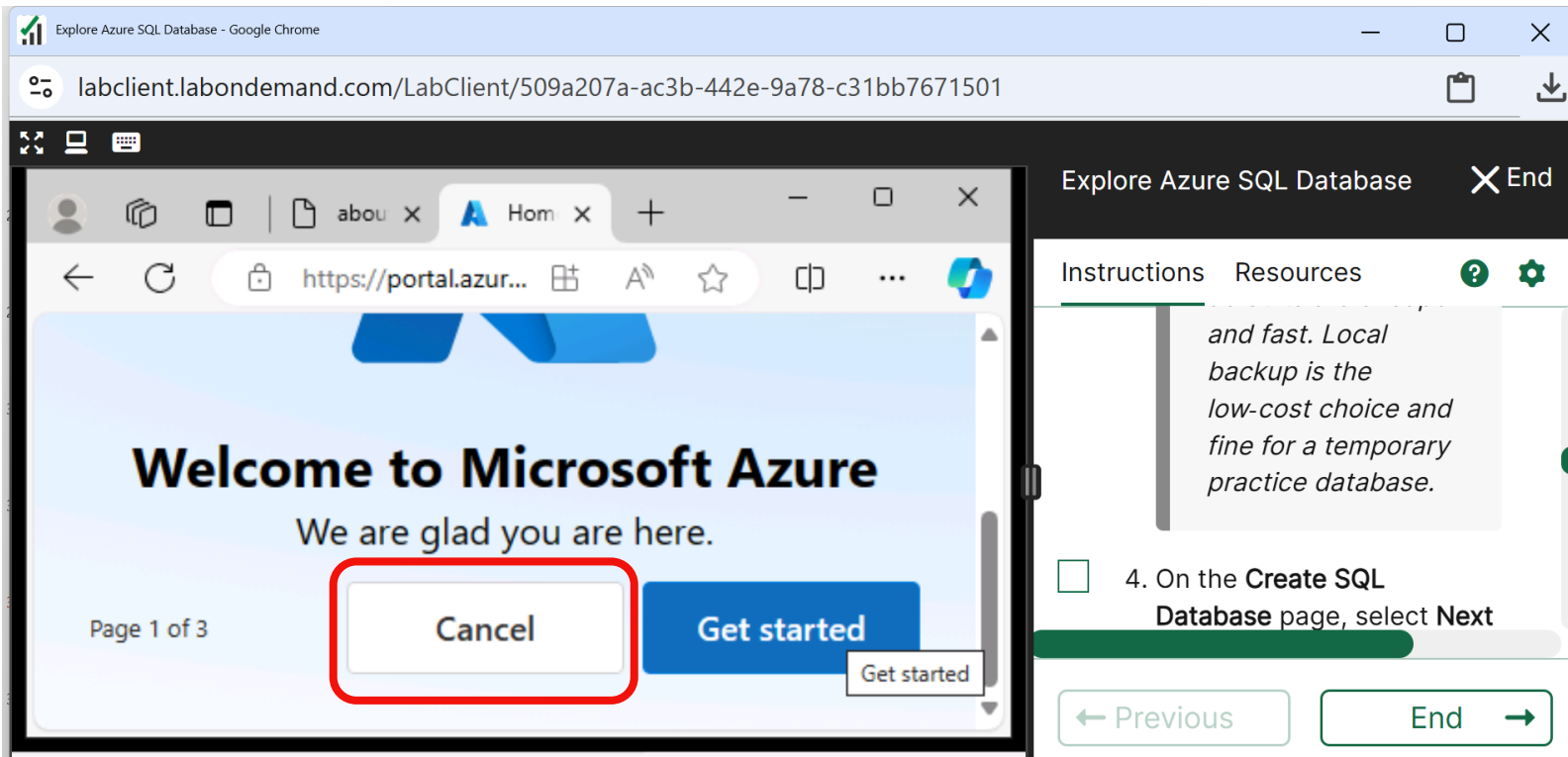
Click Yes then you see this



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

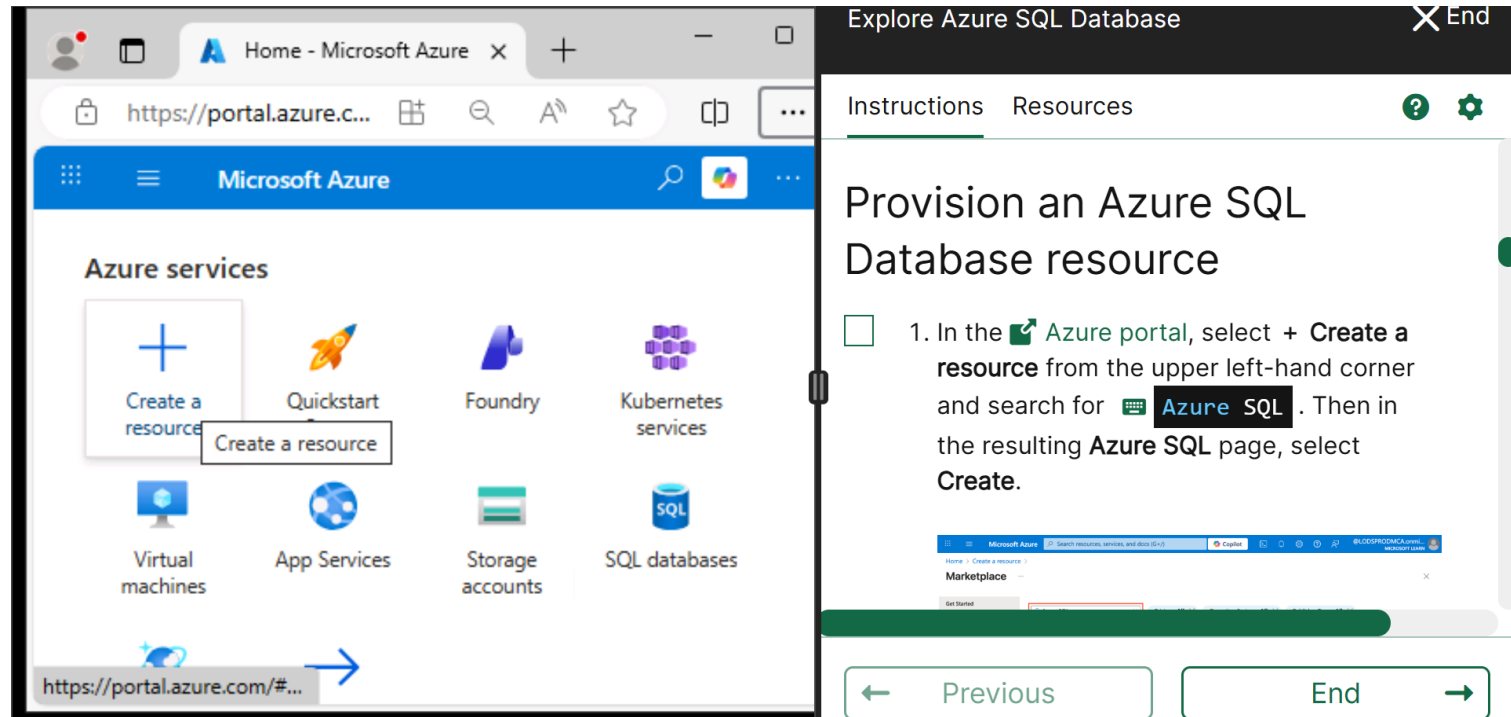
Cancel this for now



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

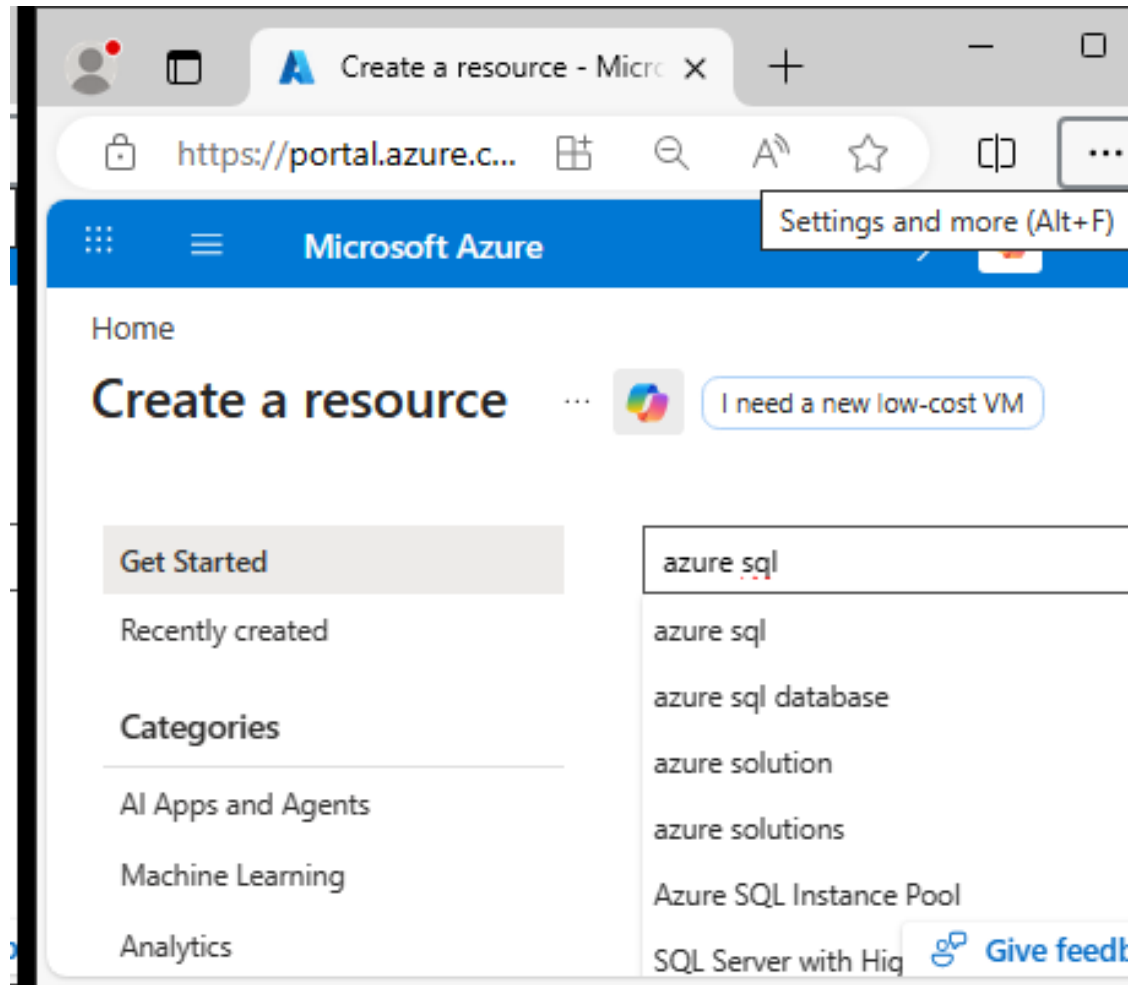
Zoom in/out to see Create a Resource and Click on it



Important: Do not open Azure Portal on your browser, instead run the virtual machine and then open the Azure Portal on the virtual machine's browser

Week 8 Tutorial Activity (Step-by-Step)

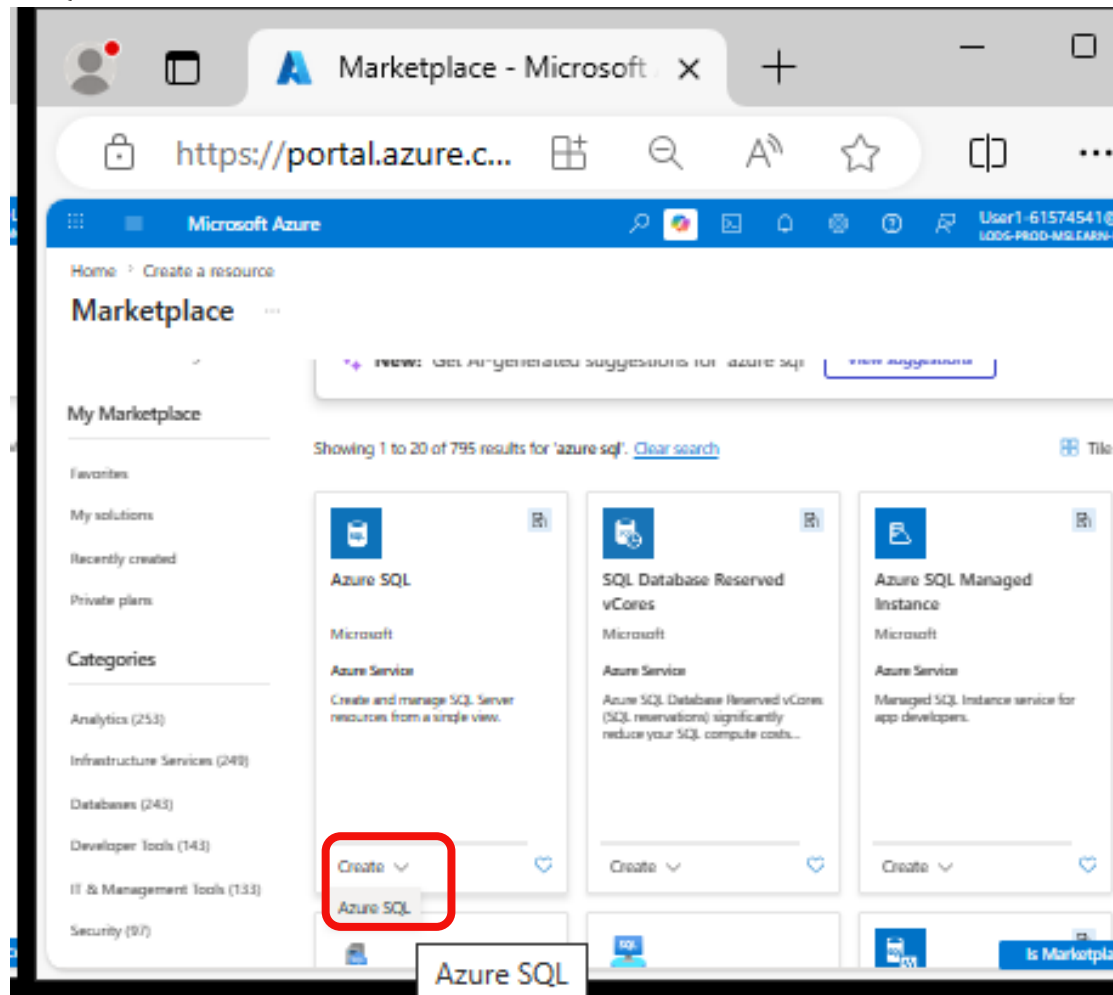
Search for Azure SQL



Week 8 Tutorial Activity (Step-by-Step)

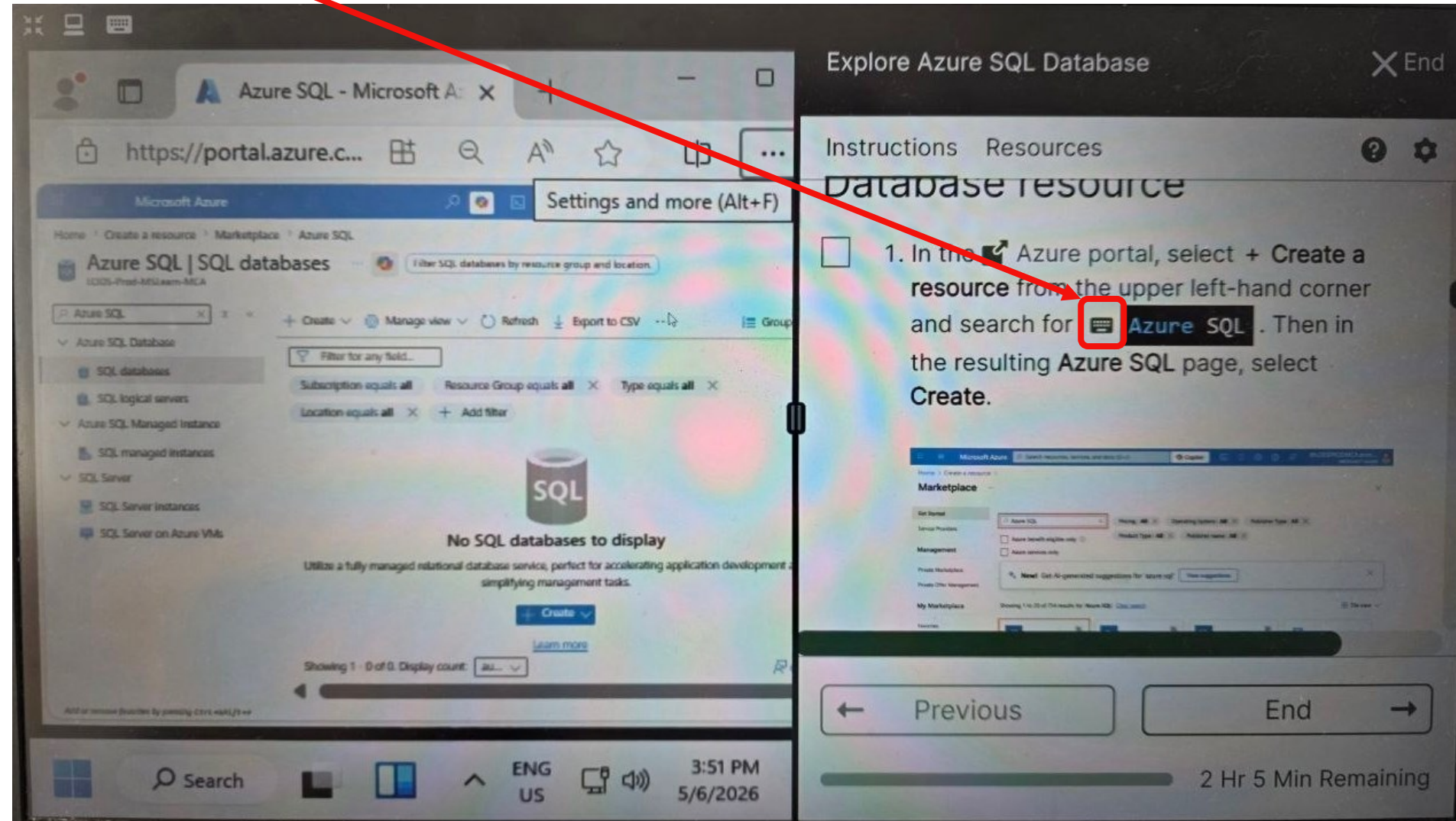
Zoom in/out to see Azure SQL

Click Azure SQL



Week 8 Tutorial Activity (Step-by-Step)

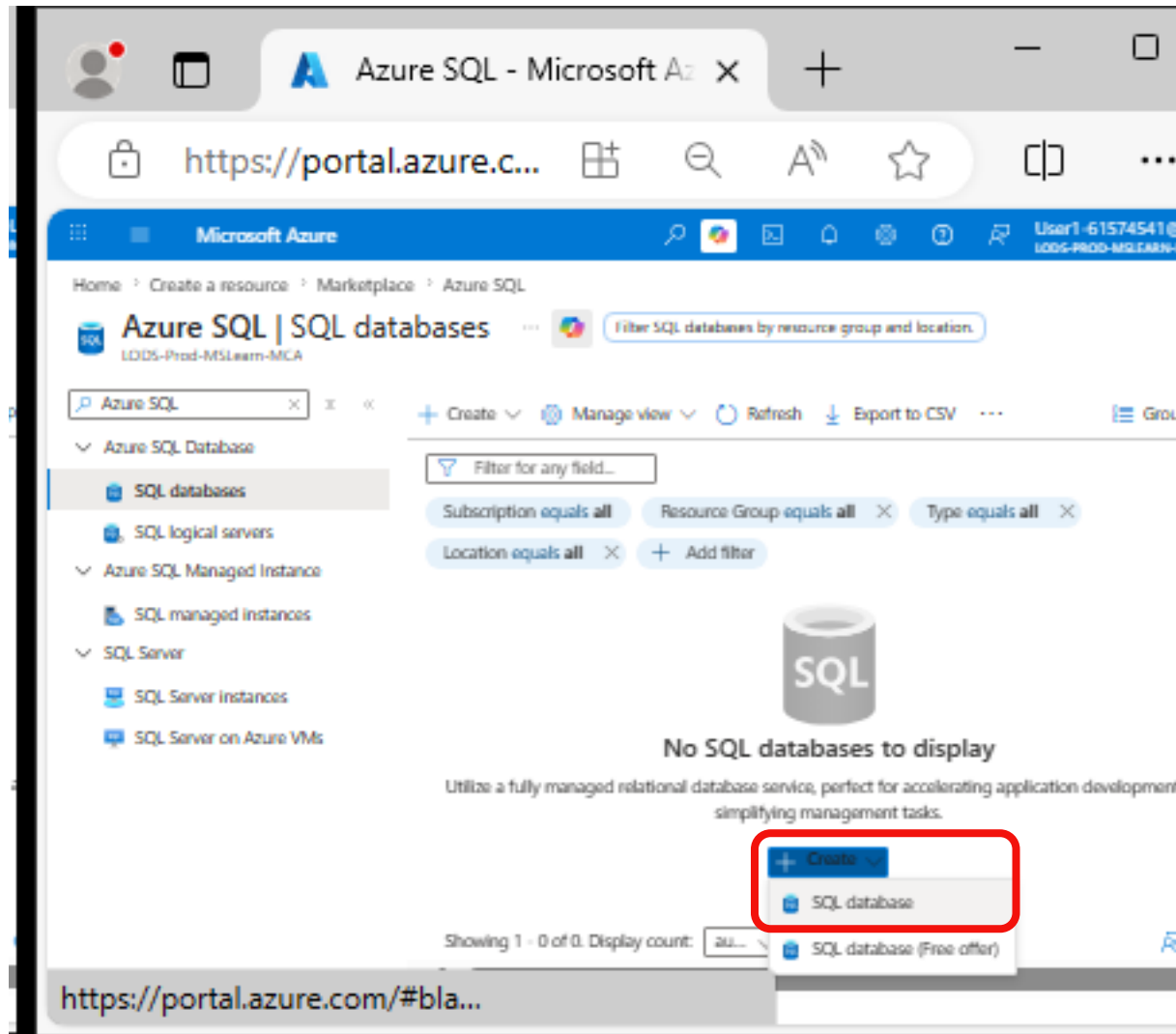
Click on type on right side



The screenshot displays the Azure portal interface. On the left, the 'Azure SQL | SQL databases' sidebar is visible, showing a list of resource types including 'SQL databases', 'SQL logical servers', 'Azure SQL Managed Instance', 'SQL managed instances', 'SQL Server', 'SQL Server instances', and 'SQL Server on Azure VMs'. A red arrow points from the text 'Click on type on right side' to the 'Type' filter dropdown in this sidebar. The main content area shows the 'Explore Azure SQL Database' page with instructions for creating a resource. The instructions state: '1. In the Azure portal, select + Create a resource from the upper left-hand corner and search for Azure SQL. Then in the resulting Azure SQL page, select Create.' Below the instructions, there is a search bar and a list of results. The bottom of the screen shows a Windows taskbar with the time 3:51 PM and date 5/6/2026.

Week 8 Tutorial Activity (Step-by-Step)

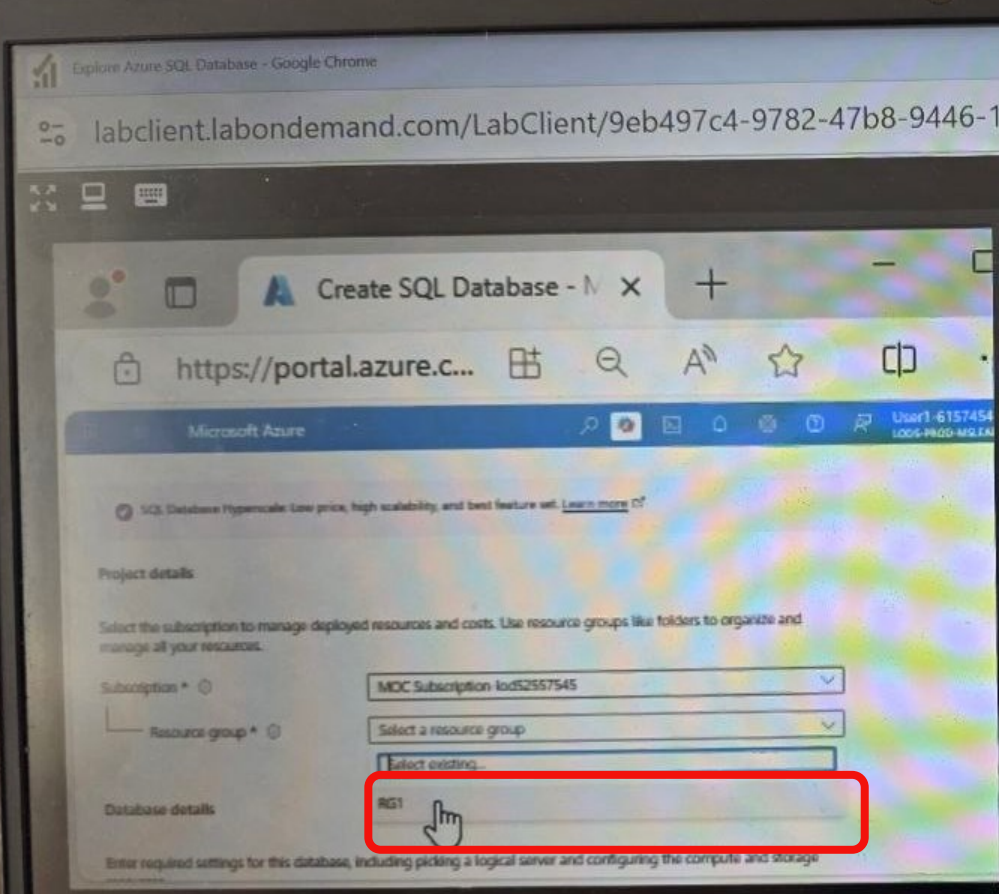
Click Create >> SQL database



Week 8 Tutorial Activity (Step-by-Step)

The subscription should be shown automatically

Choose Resource – Type: RG1



Explore Azure SQL Database

Instructions Resources

4. Enter the following values on the **Create SQL Database** page, and leave all other properties with their default setting:

- Subscription: Select your Azure subscription.
- Resource group: RG1
- Database name:

Previous End

2 Hr 1 Min Remaining

Week 8 Tutorial Activity (Step-by-Step)



- I chose default one (by clicking Type the database name automatically is written)
- You should follow Canvas: **Student ID+name** as the "database name"

labclient.labondemand.com/LabClient/9eb497c4-9782-47b8-9446-1b522da4c0e7

Explore Azure SQL Database ✕ End

Instructions Resources ? ⚙

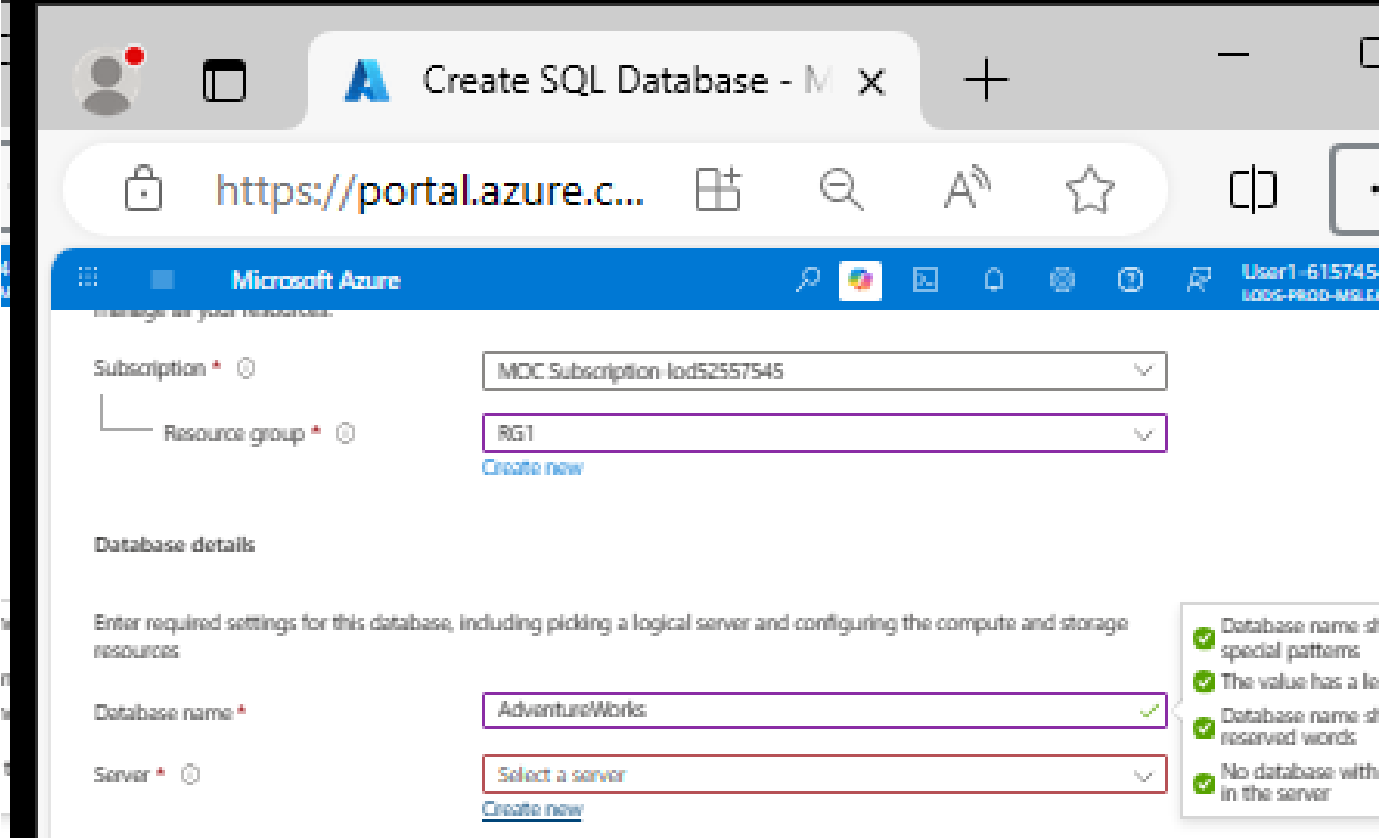
- **Subscription:** Select your Azure subscription.
- **Resource group:** RG1
- **Database name:** AdventureWorks
- **Server:** Select **Create new** and create

← Previous End →

1 Hr 58 Min Remaining

Week 8 Tutorial Activity (Step-by-Step)

Server: Create new



Microsoft Azure

Subscription * ⓘ MOC Subscription-10d52557545

Resource group * ⓘ RG1
[Create new](#)

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources.

Database name * AdventureWorks ✓

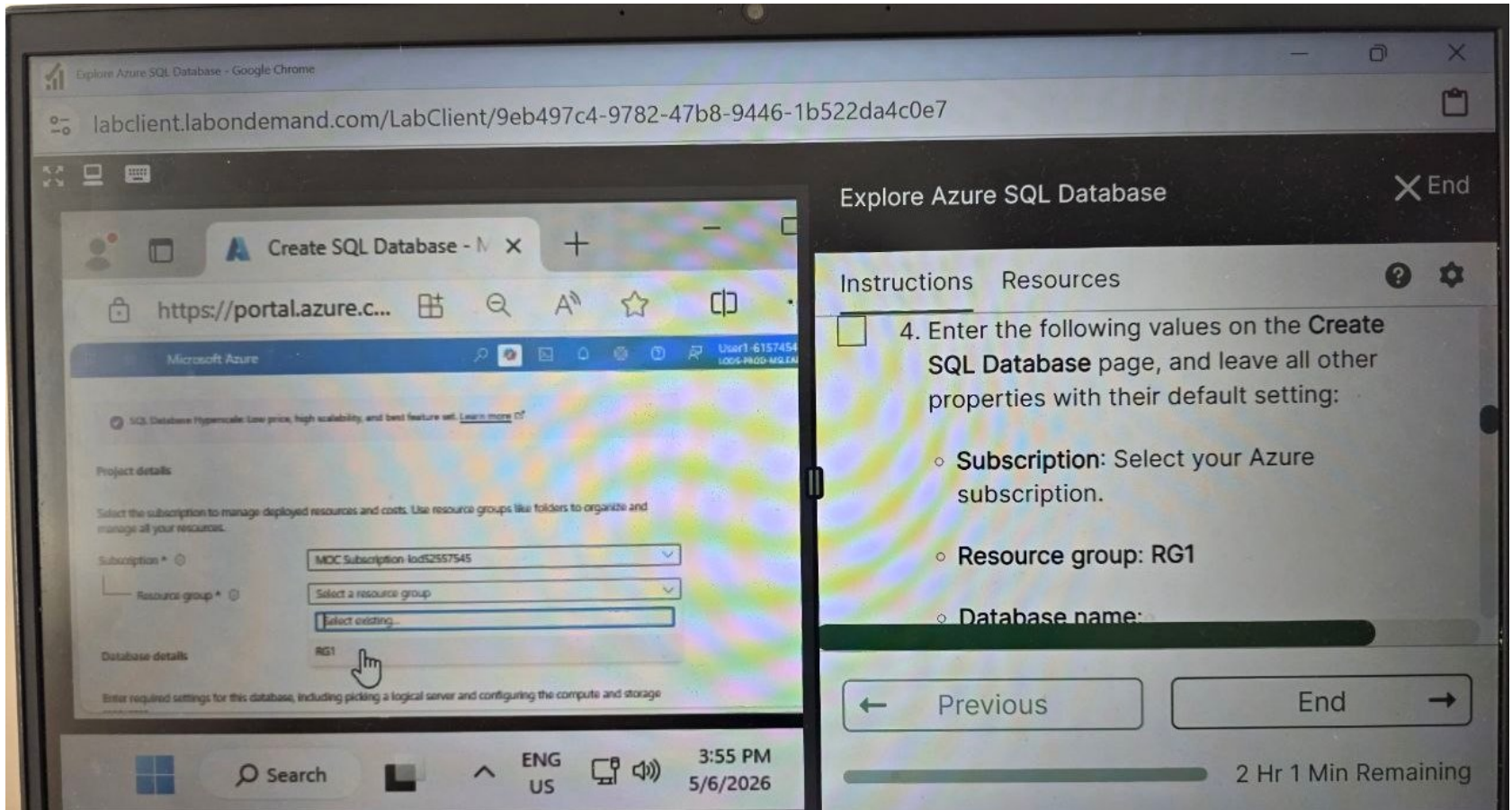
Server * ⓘ Select a server
[Create new](#)

- ✓ Database name: id special patterns
- ✓ The value has a le
- ✓ Database name: id reserved words
- ✓ No database with in the server

Make sure to use the resource group/server name and region provided on the right-hand side window.

Week 8 Tutorial Activity (Step-by-Step)

For the location, scroll down to find (US) West US



Explore Azure SQL Database

Instructions Resources

4. Enter the following values on the **Create SQL Database** page, and leave all other properties with their default setting:

- Subscription: Select your Azure subscription.
- Resource group: RG1
- Database name:

Previous End

2 Hr 1 Min Remaining

Do Not Choose West US2

Week 8 Tutorial Activity (Step-by-Step)

Use SQL Authentication

Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/868d0e6e-be99-4221-8396-a9ff8c669906

Create SQL Database Ser x

https://portal.azure.c...

Search resources, services, and docs (G+)

Copilot

User1-548281
LOGS-PROD-MCA

Authentication

Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#), using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login *

Enter server admin login

Password *

Confirm password *

Explore Azure SQL Database

End

Instructions Resources

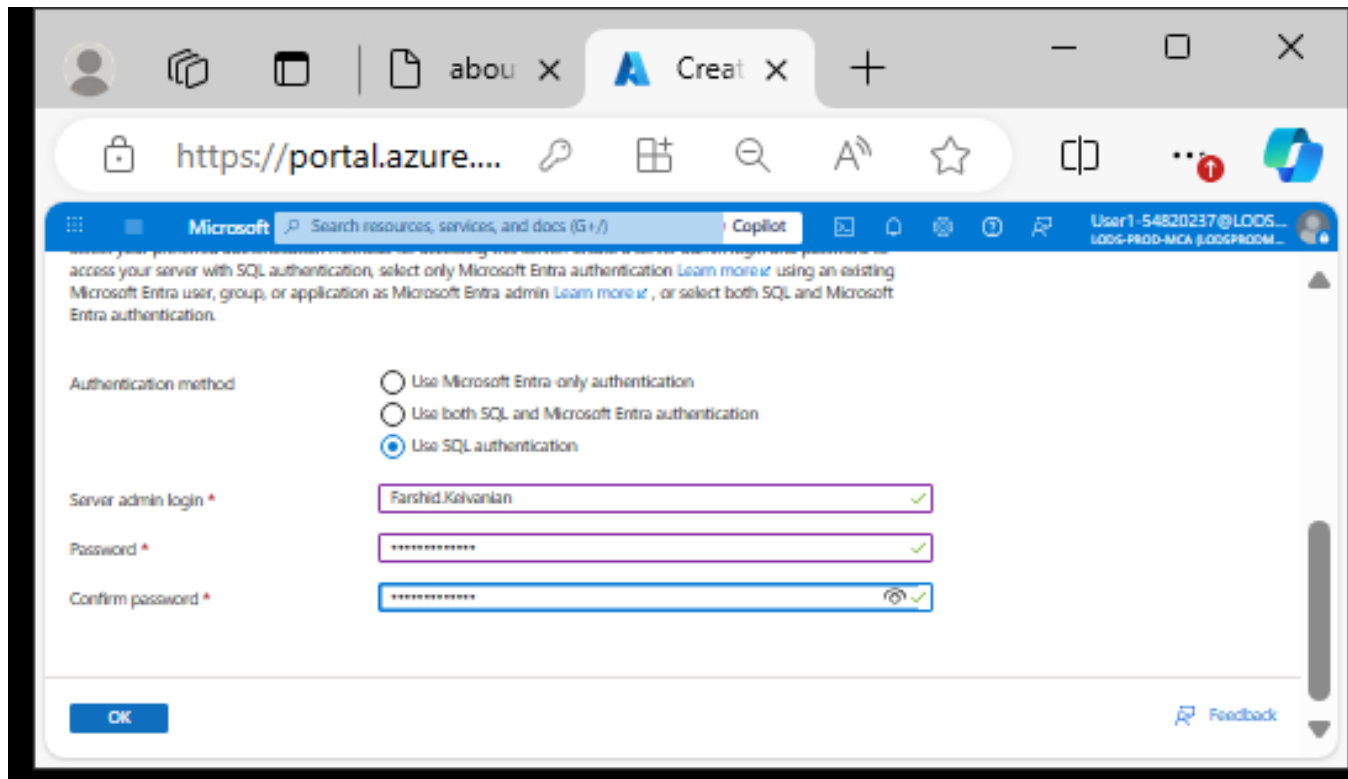
location to the same as the provided Resource group `westus`. Use **SQL authentication** and specify your name as the server admin login and a suitably complex password (remember the password - you'll need it later!)

Previous End

Week 8 Tutorial Activity (Step-by-Step)

Enter Your name as Server name and set a password (You will need it later) e.g., I typed **Farshid.Keivanian**

Click OK (remember the password)



Microsoft Search resources, services, and docs (G+J) Copilot User1-54820237@LOOS... LOOS-PROD-MCA (LOOSPROD-...)

access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login * Farshid.Keivanian ✓

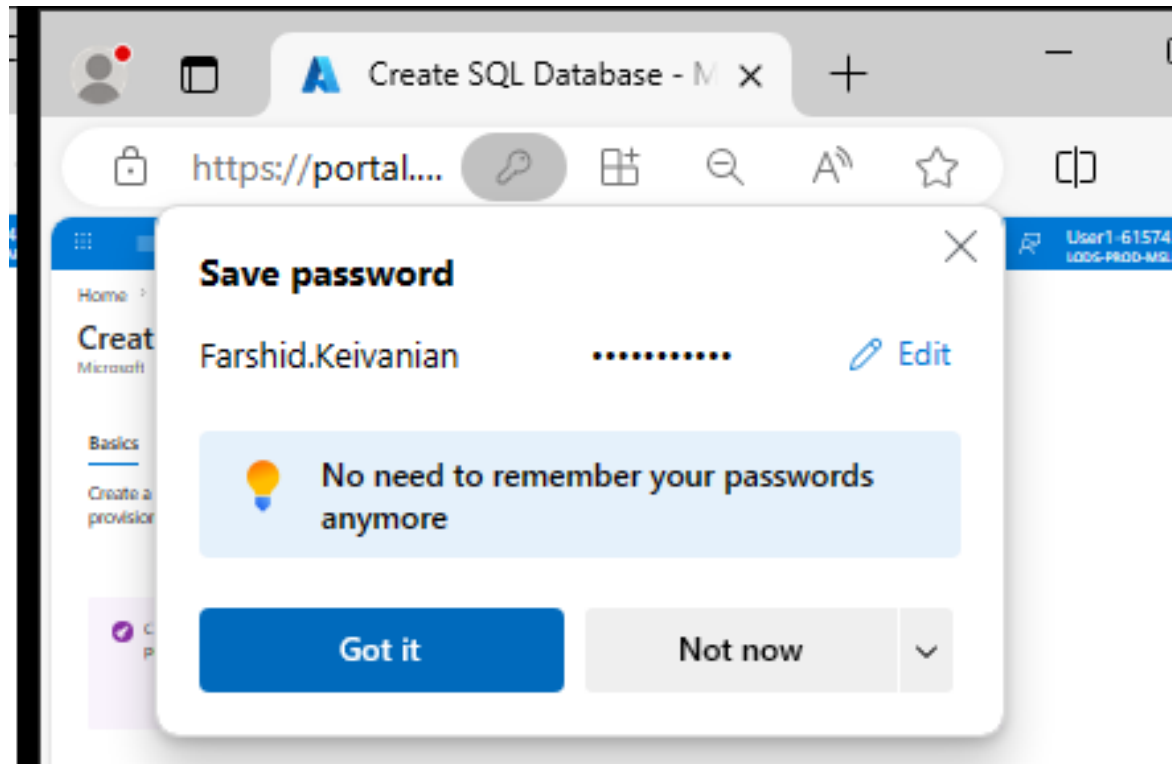
Password * ***** ✓

Confirm password * ***** ✓

OK Feedback

Week 8 Tutorial Activity (Step-by-Step)

Choose Got it



Week 8 Tutorial Activity (Step-by-Step)

Please follow the instructions on the right sidebar

Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/868d0e6e-be99-4221-8396-a9ff8c669906

Server * ⓘ

(new) sqlserver54828177 (West US)
Create new

Want to use SQL elastic pool? ⓘ

☐ Yes ☒ No

Workload environment

☒ Development
☐ Production

Compute + storage * ⓘ

General Purpose - Serverless
Standard series (Gen5), 1 vCore, 32 GB storage
Configure database

Backup storage redundancy

Explore Azure SQL Database

Instructions Resources ⓘ ⚙️

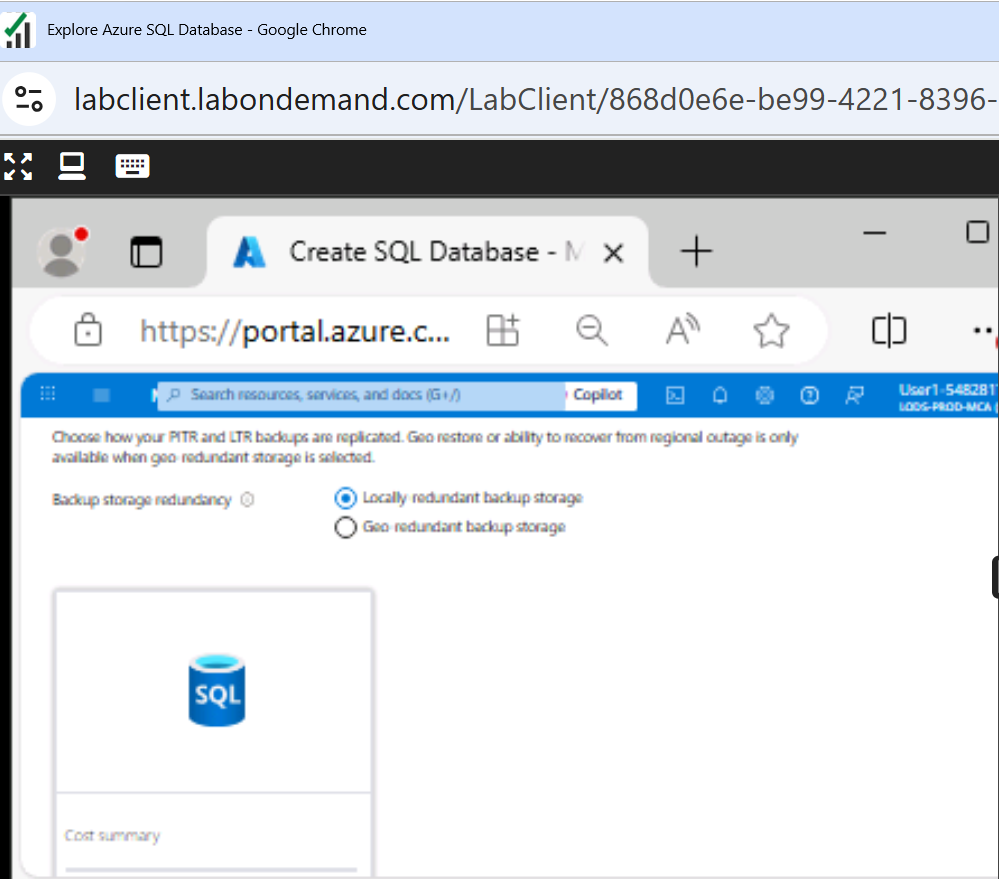
- **Want to use SQL elastic pool?:** *No*
- **Workload environment:** Development
- **Compute + storage:** Leave unchanged
- **Backup storage redundancy:** *Locally-redundant backup storage*

← Previous

End →

Week 8 Tutorial Activity (Step-by-Step)

Choose Locally-redundant backup storage



The screenshot shows a web browser window with the URL `labclient.labondemand.com/LabClient/868d0e6e-be99-4221-8396-a9ff8c669906`. The page is titled 'Explore Azure SQL Database' and shows the 'Create SQL Database' configuration page. Under 'Backup storage redundancy', the 'Locally redundant backup storage' option is selected. The 'Cost summary' section is visible at the bottom.

Explore Azure SQL Database End

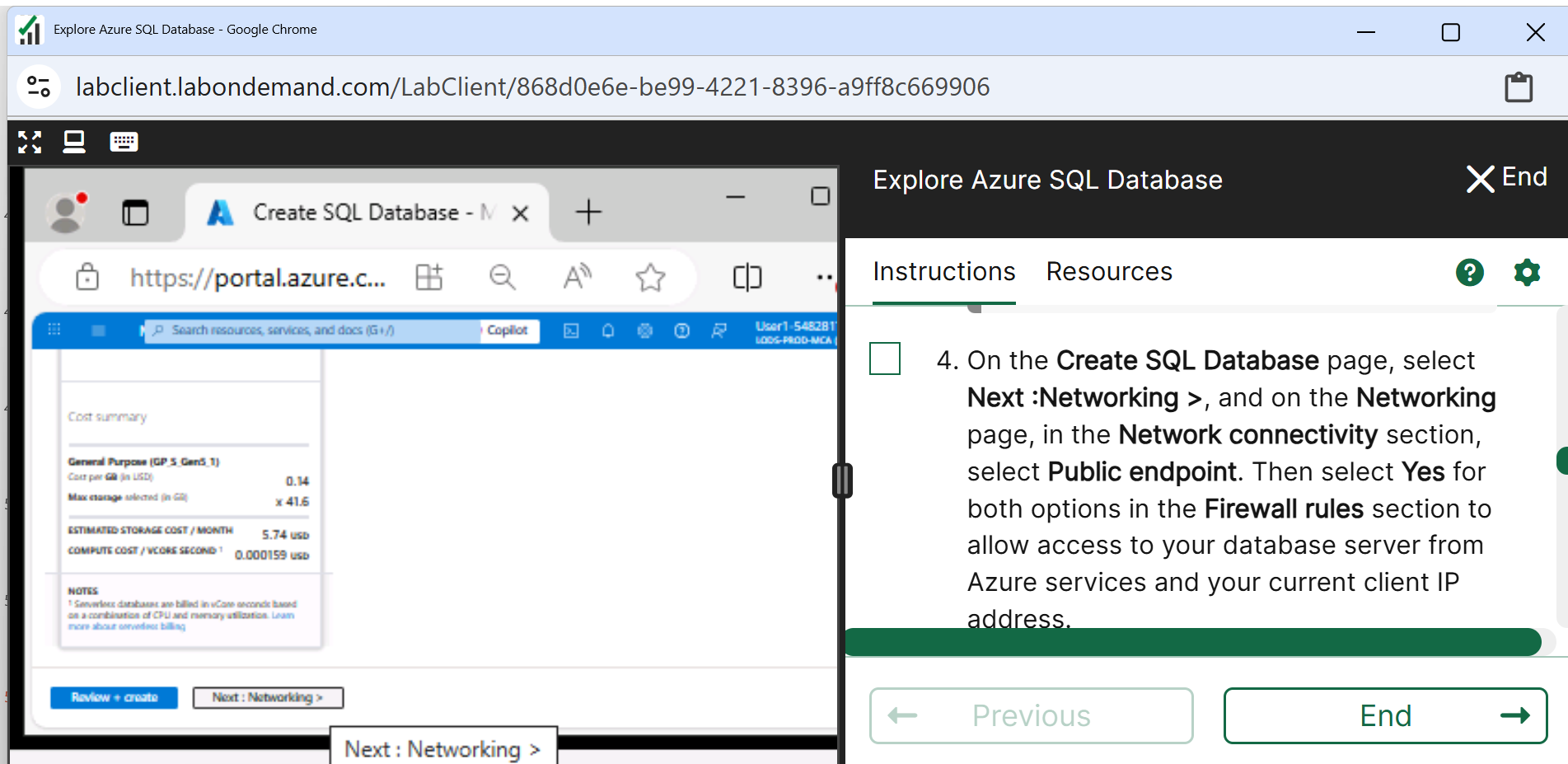
Instructions Resources ? ⚙️

☐ 4. On the **Create SQL Database** page, select **Next :Networking >**, and on the **Networking** page, in the **Network connectivity** section, select **Public endpoint**. Then select **Yes** for both options in the **Firewall rules** section to allow access to your database server from Azure services and your current client IP address.

← Previous End →

Week 8 Tutorial Activity (Step-by-Step)

Click Next: Networking



The screenshot shows a web browser window with the URL `labclient.labondemand.com/LabClient/868d0e6e-be99-4221-8396-a9ff8c669906`. The page is titled "Explore Azure SQL Database" and displays a "Cost summary" section. The summary includes the following details:

Cost summary	
General Purpose (GP, 5, Gen5, 1)	
Cost per GB (in USD)	0.14
Max storage selected (in GB)	x 41.6
ESTIMATED STORAGE COST / MONTH	5.74 usd
COMPUTE COST / VCORE SECOND ¹	0.000159 usd

Below the summary, there is a "NOTES" section with a footnote: ¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#).

At the bottom of the page, there are two buttons: "Review + create" and "Next: Networking >". The "Next: Networking >" button is highlighted with a green box.

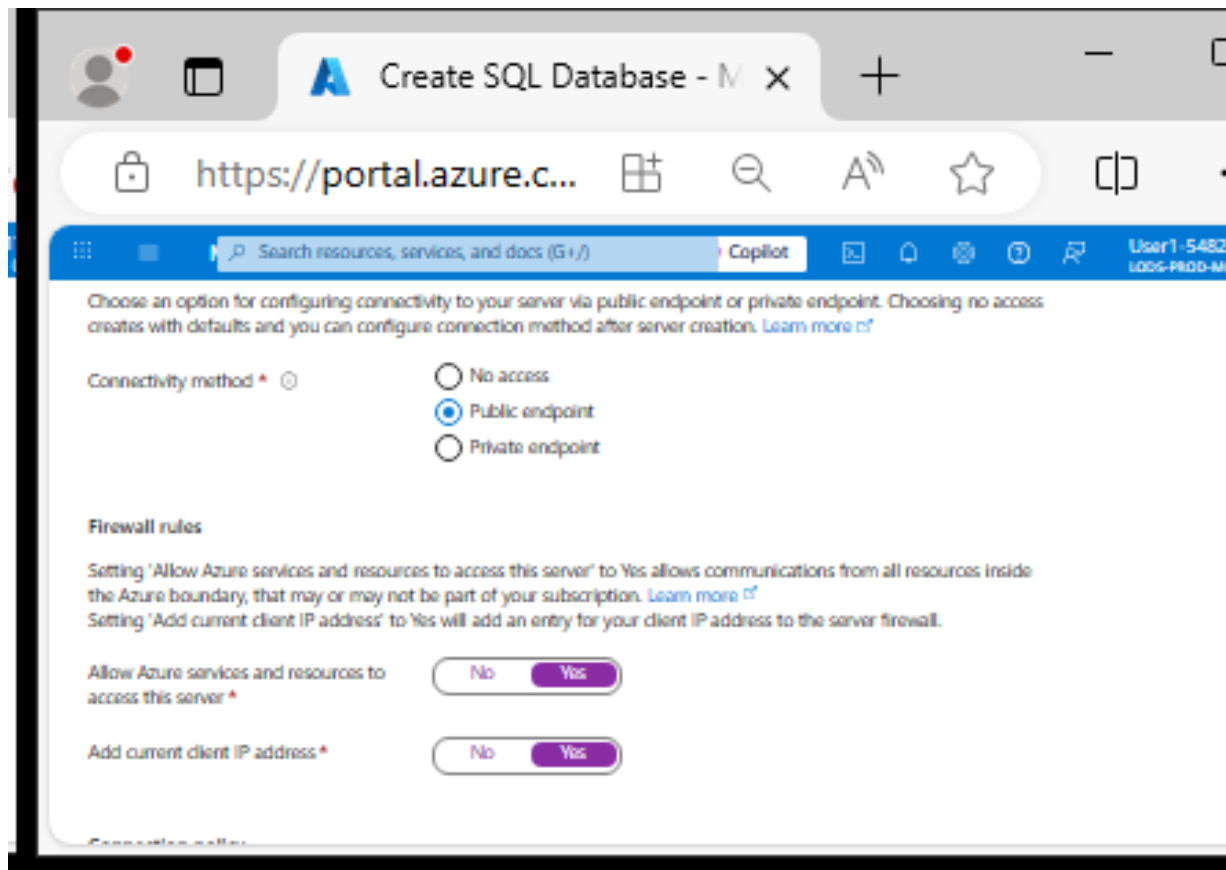
On the right side of the image, there is a sidebar titled "Explore Azure SQL Database" with a close button (X) and the word "End". It contains two tabs: "Instructions" and "Resources". The "Instructions" tab is active, showing a list of steps. Step 4 is highlighted with a green box and reads:

4. On the **Create SQL Database** page, select **Next :Networking >**, and on the **Networking** page, in the **Network connectivity** section, select **Public endpoint**. Then select **Yes** for both options in the **Firewall rules** section to allow access to your database server from Azure services and your current client IP address.

At the bottom of the sidebar, there are two buttons: "Previous" and "End". The "End" button is highlighted with a green box.

Week 8 Tutorial Activity (Step-by-Step)

Select Public endpoint, and Yes for both firewall options to allow access to your database server from Azure services and your current client IP address.



The screenshot shows the 'Create SQL Database' page in the Azure portal. The browser tab is 'Create SQL Database - M x'. The address bar shows 'https://portal.azure.c...'. The page content includes a search bar, a 'Copilot' button, and a user profile 'User1-54828'. The main section is titled 'Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. Learn more'. Under 'Connectivity method', the 'Public endpoint' radio button is selected. Below this, the 'Firewall rules' section explains that setting 'Allow Azure services and resources to access this server' to 'Yes' allows communications from all resources inside the Azure boundary, and setting 'Add current client IP address' to 'Yes' will add an entry for the client IP address to the server firewall. Both settings are currently set to 'Yes'.

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method *

☐ No access

☒ Public endpoint

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

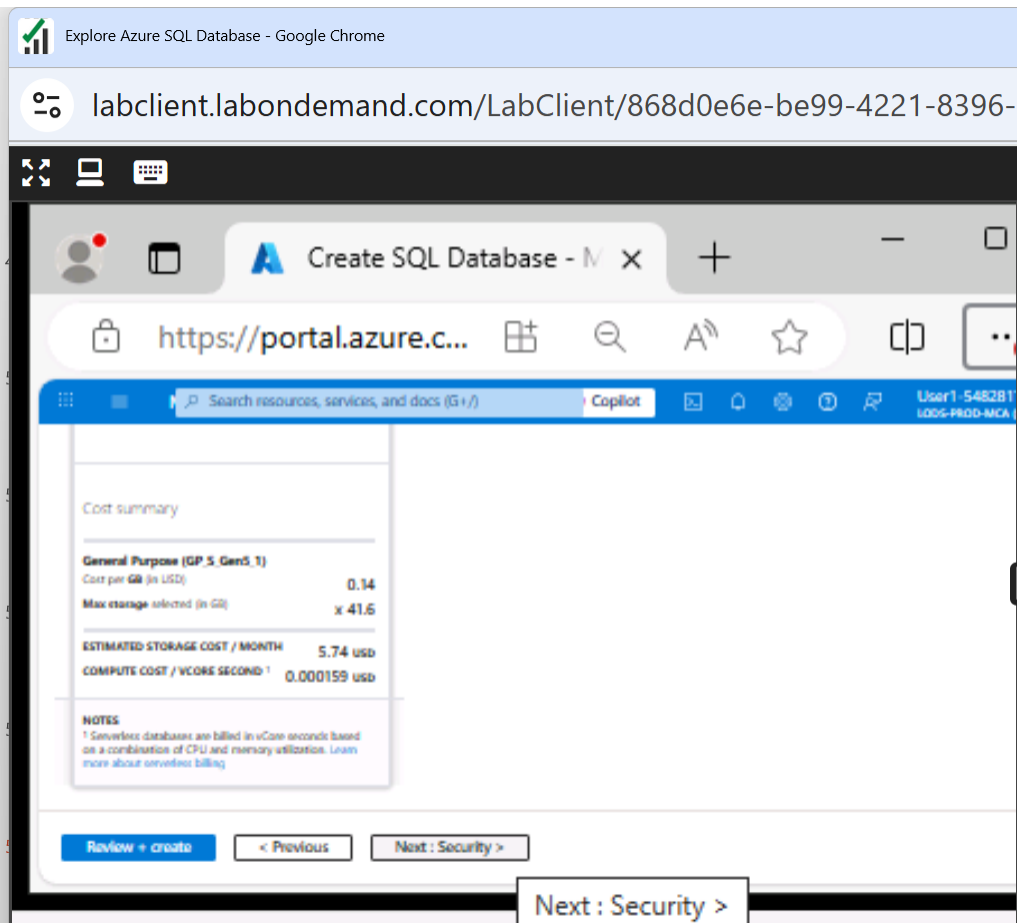
No Yes

Add current client IP address *

No Yes

Week 8 Tutorial Activity (Step-by-Step)

Scroll Down and Click on Next: Security >



Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/868d0e6e-be99-4221-8396-a9ff8c669906

Create SQL Database - M x

https://portal.azure.c...

Search resources, services, and docs (G+/f) Copilot

User1-548281
1005-PROD-MCA

Cost summary

General Purpose (GP, S, Gen5, 1)

Cost per GB (in USD) 0.14

Max storage selected (in GB) x 41.6

ESTIMATED STORAGE COST / MONTH 5.74 usd

COMPUTE COST / VCORE SECOND¹ 0.000159 usd

NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. Learn more about serverless billing

Review + create < Previous Next: Security >

Next: Security >

Explore Azure SQL Database X End

Instructions Resources ? ⚙

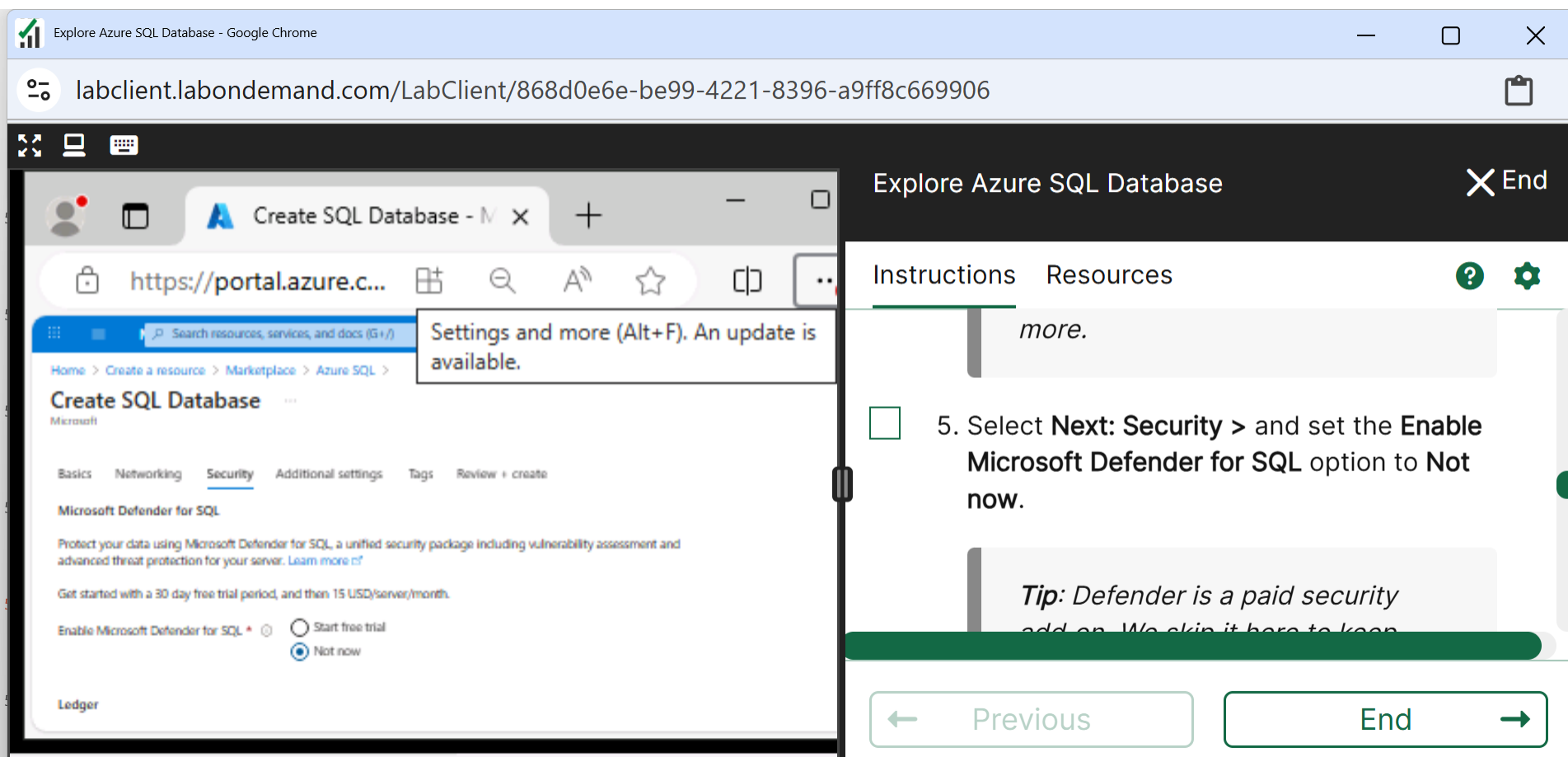
Tip: Public endpoint, allowing your IP lets you connect right away. Good for a short lab. In real projects you normally lock access down more.

☐ 5. Select **Next: Security >** and set the **Enable Microsoft Defender for SQL** option to **Not now**.

← Previous End →

Week 8 Tutorial Activity (Step-by-Step)

Microsoft Defender: Not Now



Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/868d0e6e-be99-4221-8396-a9ff8c669906

Create SQL Database - Microsoft

https://portal.azure.c...

Search resources, services, and docs (G+J)

Settings and more (Alt+F). An update is available.

Home > Create a resource > Marketplace > Azure SQL >

Create SQL Database

Microsoft

Basics Networking **Security** Additional settings Tags Review + create

Microsoft Defender for SQL

Protect your data using Microsoft Defender for SQL, a unified security package including vulnerability assessment and advanced threat protection for your server. [Learn more](#)

Get started with a 30 day free trial period, and then 15 USD/server/month.

Enable Microsoft Defender for SQL ☐ Start free trial ☒ Not now

Ledger

Explore Azure SQL Database

Instructions Resources

more.

5. Select **Next: Security >** and set the **Enable Microsoft Defender for SQL** option to **Not now**.

Tip: Defender is a paid security add-on. We skip it here to keep

Previous End

Week 8 Tutorial Activity (Step-by-Step)

Next: Additional Settings

Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/868d0e6e-be99-4221-8396-a9ff8c669906

Explore Azure SQL Database

Instructions Resources

short exercise.

6. Select **Next: Additional Settings >** and on the **Additional settings** tab, set the **Use existing data** option to **Sample** (this will create a sample database that you can explore later).

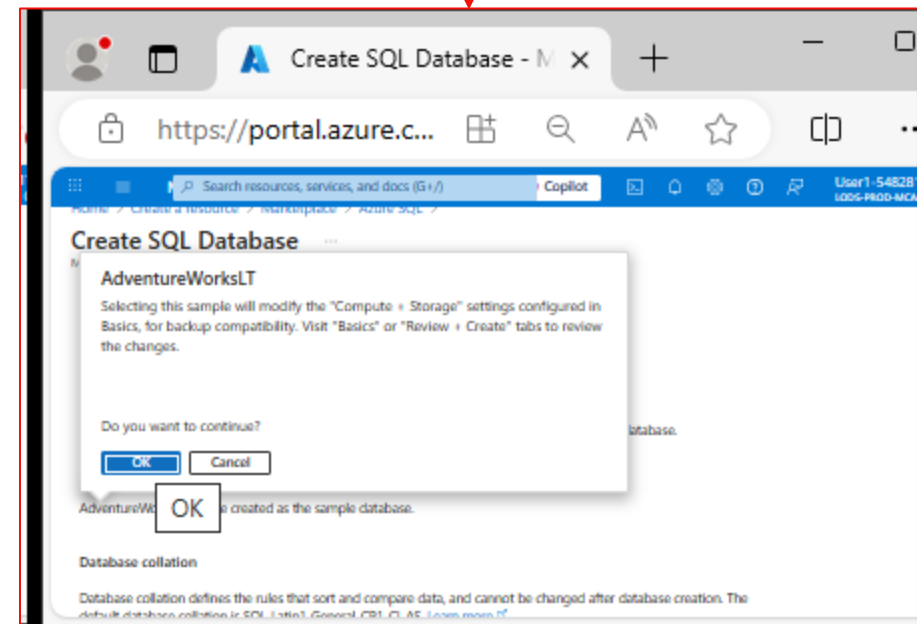
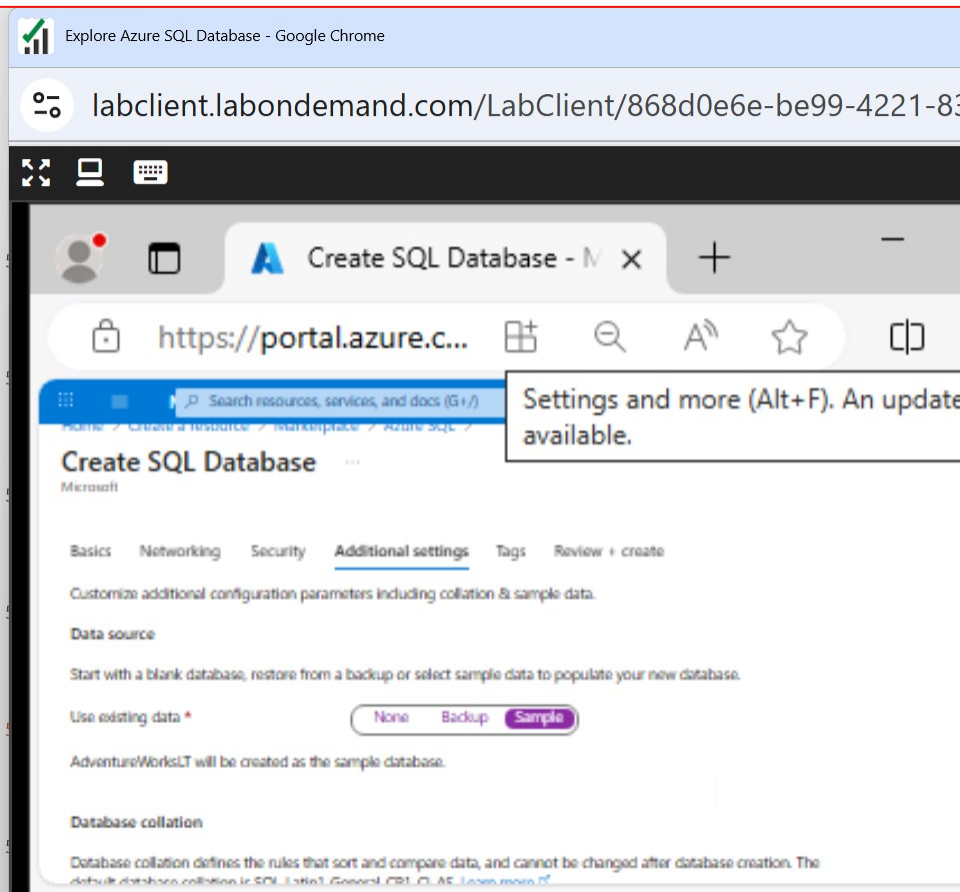
Review + create < Previous Next: Additional settings >

Next: Additional settings >

Previous End

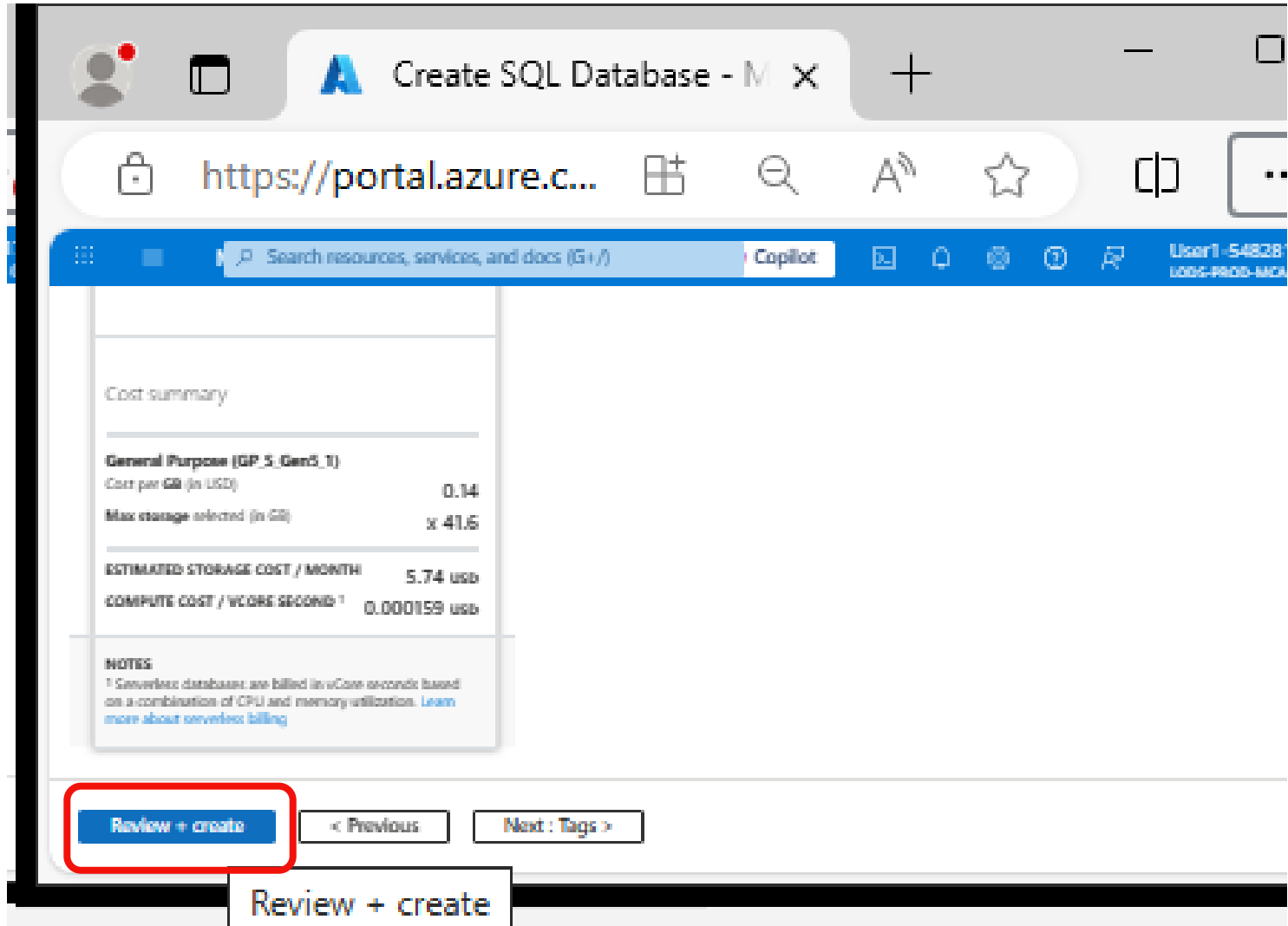
Week 8 Tutorial Activity (Step-by-Step)

- Choose Sample
- Click Ok to continue



Week 8 Tutorial Activity (Step-by-Step)

Review and Create



The screenshot shows the 'Review and Create' step of the 'Create SQL Database' wizard in the Azure portal. The browser tab is 'Create SQL Database - M x'. The address bar shows 'https://portal.azure.c...'. The search bar contains 'Search resources, services, and docs (G+/)' and 'Copilot'. The user is 'User1 -54838' with role 'USER-PROD-MCA'.

Cost summary

General Purpose (GP_S_Gen5_1)	
Cost per GB (in USD)	0.14
Max storage selected (in GB)	x 41.6
ESTIMATED STORAGE COST / MONTH 5.74 usd	
COMPUTE COST / VCORE SECOND¹ 0.000159 usd	

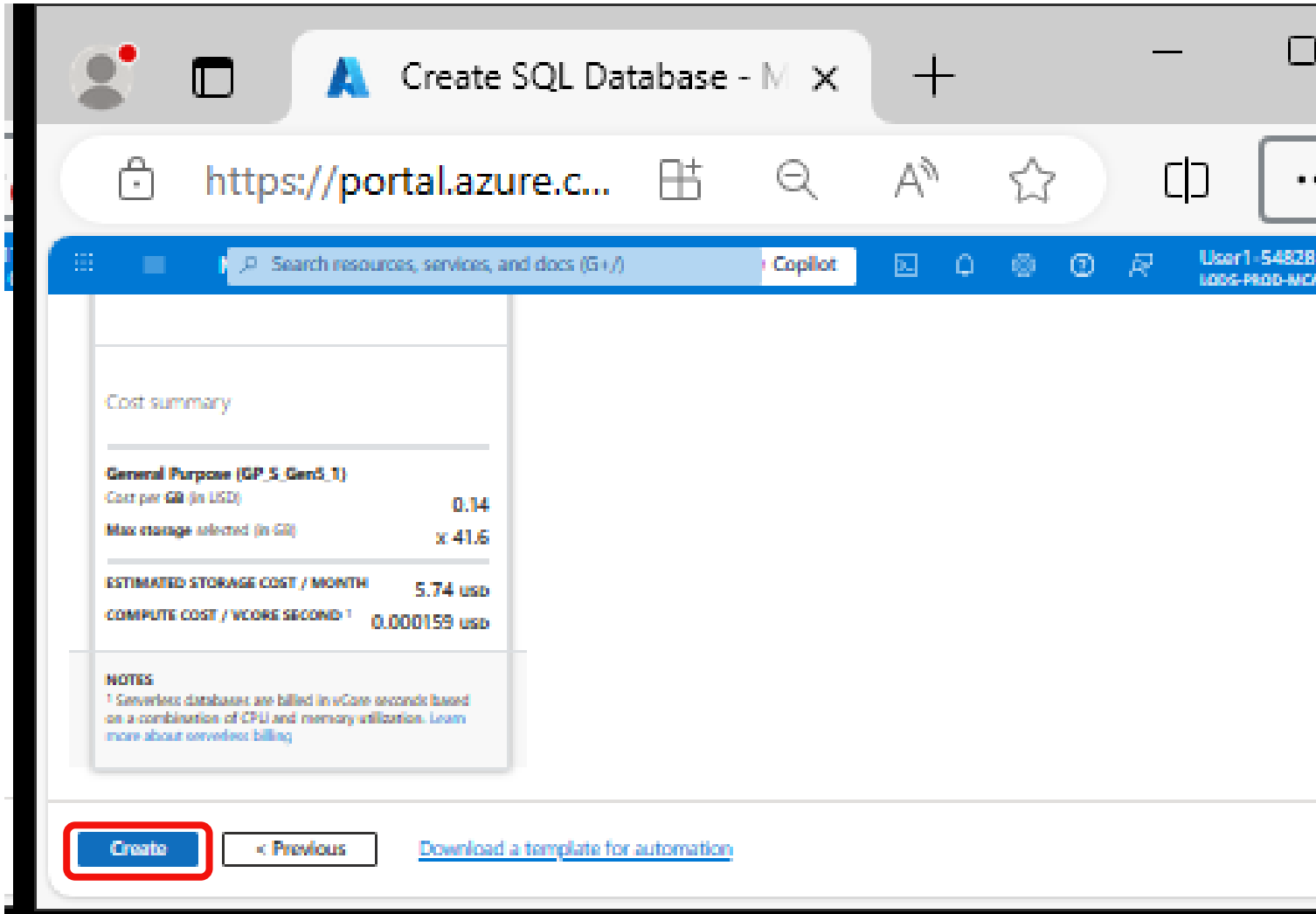
NOTES
¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

Review + create < Previous Next: Tags >

Review + create

Week 8 Tutorial Activity (Step-by-Step)

Create



The screenshot shows the Azure portal interface for creating a SQL database. The browser tab is titled 'Create SQL Database - M x'. The address bar shows 'https://portal.azure.c...'. The Azure portal header includes a search bar, a 'Copilot' button, and user information 'User1-54828' and 'LOGG-PROD-MCA'. The main content area displays a 'Cost summary' section with the following details:

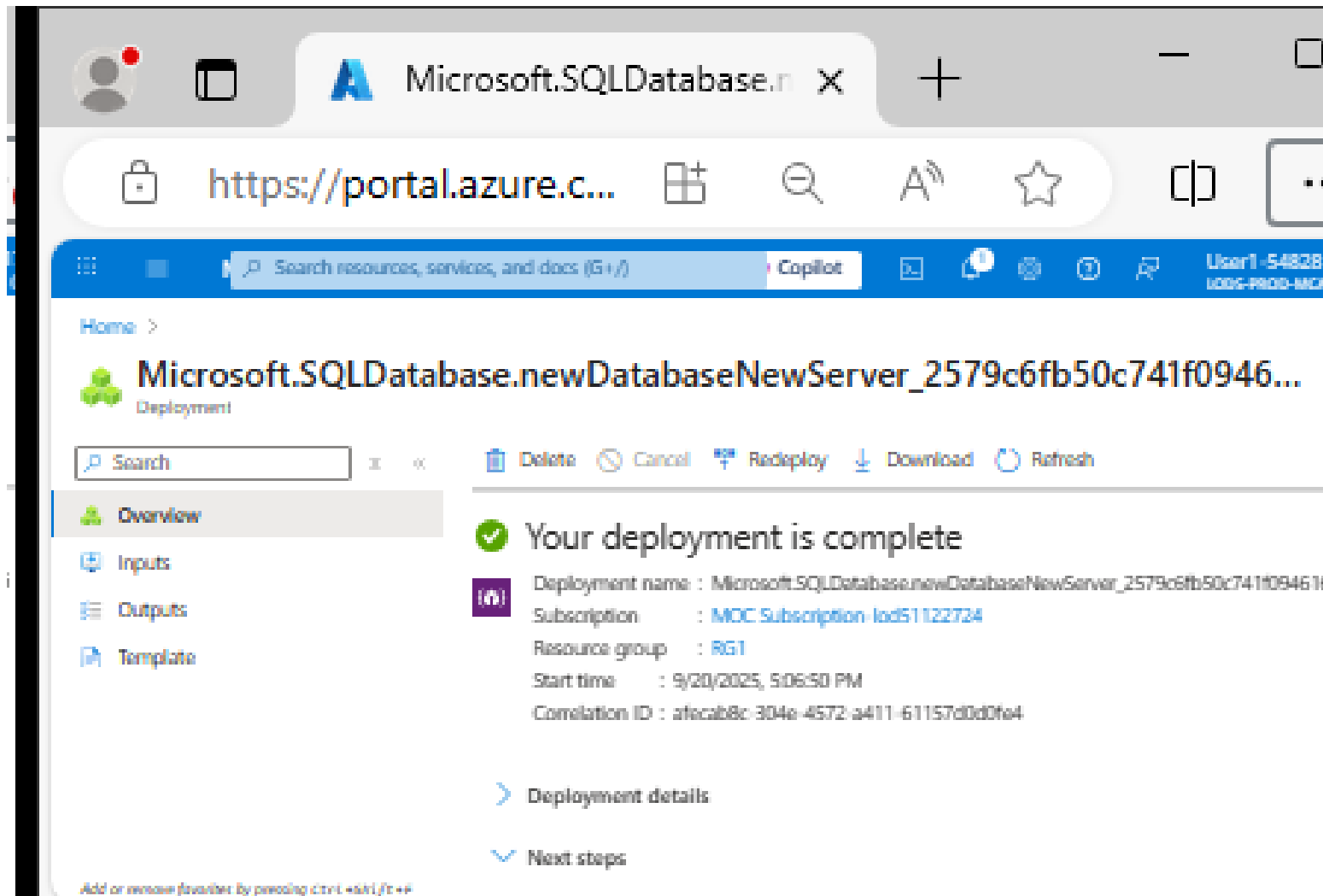
Cost summary	
General Purpose (GP_S_Gen5_1)	
Cost per GB (in USD)	0.14
Max storage selected (in GB)	x 41.6
ESTIMATED STORAGE COST / MONTH	
	5.74 usd
COMPUTE COST / vCORE SECOND¹	
	0.000159 usd

NOTES
¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

At the bottom, there are three buttons: 'Create' (highlighted with a red rectangle), '< Previous', and 'Download a template for automation'.

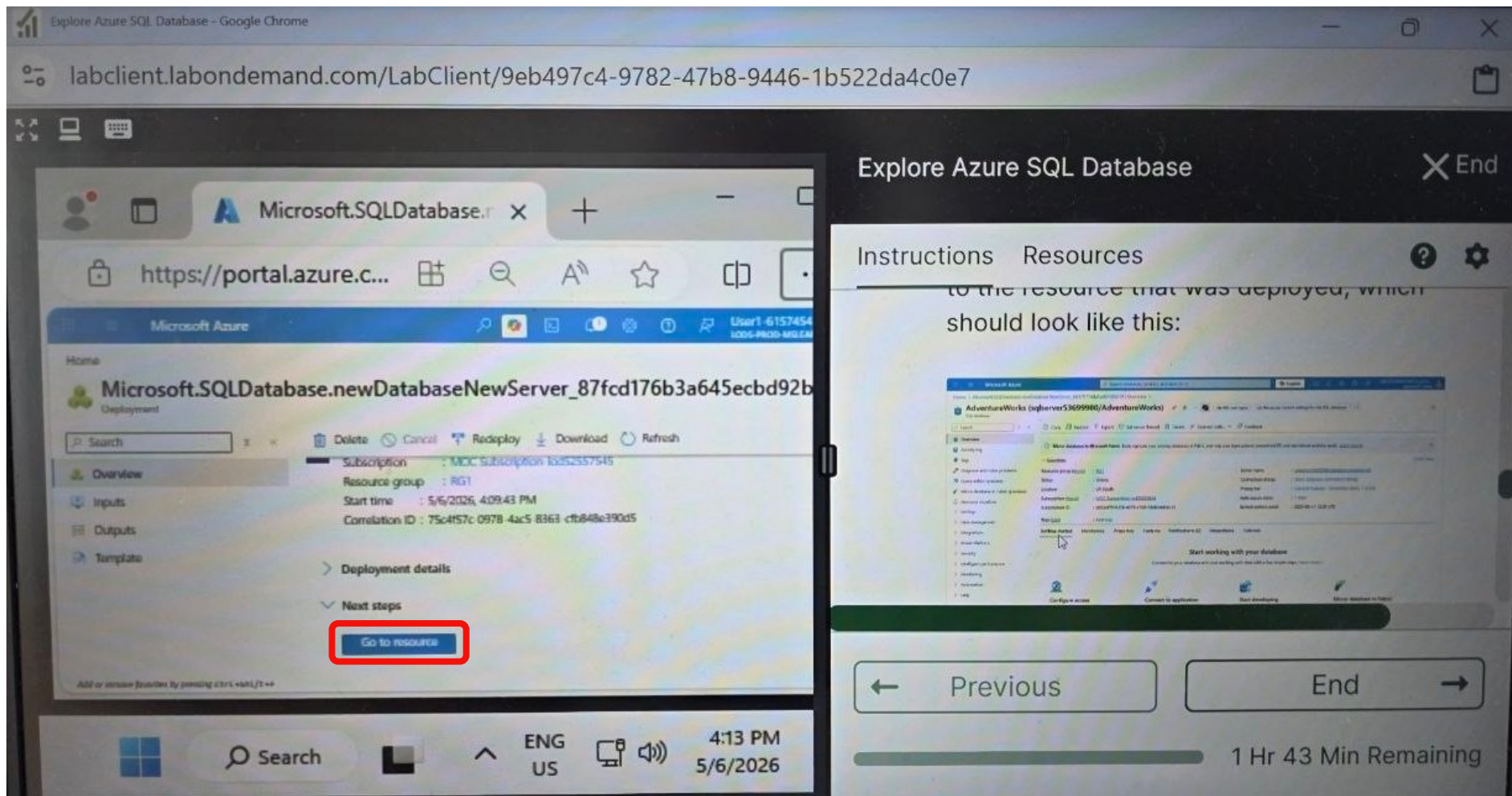
Week 8 Tutorial Activity (Step-by-Step)

Wait for deployment to complete



Week 8 Tutorial Activity (Step-by-Step)

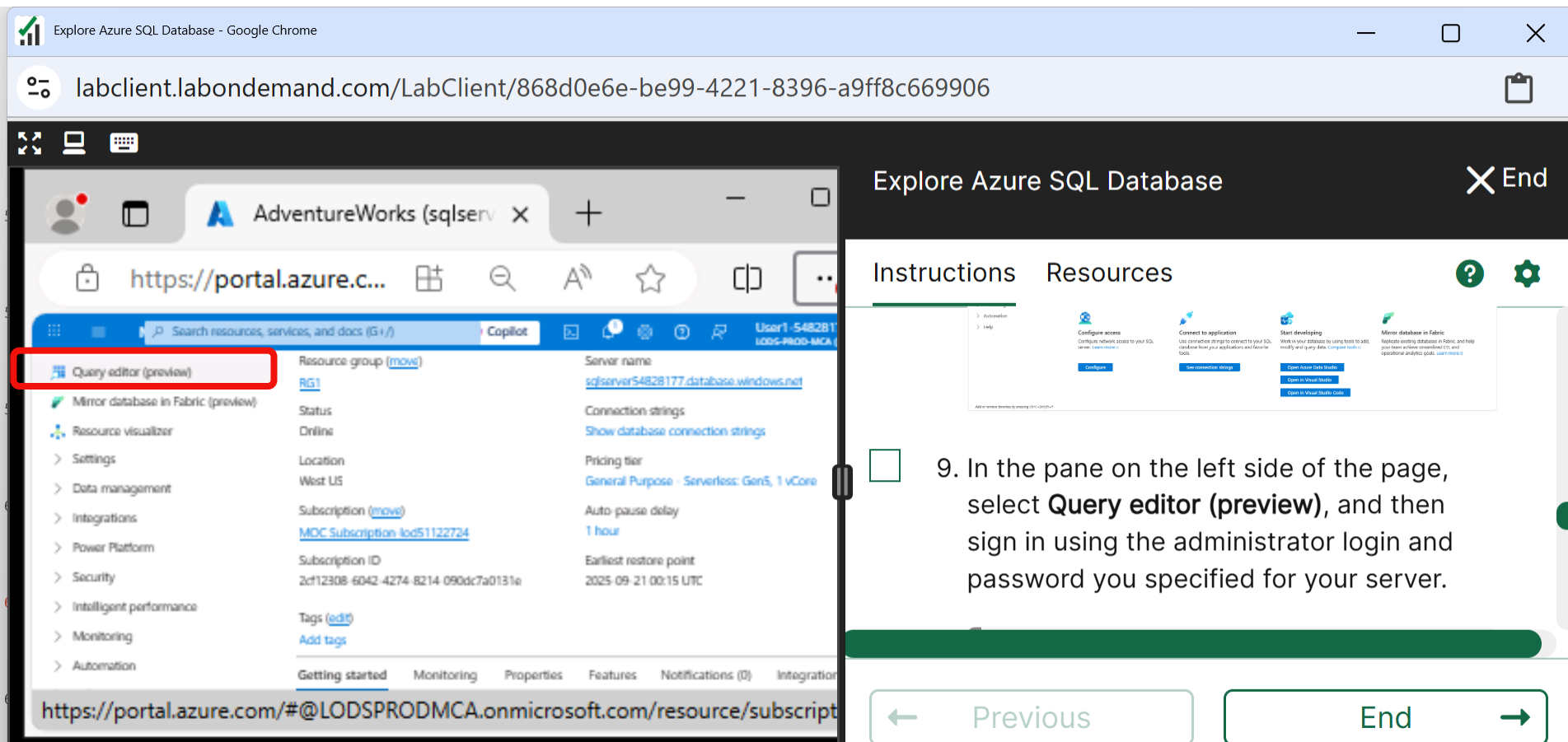
Scroll Down and Go to the Resource that is now deployed



The screenshot displays a web browser window titled "labclient.labondemand.com/LabClient/9eb497c4-9782-47b8-9446-1b522da4c0e7". The browser shows the Microsoft Azure portal for a user named "User1-6157454". The main content area displays the deployment details for a resource named "Microsoft.SQLDatabase.newDatabaseNewServer_87fcd176b3a645ecbd92b". The deployment status is "Succeeded", and the "Go to resource" button is highlighted with a red rectangle. Below the deployment details, there is a section for "Next steps" with a "Go to resource" button. To the right of the browser window, there is a video player titled "Explore Azure SQL Database". The video player shows a thumbnail of the Azure portal interface with the text "to the resource that was deployed, which should look like this:". The video player controls include a "Previous" button, an "End" button, and a progress bar showing "1 Hr 43 Min Remaining".

Week 8 Tutorial Activity (Step-by-Step)

Click on Query Editor (preview)

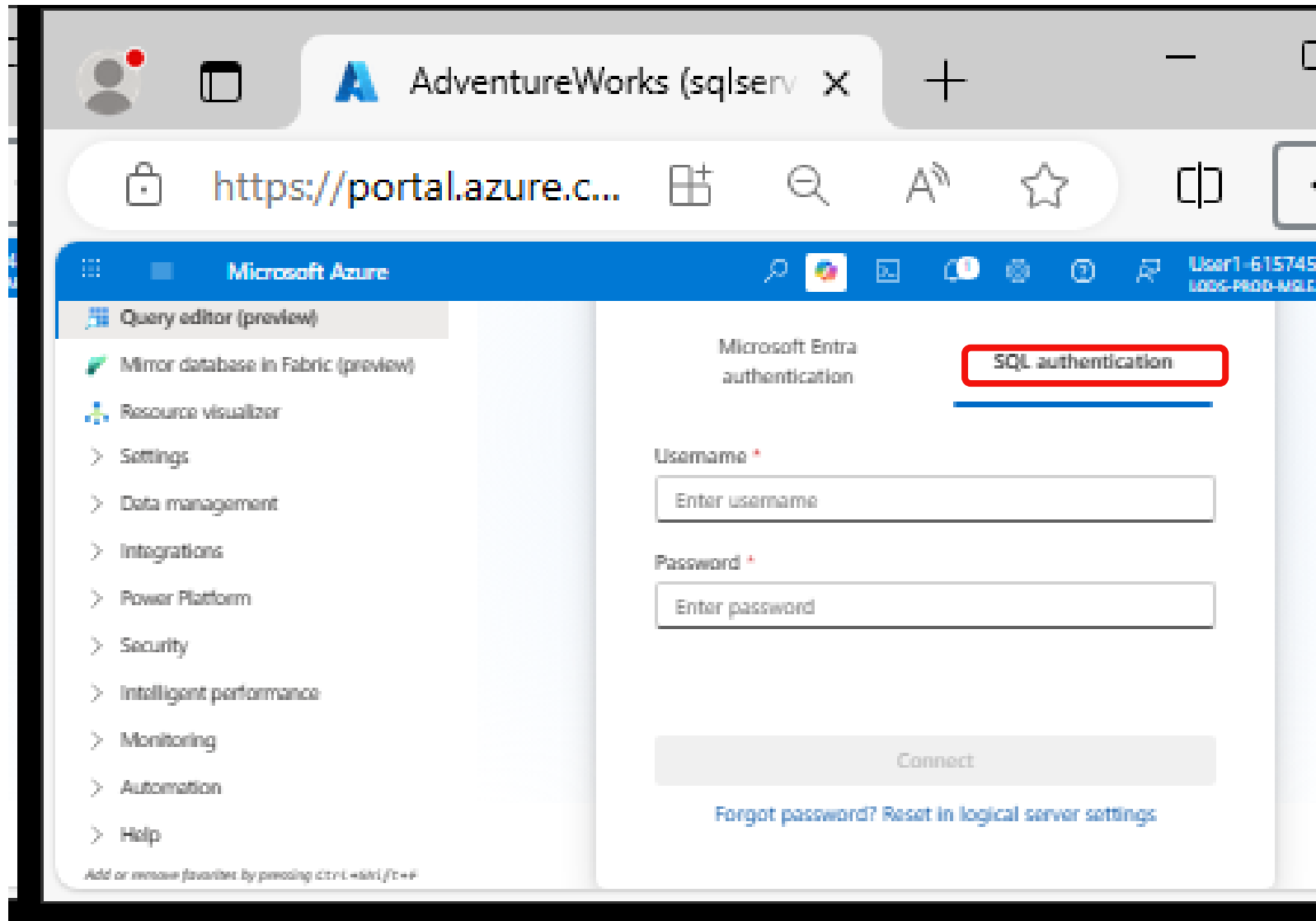


The screenshot shows the Azure portal interface. On the left, the navigation pane is open, and the 'Query editor (preview)' option is highlighted with a red box. The main pane displays the 'Explore Azure SQL Database' page, which includes sections for 'Instructions' and 'Resources'. The 'Instructions' section contains a list of steps: 'Automation', 'Help', 'Configure access', 'Connect to application', 'Start developing', and 'Mirror database in Fabric'. The 'Resources' section contains links to 'Open Azure Data Studio', 'Open in Visual Studio', and 'Open in Visual Studio Code'. Below the main pane, there are buttons for 'Previous' and 'End'.

9. In the pane on the left side of the page, select **Query editor (preview)**, and then sign in using the administrator login and password you specified for your server.

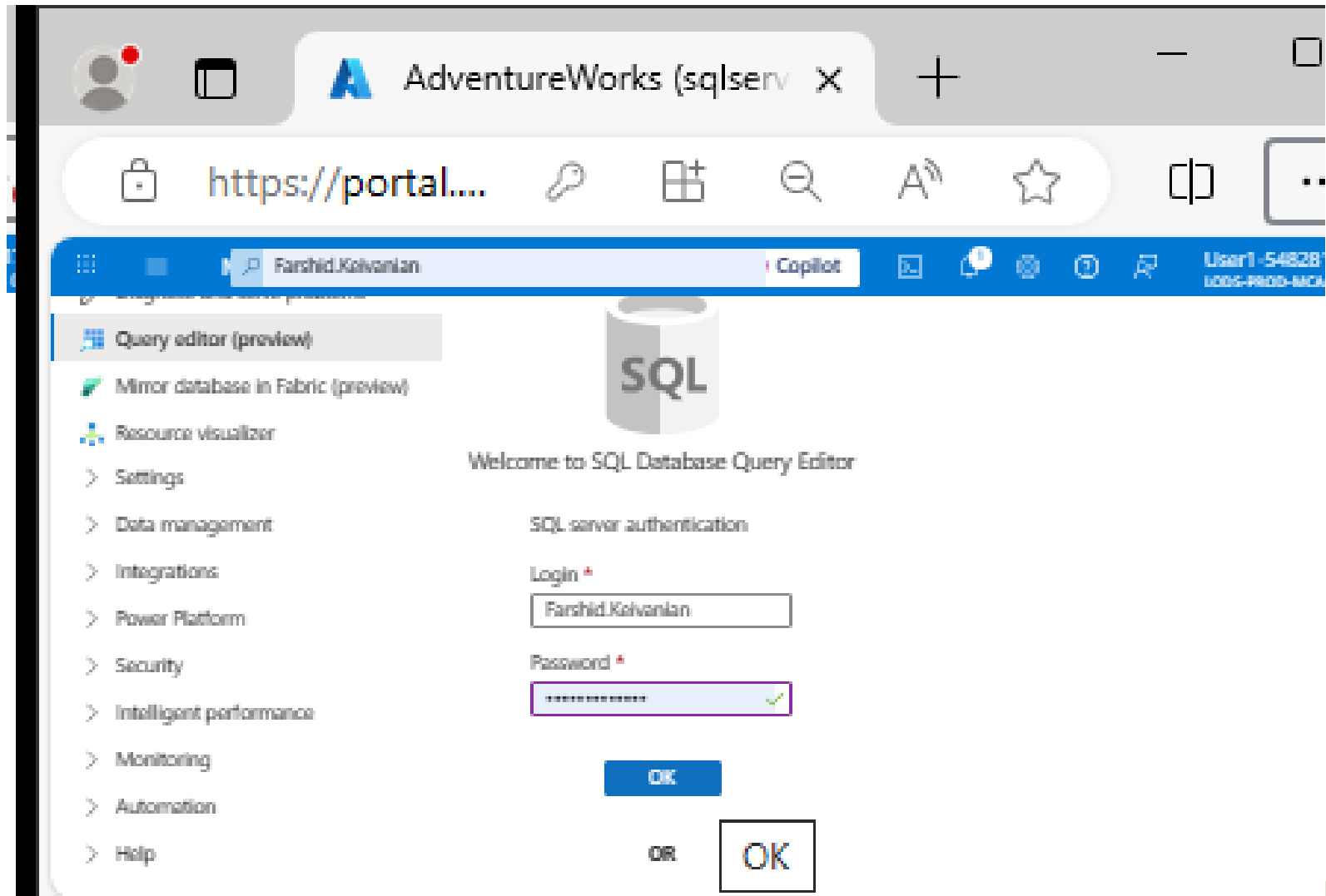
Week 8 Tutorial Activity (Step-by-Step)

Click SQL authentication



Week 8 Tutorial Activity (Step-by-Step)

Now enter your server administration log in details



AdventureWorks (sqlserv) X

https://portal....

Farshid.Kalvanian Copilot User1-548281-1005-PROD-MCA

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

Data management

Integrations

Power Platform

Security

Intelligent performance

Monitoring

Automation

Help

SQL

Welcome to SQL Database Query Editor

SQL server authentication

Login *

Farshid.Kalvanian

Password *

OK

OR

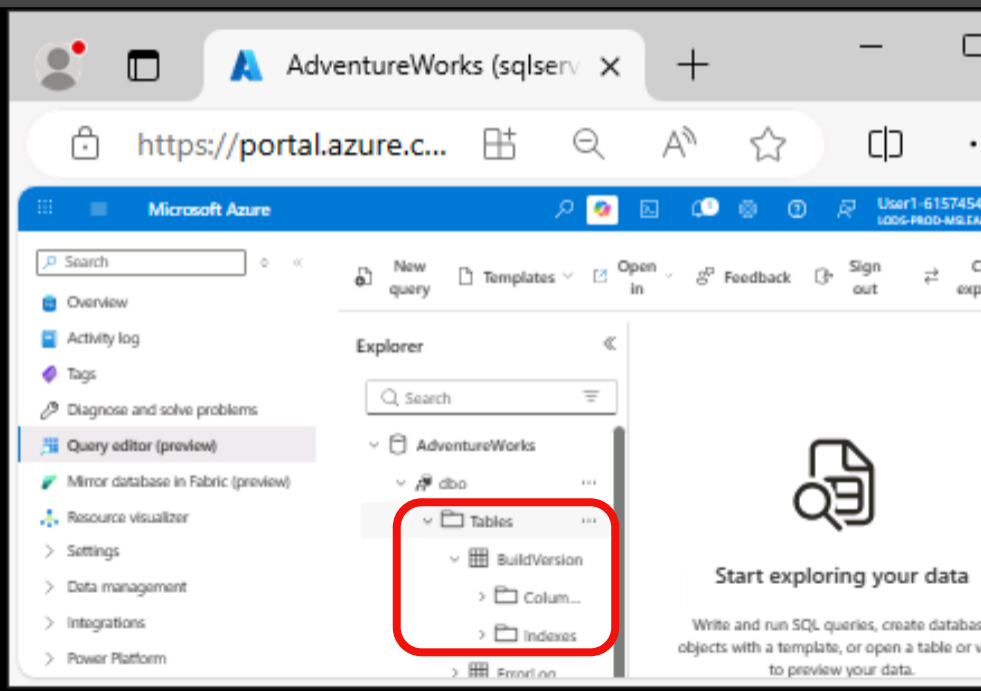
OK

Week 8 Tutorial Activity (Step-by-Step)

You should see the Tables when you expand them

Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/9eb497c4-9782-47b8-9446-1b522da4c0e7



AdventureWorks (sqlserv x)

https://portal.azure.c...

Microsoft Azure

User1-6157454
1006-PROD-MG.EA

Search

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform

New query
Templates
Open in
Feedback
Sign out

Explorer

AdventureWorks

dbo

Tables

BuildVersion

Column...

Indexes

Start exploring your data

Write and run SQL queries, create database objects with a template, or open a table or view to preview your data.

Explore Azure SQL Database

Instructions Resources

10. In the pane on the left side of the page, select **Query editor (preview)**, and then sign in using the administrator login and password you specified for your server.

Note: If an error message stating

Previous End

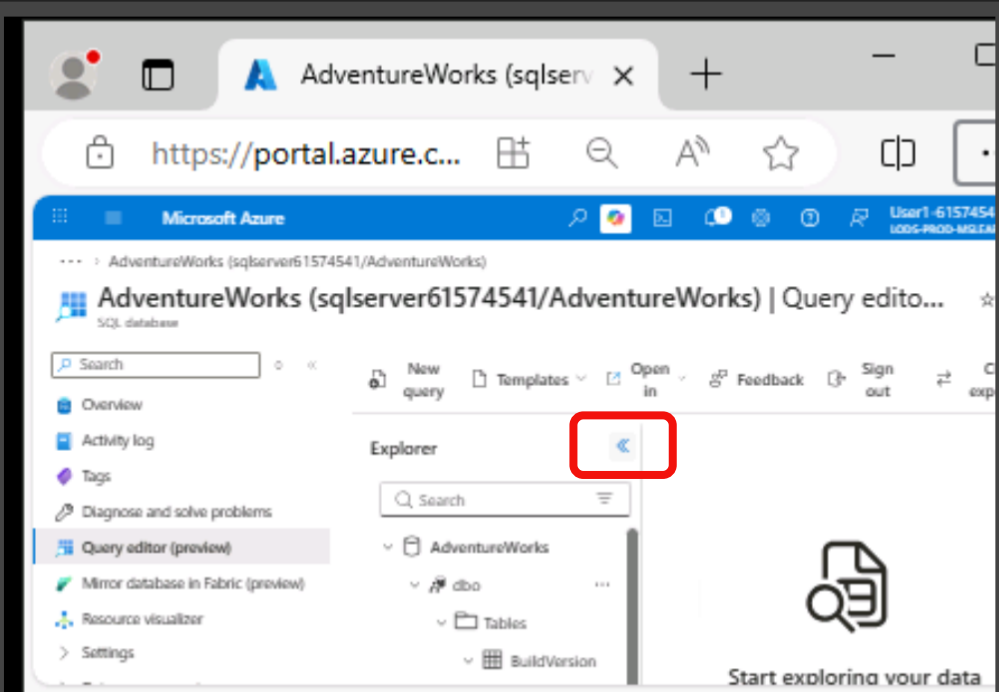
1 Hr 33 Min Remaining

Week 8 Tutorial Activity (Step-by-Step)

Click on <<

Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/9eb497c4-9782-47b8-9446-1b522da4c0e7



AdventureWorks (sqlserv x) +

https://portal.azure.c...

Microsoft Azure

User1-61574541

AdventureWorks (sqlserver61574541/AdventureWorks) | Query edito...

SQL database

Search

Overview

Activity log

Tags

Diagnose and solve problems

Query editor (preview)

Mirror database in Fabric (preview)

Resource visualizer

Settings

New query

Templates

Open in

Feedback

Sign out

Explorer

Search

AdventureWorks

dbo

Tables

BuildVersion

Start exploring your data

Explore Azure SQL Database

End

Instructions Resources

10. In the pane on the left side of the page, select **Query editor (preview)**, and then sign in using the administrator login and password you specified for your server.

Note: If an error message stating

Previous End

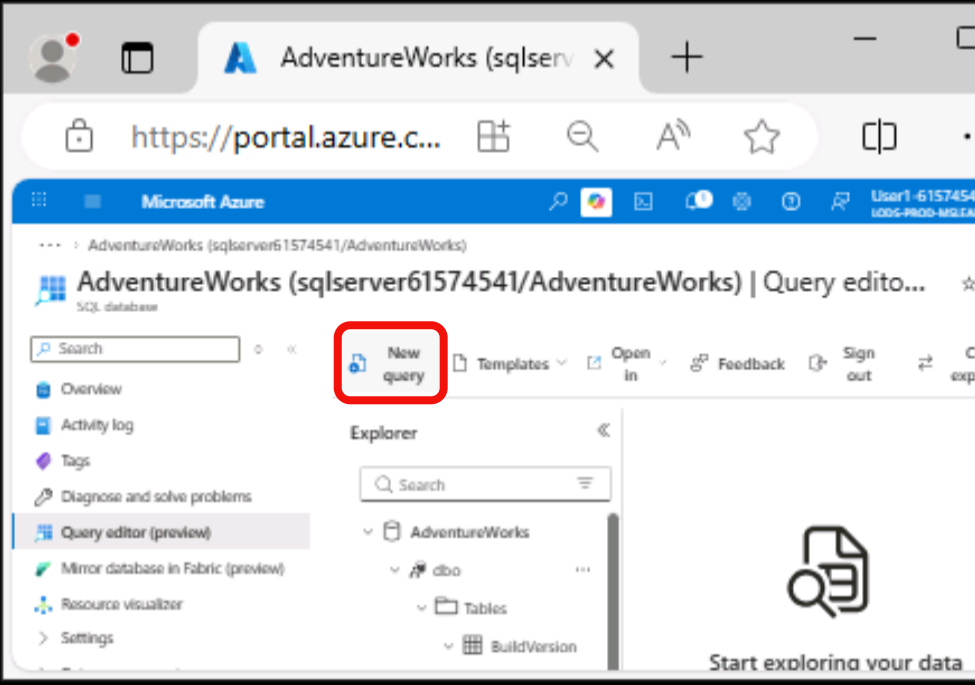
1 Hr 32 Min Remaining

Week 8 Tutorial Activity (Step-by-Step)

Click New Query

Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/9eb497c4-9782-47b8-9446-1b522da4c0e7



The screenshot shows the Azure portal interface for 'AdventureWorks (sqlserver61574541/AdventureWorks)'. The 'New query' button is highlighted with a red box. The interface includes a search bar, navigation links (Overview, Activity log, Tags, Diagnose and solve problems, Query editor (preview), Mirror database in Fabric (preview), Resource visualizer, Settings), and an Explorer pane showing the database structure (AdventureWorks, dbo, Tables, BuildVersion).

Explore Azure SQL Database

Instructions Resources

10. In the pane on the left side of the page, select **Query editor (preview)**, and then sign in using the administrator login and password you specified for your server.

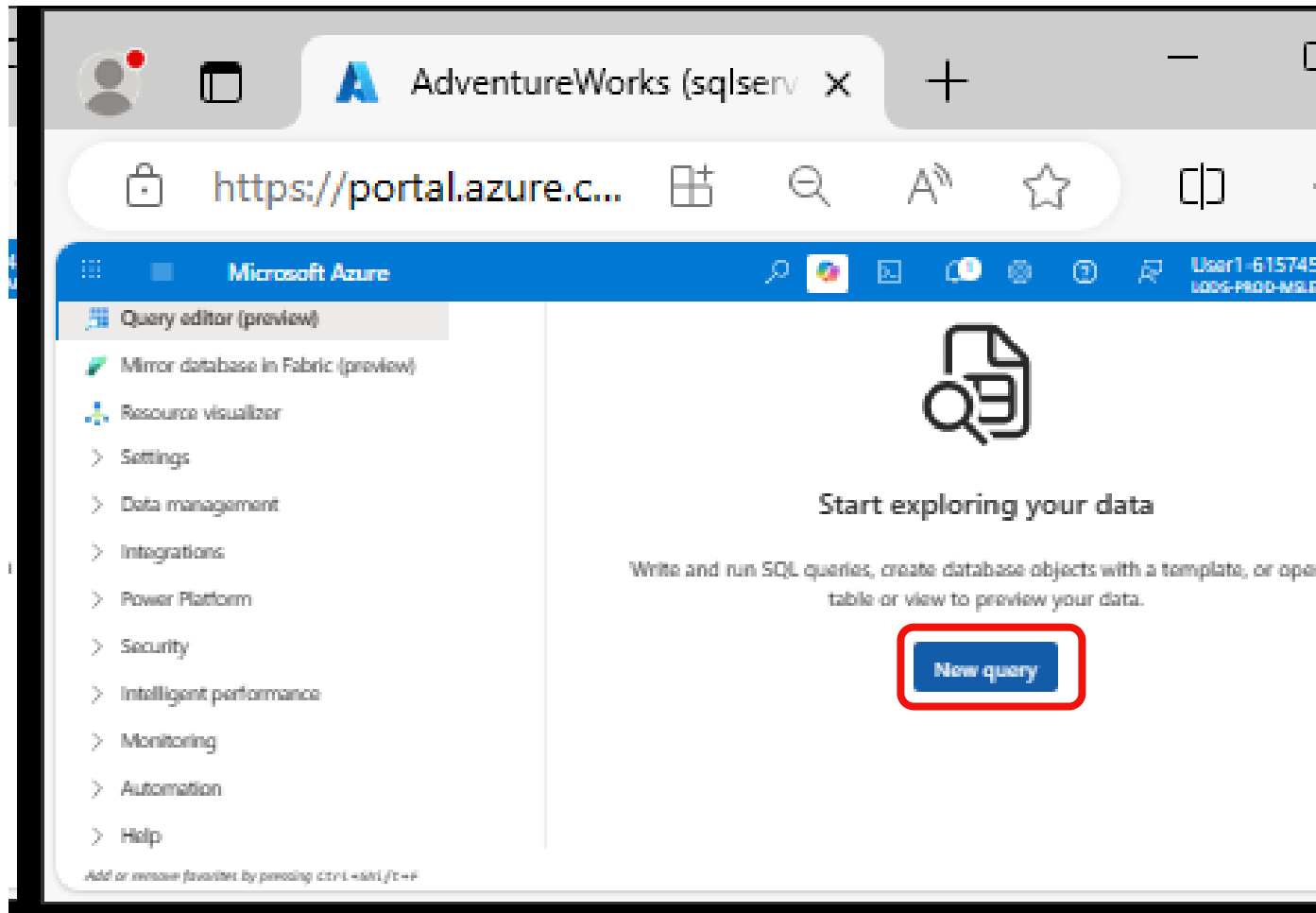
Note: If an error message stating

Previous End

1 Hr 32 Min Remaining

Week 8 Tutorial Activity (Step-by-Step)

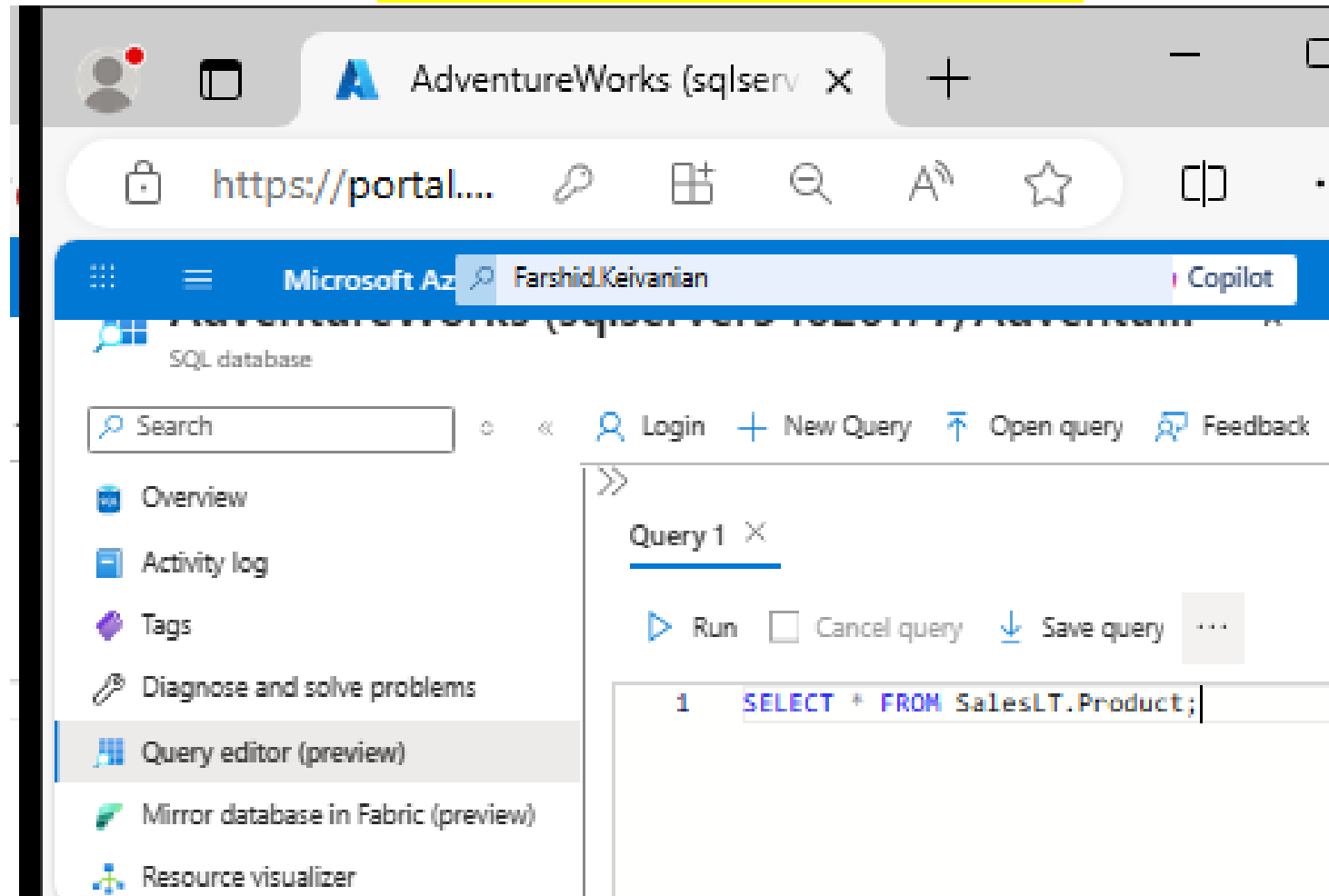
Click New Query



Week 8 Tutorial Activity (Step-by-Step)

Step 12 – Take a screenshot) In the Query 1, enter the following code: (Click on Type on right side

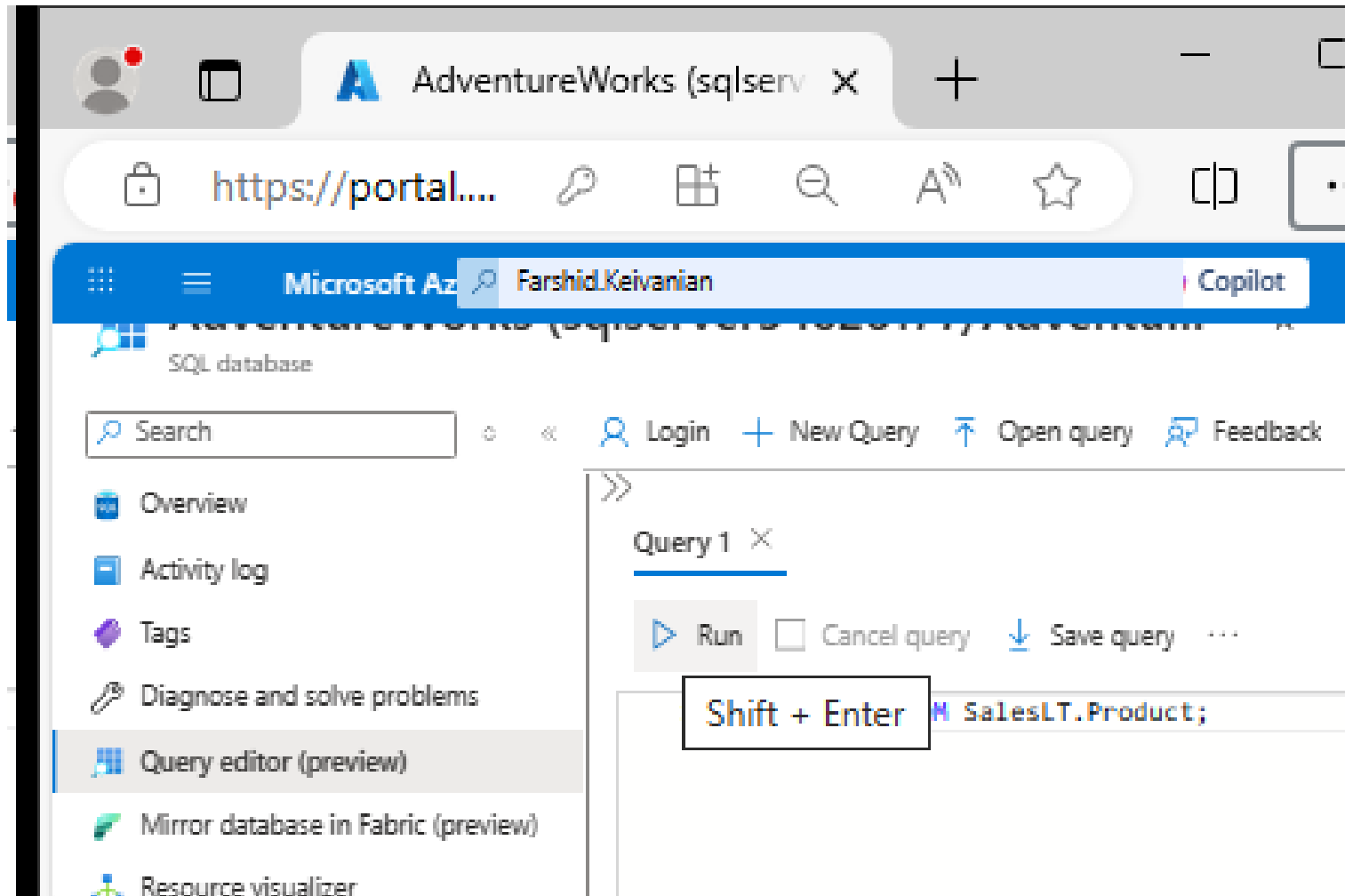
panel)



Week 8 Tutorial Activity (Step-by-Step)

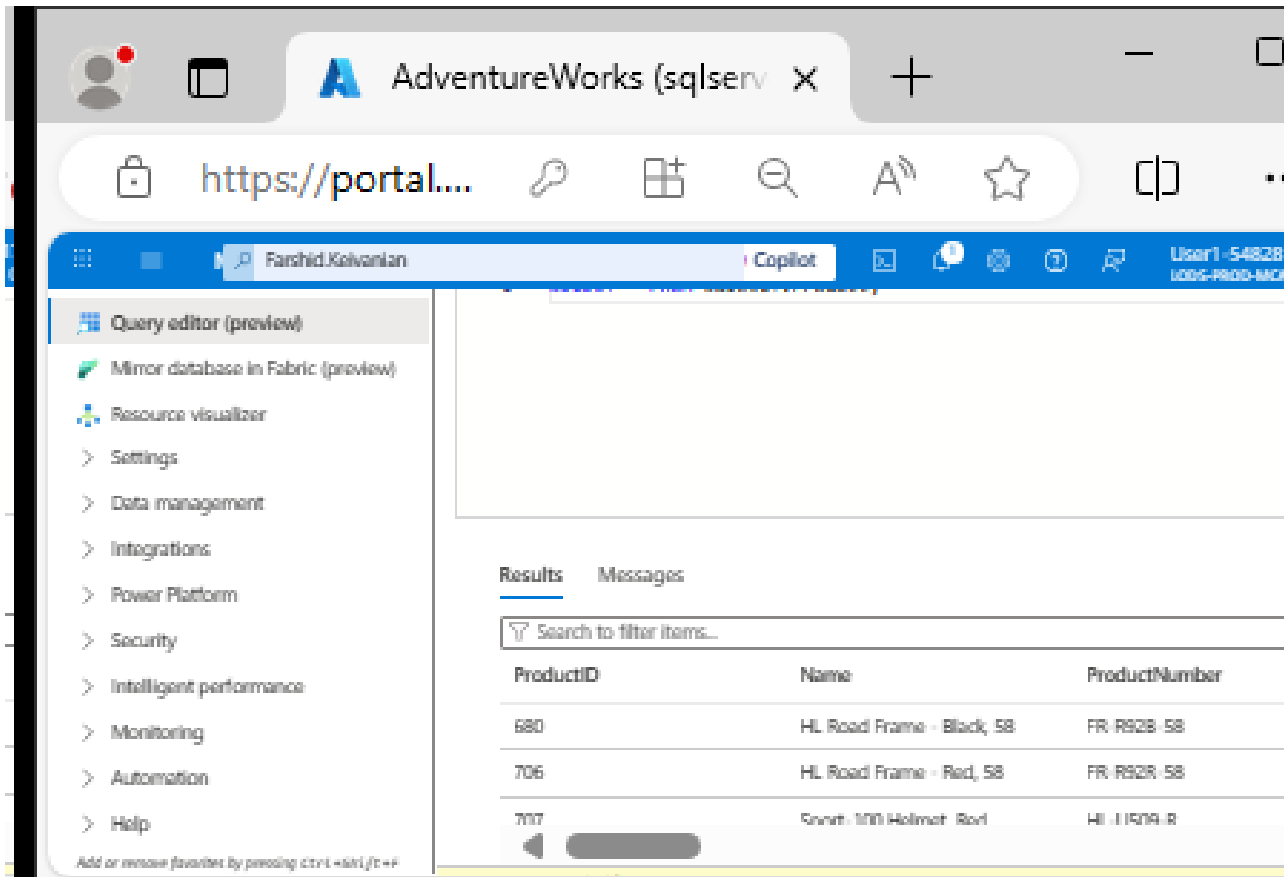
Step 12 – Take a screenshot) Shift + Enter (Or click

Run)



Week 8 Tutorial Activity (Step-by-Step)

Step 12 – Take a screenshot) It will show all columns for all rows

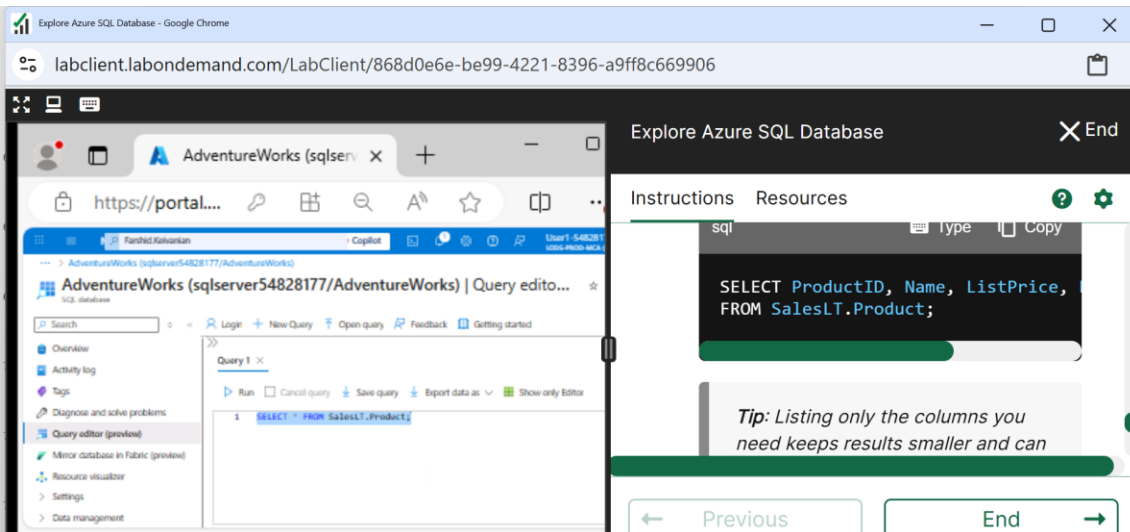


The screenshot shows the Microsoft Fabric portal interface. The left sidebar contains a navigation menu with the following items: Query editor (preview), Mirror database in Fabric (preview), Resource visualizer, Settings, Data management, Integrations, Power Platform, Security, Intelligent performance, Monitoring, Automation, and Help. The main area displays a query editor with a table of results. The table has three columns: ProductID, Name, and ProductNumber. The results are as follows:

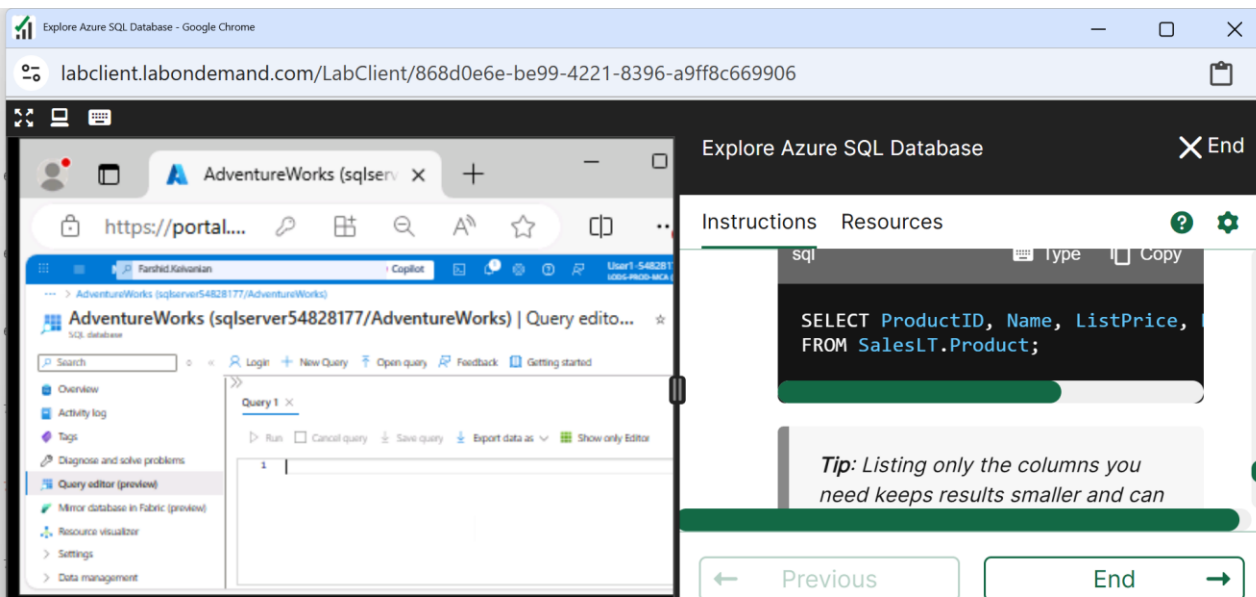
ProductID	Name	ProductNumber
680	HL Road Frame - Black, 58	FR-R928-58
706	HL Road Frame - Red, 58	FR-R928-58
707	Scout 100 Helmet - Red	HL-I15714-R

Week 8 Tutorial Activity (Step-by-Step)

Select All and remove



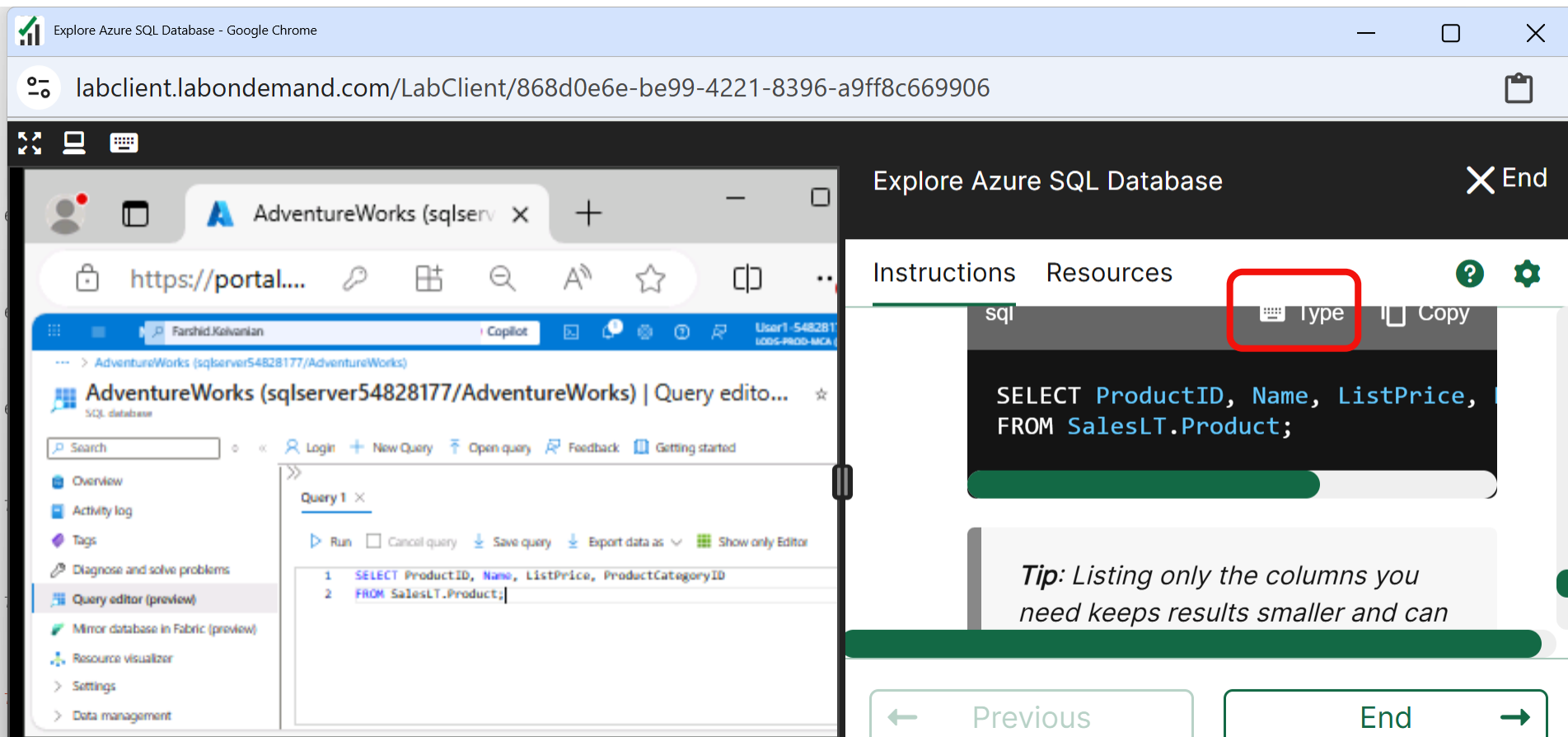
The screenshot shows the 'Explore Azure SQL Database' interface. On the left, the 'Query editor' is open, displaying a SQL query: `SELECT * FROM SalesLT.Product;`. The query is highlighted with a blue selection bar. On the right, the 'Instructions' panel shows the same query: `SELECT ProductID, Name, ListPrice, FROM SalesLT.Product;`. A red arrow points from the 'Query editor' to the 'Instructions' panel.



The screenshot shows the 'Explore Azure SQL Database' interface. On the left, the 'Query editor' is open, displaying a SQL query: `SELECT * FROM SalesLT.Product;`. The query is highlighted with a blue selection bar. On the right, the 'Instructions' panel shows the same query: `SELECT ProductID, Name, ListPrice, FROM SalesLT.Product;`. A red arrow points from the 'Query editor' to the 'Instructions' panel.

Week 8 Tutorial Activity (Step-by-Step)

Step 13 – Take a screenshot) Click on Type to enter this Query



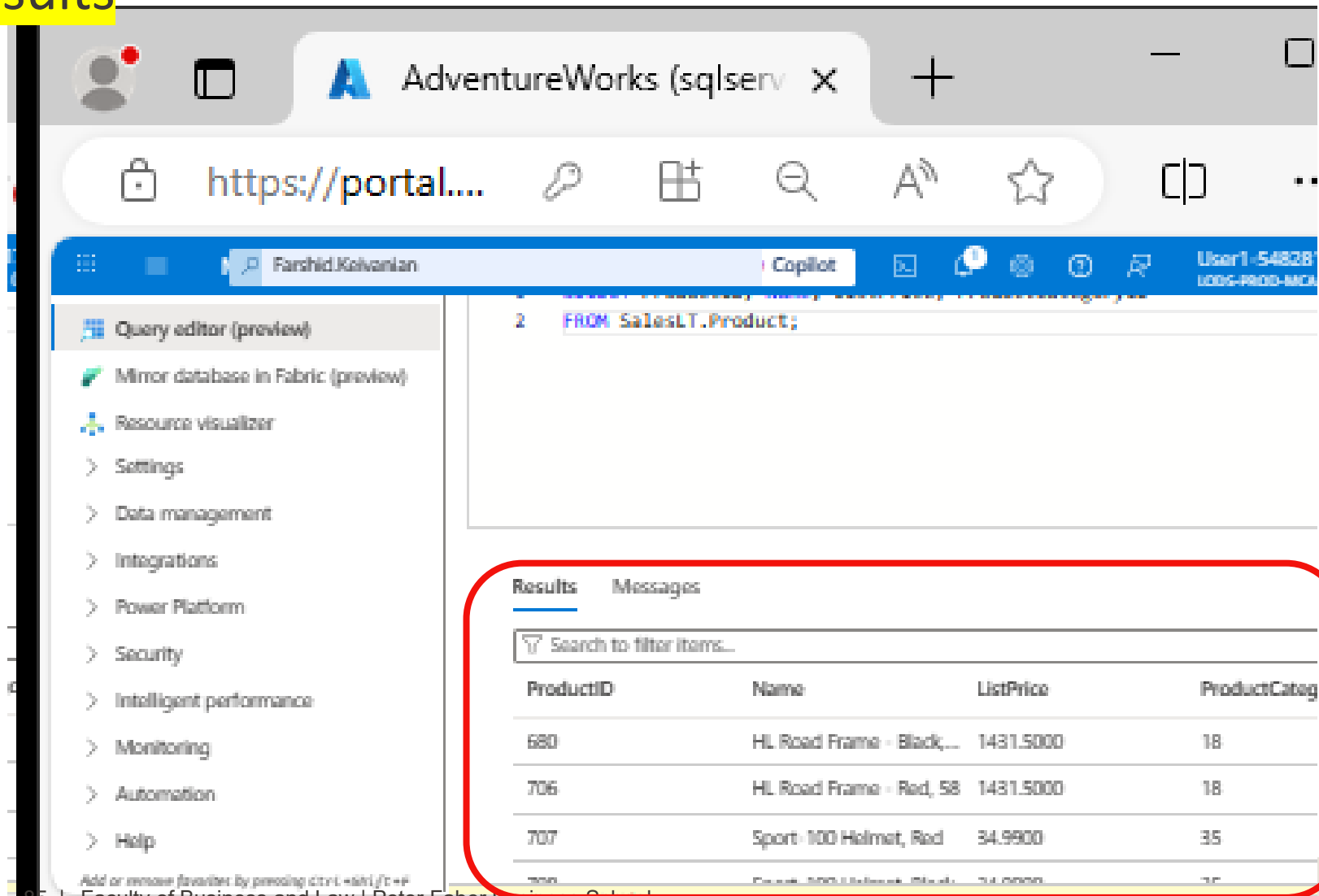
The screenshot displays the 'Explore Azure SQL Database' web application. The main interface shows the 'AdventureWorks (sqlserv...)' database selected. The 'Query editor (preview)' tab is active, showing a query editor with the following SQL code:

```
1 SELECT ProductID, Name, ListPrice, ProductCategoryID
2 FROM SalesLT.Product;
```

A modal window titled 'Explore Azure SQL Database' is open, showing the 'Instructions' and 'Resources' tabs. The 'Type' button is highlighted with a red box. The modal also displays the same SQL query and a tip: 'Tip: Listing only the columns you need keeps results smaller and can...'. The modal includes a 'Copy' button and a 'Previous' button.

Week 8 Tutorial Activity (Step-by-Step)

Step 13 – Take a screenshot) You will see the results



The screenshot shows the Microsoft Fabric portal interface. The browser tab is titled 'AdventureWorks (sqlserv...'. The address bar shows 'https://portal....'. The user is logged in as 'User1-548281' with role 'LODS-PROD-MCA'. The left sidebar contains a navigation menu with items like 'Query editor (preview)', 'Mirror database in Fabric (preview)', 'Resource visualizer', 'Settings', 'Data management', 'Integrations', 'Power Platform', 'Security', 'Intelligent performance', 'Monitoring', 'Automation', and 'Help'. The main area displays a SQL query in the editor:

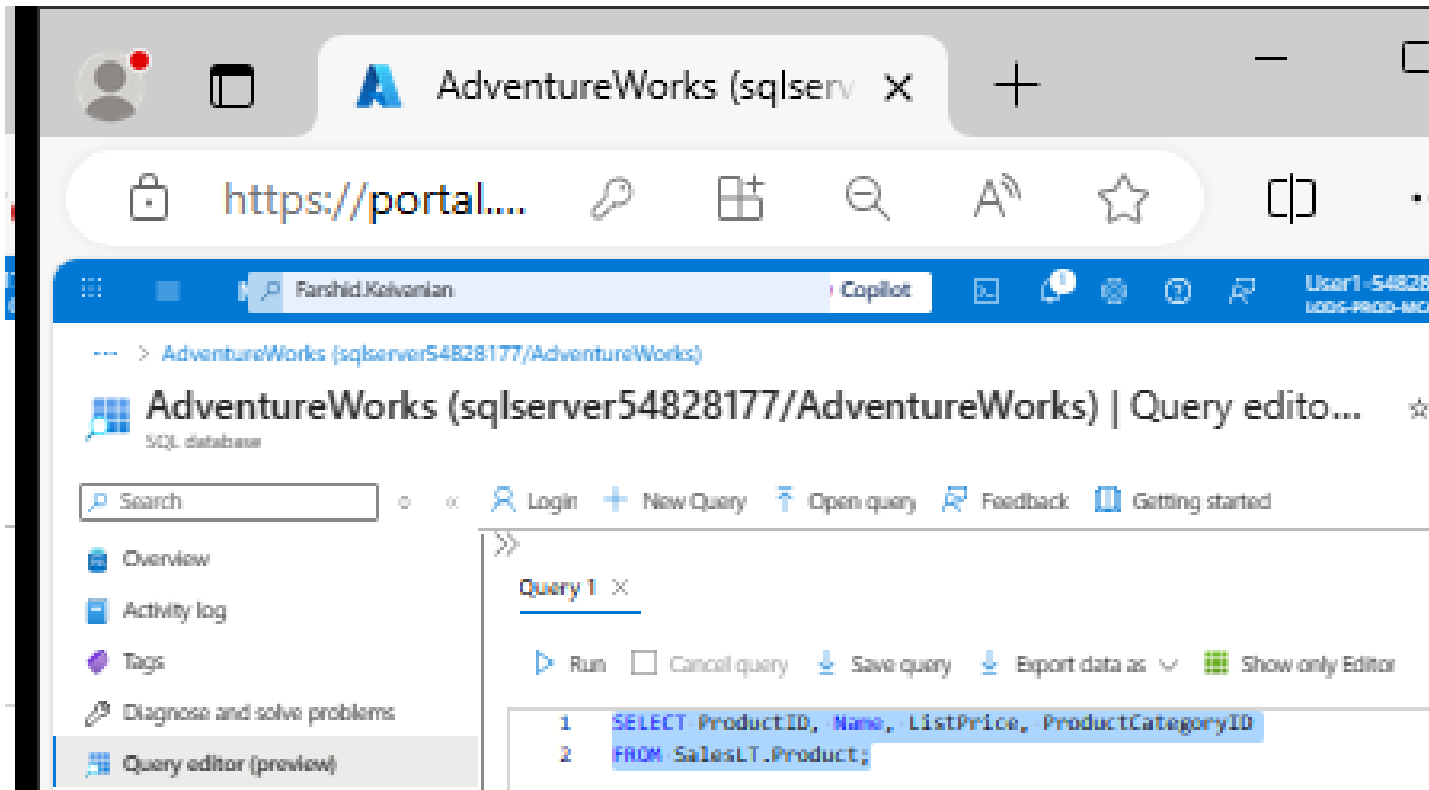
```
2 FROM SalesLT.Product;
```

Below the query editor, the 'Results' tab is active, showing a table of data. The table has columns: ProductID, Name, ListPrice, and ProductCateg. The results are:

ProductID	Name	ListPrice	ProductCateg
680	HL Road Frame - Black,...	1431.5000	18
706	HL Road Frame - Red, 58	1431.5000	18
707	Sport-100 Helmet, Red	34.9900	35

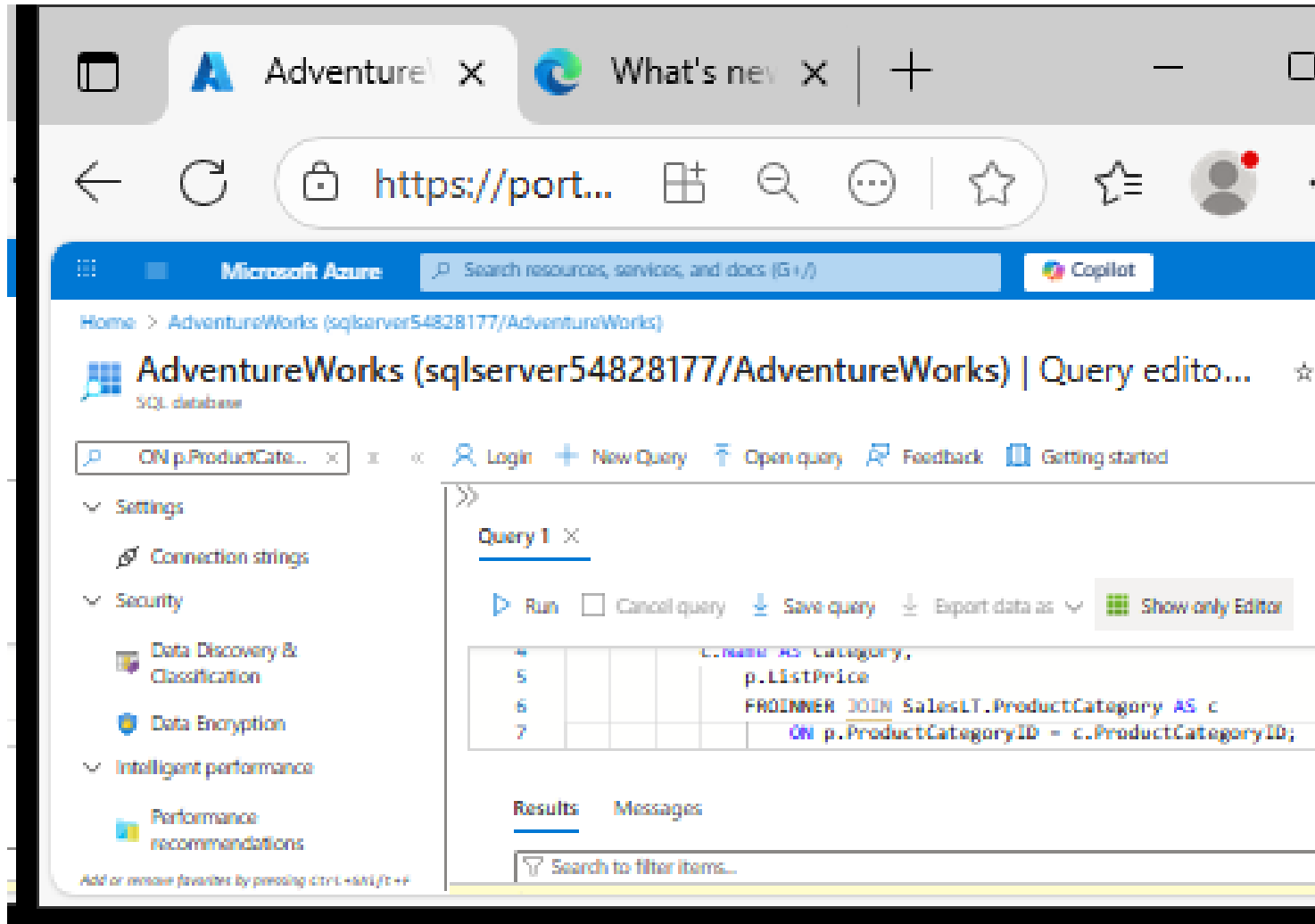
Week 8 Tutorial Activity (Step-by-Step)

Select all and remove them



Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run



The screenshot shows the Microsoft Azure portal interface for the AdventureWorks database. The browser tabs include 'AdventureWorks' and 'What's new'. The address bar shows 'https://port...'. The page title is 'AdventureWorks (sqlserver54828177/AdventureWorks) | Query editor...'. The left sidebar contains a navigation menu with options like Settings, Connection strings, Security, Data Discovery & Classification, Data Encryption, Intelligent performance, and Performance recommendations. The main area displays a query editor for 'Query 1' with the following SQL code:

```
SELECT c.Name AS Category,
       p.ListPrice
FROM SalesLT.ProductCategory AS c
JOIN SalesLT.Product AS p
ON p.ProductCategoryID = c.ProductCategoryID;
```

Below the query editor, there are tabs for 'Results' and 'Messages'. The 'Results' tab is active, showing a search bar with the text 'Search to filter items...'.

Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

What this query is doing

It is combining two tables to show product details along with their category names.

Step 14 – Take a screenshot) Click Type and Run

The two tables involved

1. Product table (SalesLT.Product)

ProductID	Name	ProductCategoryID	ListPrice
1	Bike	10	500
2	Helmet	20	50
3	Gloves	20	25

Step 14 – Take a screenshot) Click Type and Run

2. Category table (SalesLT.ProductCategory)

ProductCategoryID	Name
10	Vehicles
20	Accessories

Step 14 – Take a screenshot) Click Type and Run

What INNER JOIN does

It matches rows from both tables where:

$p.ProductCategoryID = c.ProductCategoryID$

In simple words:

- Find the category ID in the Product table
- Match it with the same ID in the Category table
- Combine the information into one row

Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Choose these columns:

- Product ID
- Product name (renamed as ProductName)
- Category name (from the other table)
- Price

Week 8 Tutorial Activity (Step-by-Step)

Step 14 – Take a screenshot) Click Type and Run

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

We are asking the database to show these columns:

- p.ProductID → the product's ID number
- p.Name AS ProductName → the product's name (renamed to ProductName so it's clearer)
- c.Name AS Category → the category name (renamed to Category)
- p.ListPrice → the product's price

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Start with the Product table

Give it a short name p (alias)

Look in the Product table (short name = p).

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Bring in the Category table

Give it alias c

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Match rows where both tables have the same category ID

Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

Combine the Product table (p) with the ProductCategory table (c).

- INNER JOIN means: only show rows where the ProductCategoryID matches in both tables.
- This lets us see each product together with its category name.

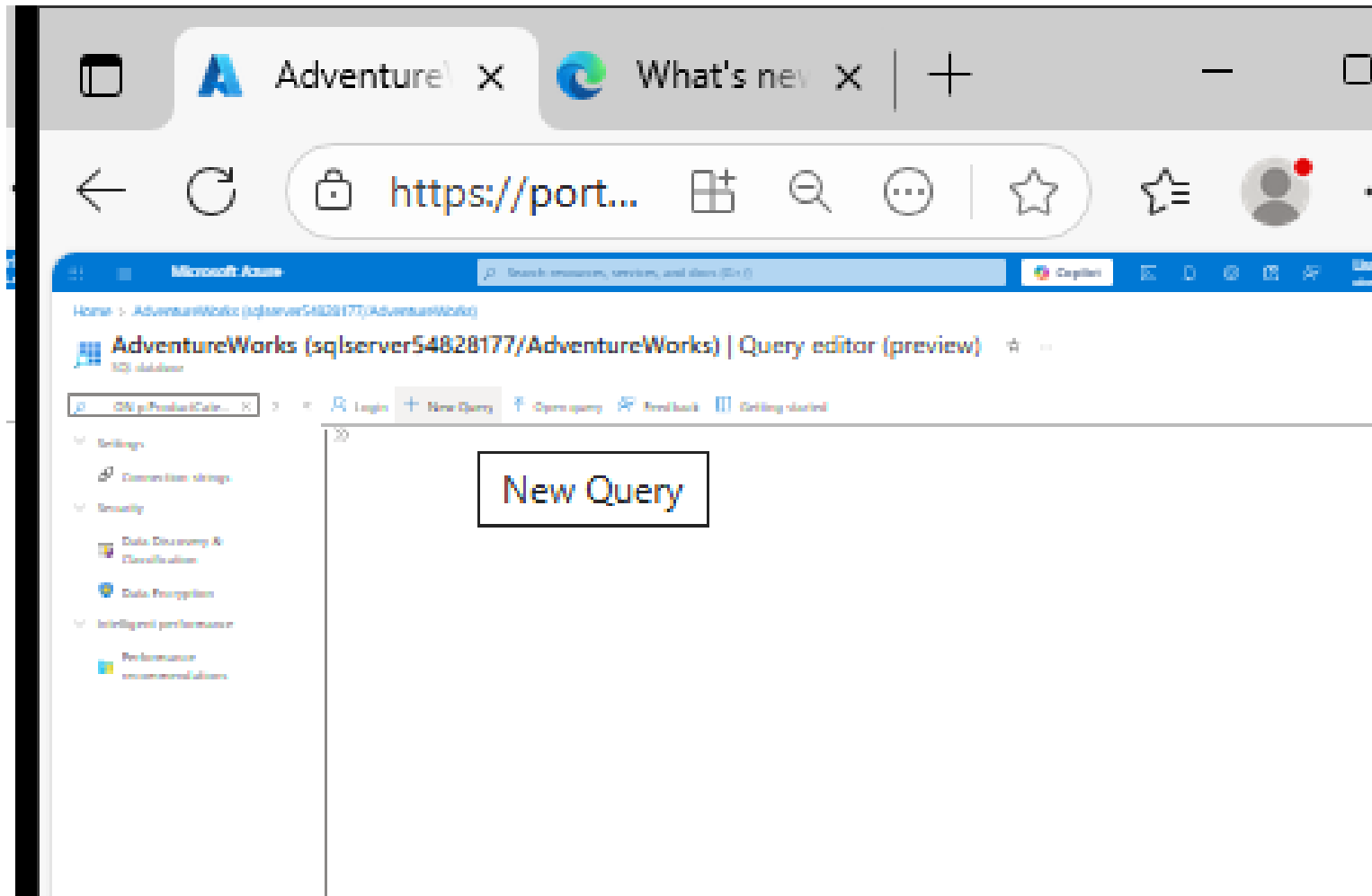
Week 8 Tutorial Activity (Step-by-Step)

```
SELECT
    p.ProductID,
    p.Name AS ProductName,
    c.Name AS Category,
    p.ListPrice
FROM SalesLT.Product AS p
INNER JOIN SalesLT.ProductCategory AS c
    ON p.ProductCategoryID = c.ProductCategoryID;
```

In summary, show me each product, its price, and the name of its category by matching the category IDs from both tables.

Week 8 Tutorial Activity (Step-by-Step)

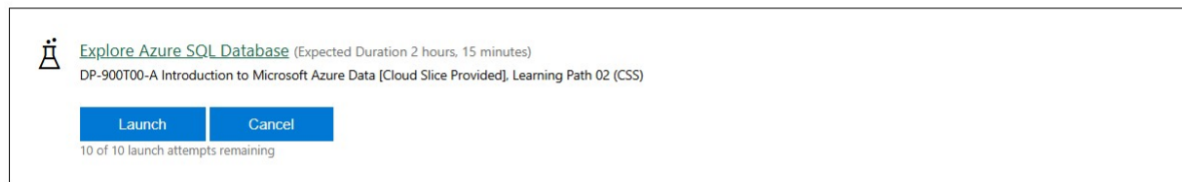
New Query



Week 8 Tutorial Activity (Step-by-Step)

- Query 1: Write a new query to retrieve the CustomerID, Title, last name, and Company Name of Customers whose Customer ID is in the following list: (1, 3, 5, 20, 45), sorted by Customer last name in reverse alphabetical order.

- Scroll down to reach the "Explore Azure SQL Database" lab. Then, click on the 'Launch' button.



- The Virtual Lab will begin loading in a new window (Screenshot below). This will take 1-2 minutes. Follow the on-screen instructions in the right-hand side window of the virtual lab to log in and complete the tasks as instructed. In step 3, use your Student ID+name as the "database name". **Make sure to use the resource group/server name and region provided on the right-hand side window.**
- Take a screenshot after completing steps 12, 13 and 14.
- Query 1: Write a new query to retrieve the CustomerID, Title, last name, and Company Name of Customers whose Customer ID is in the following list: (1, 3, 5, 20, 45), sorted by Customer last name in reverse alphabetical order.
- Query 2: Write a new query to retrieve a list of sales order revisions

Week 8 Tutorial Activity (Step-by-Step)

Query 1 – Take a screenshot) Type this

```
SELECT CustomerID, Title, LastName, CompanyName
FROM SalesLT.Customer
WHERE CustomerID IN (1, 3, 5, 20, 45)
ORDER BY LastName DESC;
```

SELECT

Choose these columns:

- CustomerID
- Title
- LastName
- CompanyName

Week 8 Tutorial Activity (Step-by-Step)

Query 1 – Take a screenshot) Type this

```
SELECT CustomerID, Title, LastName, CompanyName  
FROM SalesLT.Customer  
WHERE CustomerID IN (1, 3, 5, 20, 45)  
ORDER BY LastName DESC;
```

WHERE CustomerID IN (1, 3, 5, 20, 45)

This means:

- Only include customers with these IDs
- Ignore all others

Week 8 Tutorial Activity (Step-by-Step)

Query 1 – Take a screenshot) Type this

```
SELECT CustomerID, Title, LastName, CompanyName  
FROM SalesLT.Customer  
WHERE CustomerID IN (1, 3, 5, 20, 45)  
ORDER BY LastName DESC;
```

Selected rows with CustomerID: 1, 3, 5, 20, 45

ORDER BY LastName DESC;

Sort by LastName

- DESC = descending (Z → A)

Week 8 Tutorial Activity (Step-by-Step)

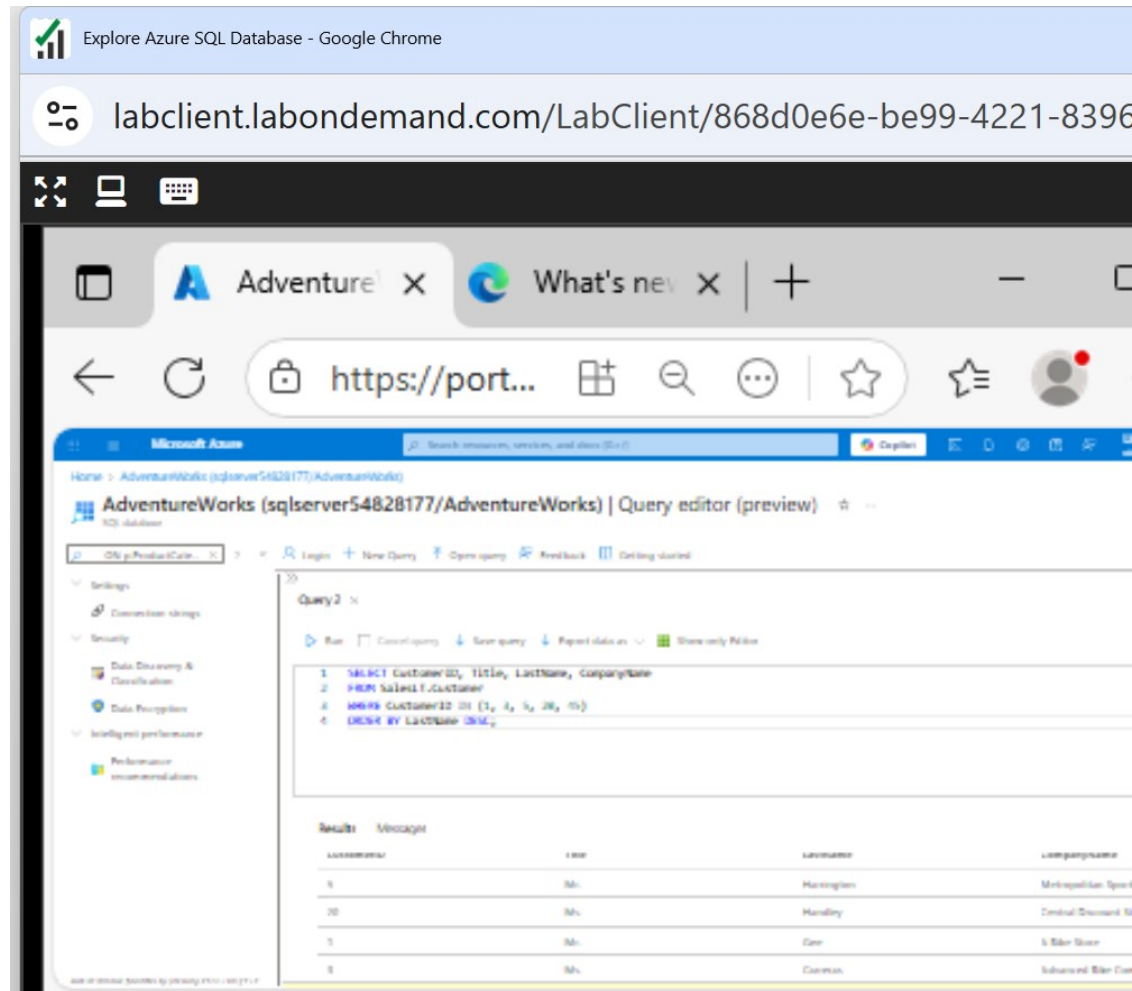
Query 1 – Take a screenshot) Type this

```
SELECT CustomerID, Title, LastName, CompanyName  
FROM SalesLT.Customer  
WHERE CustomerID IN (1, 3, 5, 20, 45)  
ORDER BY LastName DESC;
```

In summary, show specific customers (IDs 1, 3, 5, 20, 45) and sort them by last name in reverse alphabetical order.

Week 8 Tutorial Activity (Step-by-Step)

Query 1 – Take a screenshot) Run and See the results



Explore Azure SQL Database - Google Chrome

labclient.labondemand.com/LabClient/868d0e6e-be99-4221-8396

AdventureWorks | Query editor (preview)

Query2

```
1 SELECT CustomerID, Title, LastName, CompanyName
2 FROM SALESLT.Customer
3 WHERE CustomerID IN (1, 2, 3, 20, 15)
4 ORDER BY LastName DESC
```

Results

CustomerID	Title	LastName	CompanyName
1	Mr.	Harrington	Mountain Lakes Sport
20	Ms.	Hardley	Central Discount S
1	Mr.	Lee	Solar Shade
3	Ms.	Common	Suburban Solar Co

Query 2: Write a new query to retrieve a list of sales order revisions

- The SalesLT.SalesOrderHeader table contains records of sales orders.

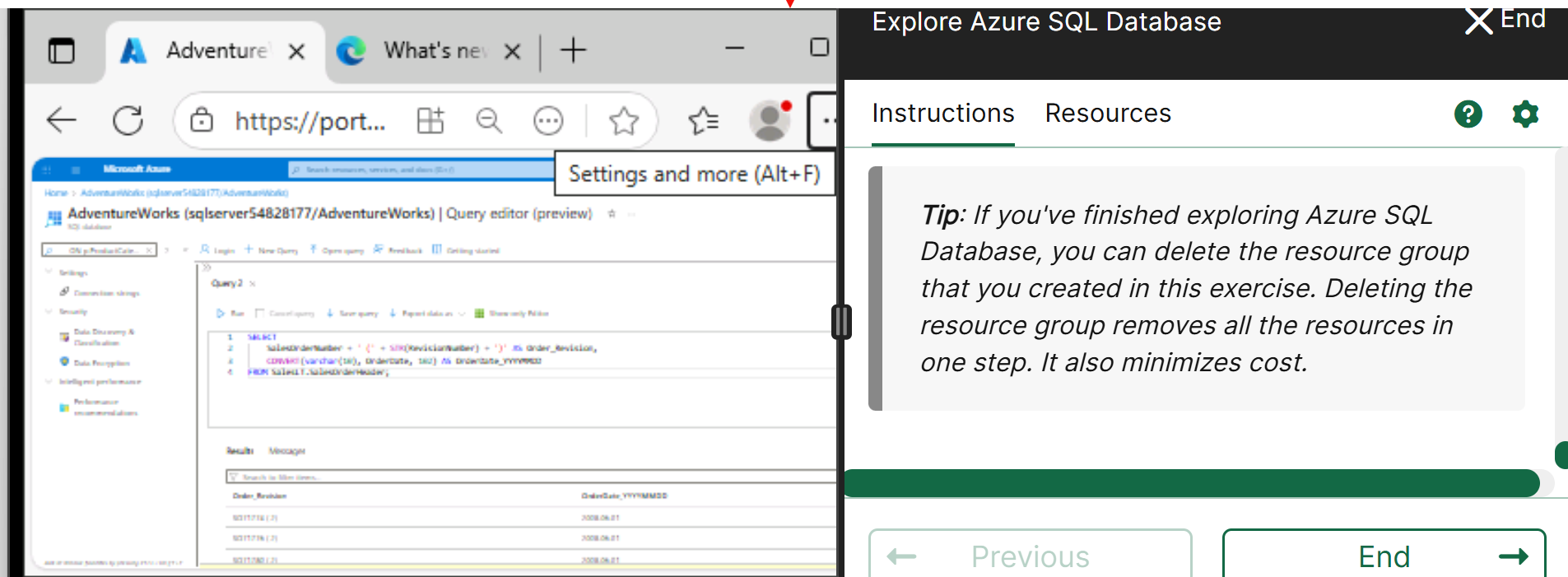
You have been asked to retrieve data for a report that shows:

- The sales order number and revision number in the format () – for example, SO71774 (2).
- The order date converted to ANSI standard *102* format (*yyyy.mm.dd* – for example, *2015.01.31*).
- Hint: Use the STR function to convert the number data type to String.
- Take a screenshot showing both the SQL code of the query and its results.

Week 8 Tutorial Activity (Step-by-Step)

SELECT

```
SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD  
FROM SalesLT.SalesOrderHeader;
```



AdventureWorks (sqlserver54828177/AdventureWorks) | Query editor (preview)

Query 2

```
1 SELECT  
2 SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
3 CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD  
4 FROM SalesLT.SalesOrderHeader;
```

Results Message

Order_Revision	OrderDate_YYYYMMDD
5011718 (2)	2008.06.01
5011719 (2)	2008.06.01
5011720 (2)	2008.06.01

Explore Azure SQL Database

Instructions Resources

Tip: If you've finished exploring Azure SQL Database, you can delete the resource group that you created in this exercise. Deleting the resource group removes all the resources in one step. It also minimizes cost.

Previous End

```
SELECT  
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD  
FROM SalesLT.SalesOrderHeader;
```

What this query does

It:

1. Combines order number + revision number into one column
2. Formats the date into a specific style (YYYY.MM.DD)

SELECT

```
SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
```

FROM SalesLT.SalesOrderHeader;

Step 1: Creating Order + Revision

SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS
Order_Revision

What's happening:

- Take SalesOrderNumber → e.g. SO123
- Add " ("

SELECT

```
SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
```

FROM SalesLT.SalesOrderHeader;

- Convert RevisionNumber (number → text) using STR()
- Add ")"

```
SELECT  
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD  
FROM SalesLT.SalesOrderHeader;
```

Step 2: Formatting the date

CONVERT(varchar(10), OrderDate, 102) AS
OrderDate_YYYYMMDD

This means:

- Convert the date into text (varchar)
- Use format 102

Week 8 Tutorial Activity (Step-by-Step)

SELECT

```
SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
```

FROM SalesLT.SalesOrderHeader;

Format 102 =

YYYY.MM.DD

Example:

- 2026-04-20 → 2026.04.20

```
SELECT  
    SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
    CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD  
FROM SalesLT.SalesOrderHeader;
```

First column

Join order number and revision number into one string

Second column

Change the date into a readable text format (YYYY.MM.DD)

FROM

Get data from the SalesOrderHeader table

SELECT

```
SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
```

FROM SalesLT.SalesOrderHeader;

In summary, show each order as 'OrderNumber (Revision)' and display the date in YYYY.MM.DD format.

SELECT

```
SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
```

FROM SalesLT.SalesOrderHeader;

- **SalesOrderNumber + ' (' + STR(RevisionNumber) + ')':**
 - Joins the order number with the revision number.
 - Example: SO71774 (2).
 - STR() changes the RevisionNumber (a number) into text so it can be joined.
- Renamed as Order_Revision.

SELECT

```
SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
```

FROM SalesLT.SalesOrderHeader;

- **CONVERT(varchar(10), OrderDate, 102):**
 - Converts the date into the format yyyy.mm.dd.
 - Example: 2015.01.31.
 - Renamed as OrderDate_YYYYMMDD.

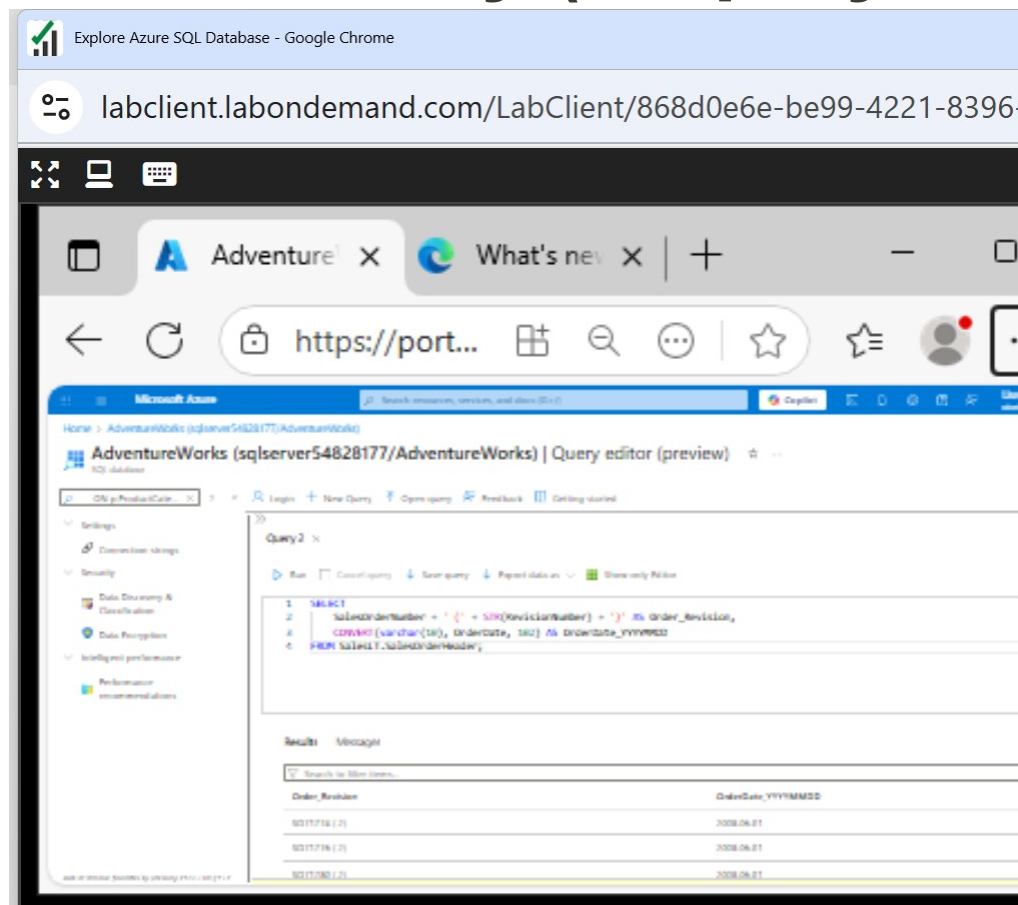
SELECT

```
SalesOrderNumber + ' (' + STR(RevisionNumber) + ')' AS Order_Revision,  
CONVERT(varchar(10), OrderDate, 102) AS OrderDate_YYYYMMDD
```

FROM SalesLT.SalesOrderHeader;

- **FROM SalesLT.SalesOrderHeader:**
 - Tells SQL to pull the data from the SalesLT.SalesOrderHeader table.

Week 8 Tutorial Activity (Step-by-Step)

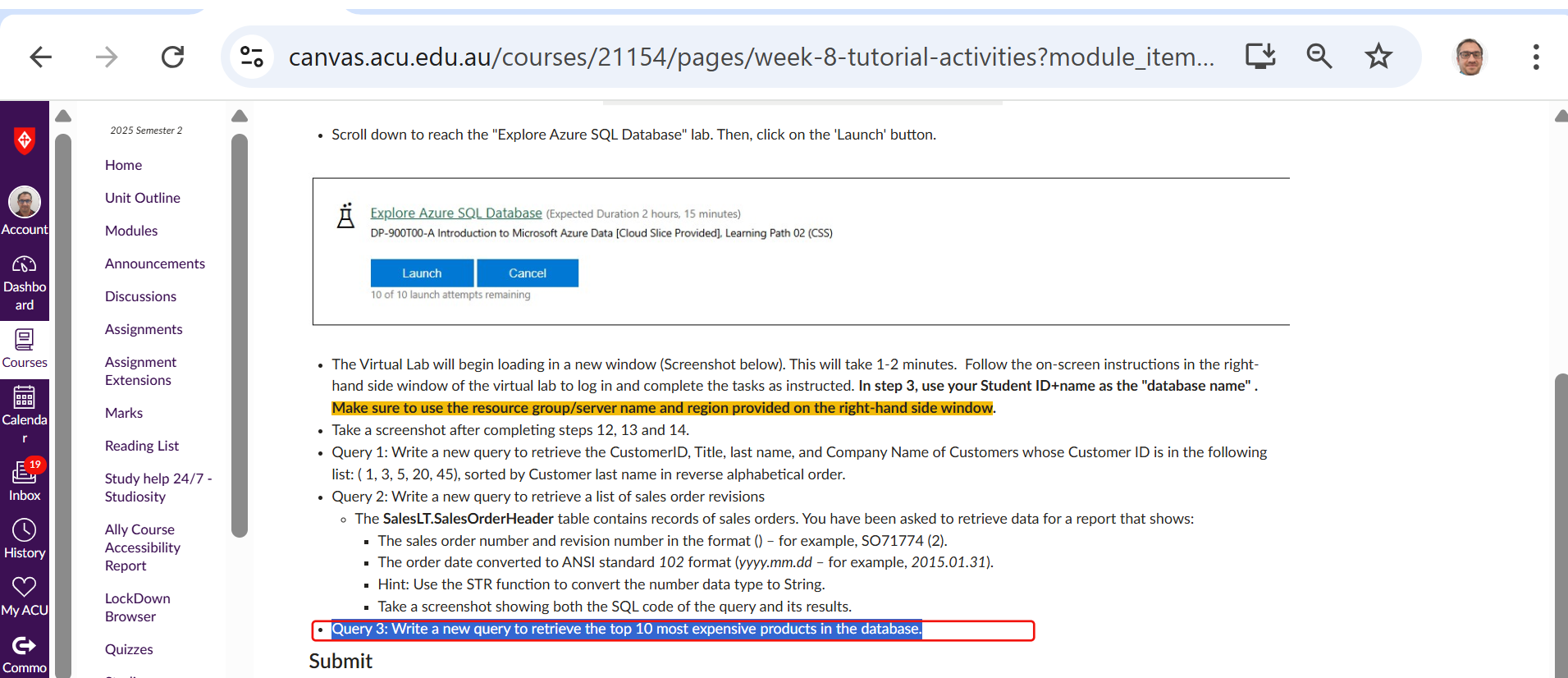


The result should be two columns:

1. `_YYYYMOrder_Revision` (e.g., SO71774 (2))
2. `OrderDateMDD` (e.g., 2015.01.31)

Week 8 Tutorial Activity (Step-by-Step)

Query 3: Write a new query to retrieve the top 10 most expensive products in the database.



The screenshot shows the Canvas LMS interface. The left sidebar contains navigation links: Home, Unit Outline, Modules, Announcements, Discussions, Assignments, Assignment Extensions, Marks, Reading List, Study help 24/7 - Studiosity, Ally Course Accessibility Report, LockDown Browser, Quizzes, and a search bar. The main content area displays a virtual lab launch for 'Explore Azure SQL Database' (Expected Duration 2 hours, 15 minutes). Below the lab title, there are 'Launch' and 'Cancel' buttons. A message indicates '10 of 10 launch attempts remaining'. The lab instructions list several tasks, including writing a new query to retrieve the top 10 most expensive products in the database. The 'Launch' button is highlighted with a red box.

2025 Semester 2

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Launch

Cancel

10 of 10 launch attempts remaining

- Scroll down to reach the "Explore Azure SQL Database" lab. Then, click on the 'Launch' button.
- The Virtual Lab will begin loading in a new window (Screenshot below). This will take 1-2 minutes. Follow the on-screen instructions in the right-hand side window of the virtual lab to log in and complete the tasks as instructed. In step 3, use your Student ID+name as the "database name". **Make sure to use the resource group/server name and region provided on the right-hand side window.**
- Take a screenshot after completing steps 12, 13 and 14.
- Query 1: Write a new query to retrieve the CustomerID, Title, last name, and Company Name of Customers whose Customer ID is in the following list: (1, 3, 5, 20, 45), sorted by Customer last name in reverse alphabetical order.
- Query 2: Write a new query to retrieve a list of sales order revisions
 - The **SalesLT.SalesOrderHeader** table contains records of sales orders. You have been asked to retrieve data for a report that shows:
 - The sales order number and revision number in the format () – for example, SO71774 (2).
 - The order date converted to ANSI standard 102 format (yyyy.mm.dd – for example, 2015.01.31).
 - Hint: Use the STR function to convert the number data type to String.
 - Take a screenshot showing both the SQL code of the query and its results.
- **Query 3: Write a new query to retrieve the top 10 most expensive products in the database.**

Submit

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products in the database.

What this query does

It shows the 10 most expensive products (that actually have a price).

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products in the database.

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

Step 1: WHERE clause

WHERE ListPrice IS NOT
NULL

This means:

- Ignore products with no price (NULL)
- Only keep products that have a price

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products in the database.

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

Step 2: ORDER BY

ORDER BY ListPrice
DESC;

Sort by price:

- DESC = highest → lowest

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products in the database.

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

Step 3: TOP (10)

SELECT TOP (10)

- Only show the first 10 rows
- Since we sorted highest first, this gives:
- Top 10 most expensive products

Query 3 – Take a screenshot: Write a new query to retrieve the top 10 most expensive products

```
SELECT TOP (10)
    ProductID,
    Name,
    ListPrice
FROM SalesLT.Product
WHERE ListPrice IS NOT NULL
ORDER BY ListPrice DESC;
```

SELECT TOP (10)

Only show the first 10 results

WHERE

Ignore products with no price

ORDER BY DESC

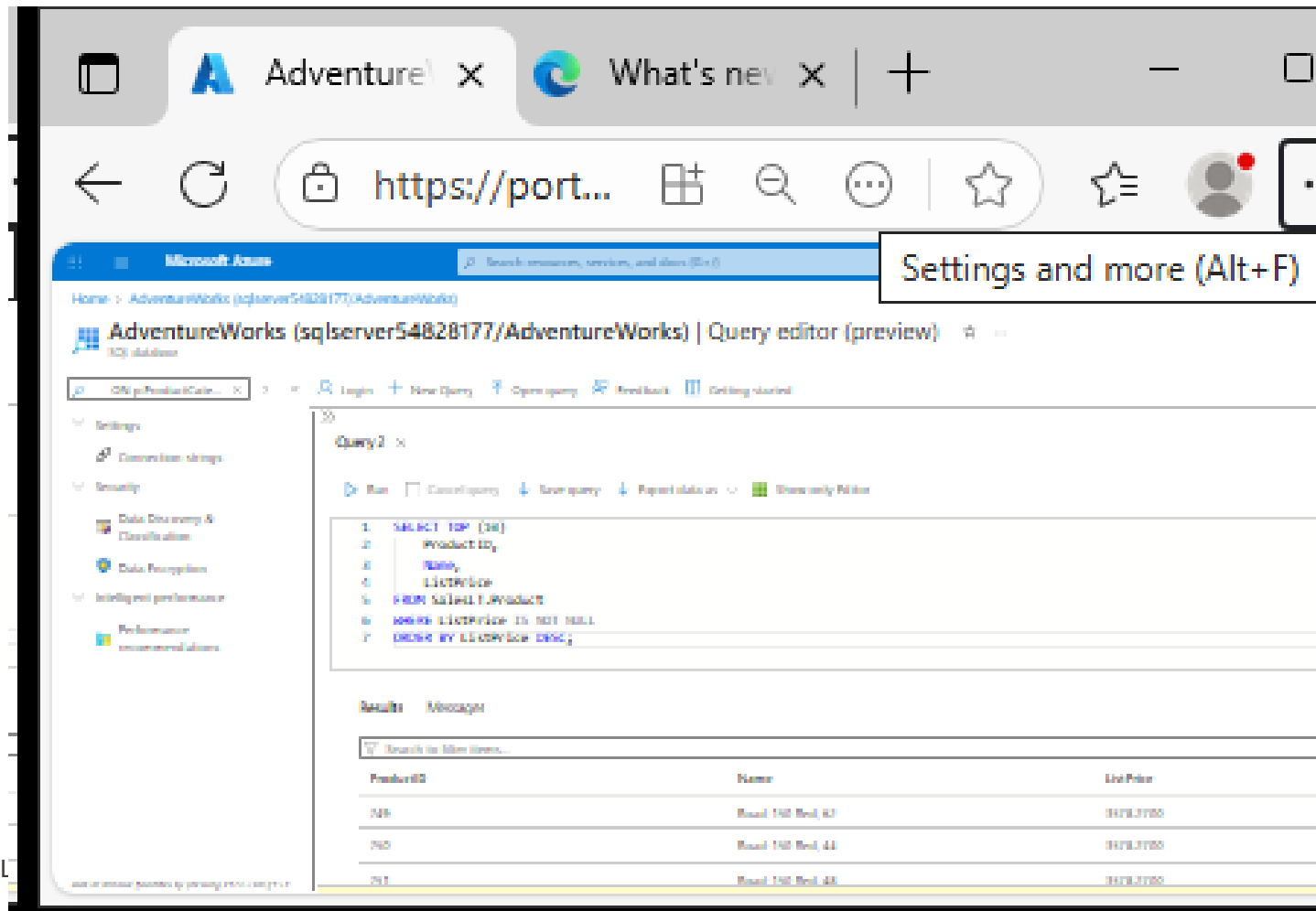
Sort from most expensive to
cheapest

One-line summary

Show the 10 most expensive
products that have a price.

Week 8 Tutorial Activity (Step-by-Step)

The results lists the 10 most expensive products in the database, starting with the highest price at the top. (Take a screenshot)



The screenshot shows the Microsoft Azure portal interface. The browser tabs include 'Adventure' and 'What's new'. The address bar shows 'https://port...'. The page title is 'AdventureWorks (sqlserver54828177/AdventureWorks) | Query editor (preview)'. The left sidebar shows navigation options like 'Settings', 'Connection strings', 'Security', 'Data Discovery & Classification', 'Data Protection', 'Intelligent performance', and 'Performance recommendations'. The main area displays a SQL query in the 'Query2' editor:

```
1 SELECT TOP (10)  
2     ProductID  
3     Name,  
4     ListPrice  
5 FROM SalesLT.Product  
6 WHERE ListPrice IS NOT NULL  
7 ORDER BY ListPrice DESC;
```

Below the query editor, the 'Results' tab is active, showing a table with 3 columns: ProductID, Name, and ListPrice. The table contains 3 rows of data:

ProductID	Name	ListPrice
749	Road 140 Road, 63	1479.0000
740	Road 140 Road, 44	1479.0000
751	Road 140 Road, 48	1479.0000

Submit

Paste the screenshots of completing steps 12, 13 and 14, Query 1, Query 2, and Query 3 (6 screenshots) in a single Word/pdf file and submit using the assignment module 8 submission link. The screenshots must show both the SQL code of the query and its results.

Quick fixes (common issues)

- “Customers table not found” → Use the right name: SalesLT.Customer (not Customers)
- Login fails in Query editor → Add your client IP in Networking
- No sample tables → You likely missed AdventureWorksLT; use SELECT 1; or create a simple table to test

What the answers should show (in words)

- Q1: Only those 5 CustomerIDs, sorted by LastName descending
- Q2: Orders with their RevisionNumber (and a neat date), or a count per revision
- Q3: The 10 highest ListPrice products, from most to less expensive

- **What did we learn?**
 - Difference: Access vs Azure SQL.
 - Provision DBs in the cloud.
 - Write SQL queries.
- **Why important?**
 - Skills → Azure Fundamentals Certification.
 - Real jobs in data engineering, DB admin.

1. **Q:** Why can't Woolworths or Qantas run their operations on MS Access?

A: Access can't handle thousands of concurrent users & terabytes of data.

2. **Q:** Debate: Should Australian universities keep data on campus servers or move to Azure Cloud?

For: Cloud = secure, scalable, cheaper long-term.

Against: Privacy risks, ongoing cost, internet dependency.

3. Q: If Azure SQL is PaaS, what control do we lose vs IaaS?

A: We lose full control of the server (patching, OS-level settings).

- *Have a Great Learning Day!*
- Feel free to reach out with any questions!
- Dr. Farshid Keivanian